MAMMA ADA CENTRE

In Collaboration with FACM ITALY

Class 11

Innovate Today Lead Tomorrow

From Ideas To Global Recognition

GAIO

Global AI Olympiad

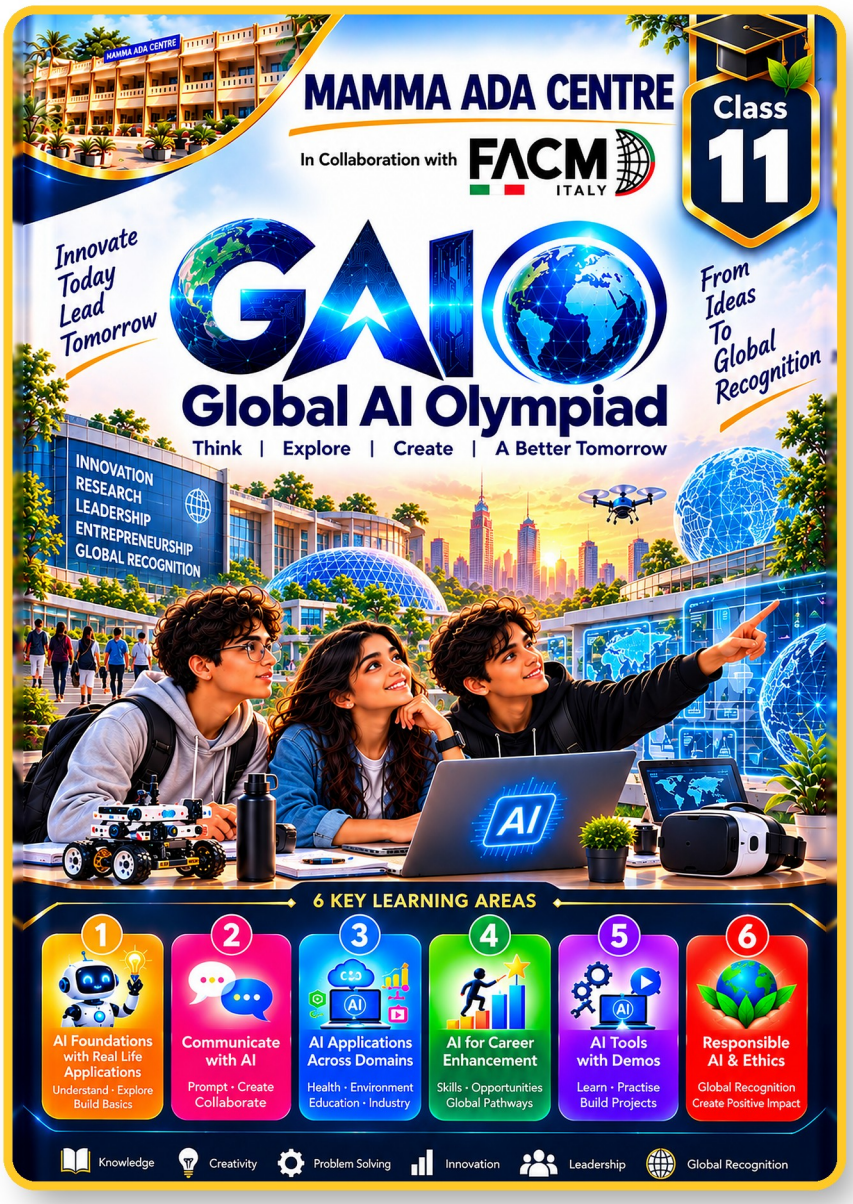
Think | Explore | Create | A Better Tomorrow

6 KEY LEARNING AREAS

1. **AI Foundations with Real Life Applications**
   Understand · Explore
   Build Basics
2. **Communicate with AI**
   Prompt · Create
   Collaborate
3. **AI Applications Across Domains**
   Health · Environment
   Education · Industry
4. **AI for Career Enhancement**
   Skills · Opportunities
   Global Pathways
5. **AI Tools with Demos**
   Learn · Practise
   Build Projects
6. **Responsible AI & Ethics**
   Global Recognition
   Create Positive Impact

Knowledge · Creativity · Problem Solving · Innovation · Leadership · Global Recognition

This book belongs to

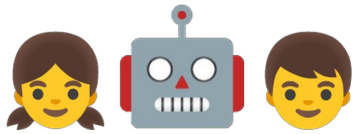


Name: _____

Class & Section: _____

School: _____

📖 PREFACE



Artificial Intelligence is shaping how we learn, communicate, travel, farm, heal and create. As senior students, you are ready not only to use AI tools, but to understand how they work, question their answers and use them responsibly. This book is designed to build exactly those skills.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of Class 11 learners.

Every topic opens with a clear introduction and continues with "Tech Talk", a group chat in which Riya, Kabir and their AI mentor Bolt explore ideas through real-life examples, then a detailed "Concept Explained" chapter. A "How it works" flow, an India Spotlight, a hands-on Demo Lab with real AI tools, an "AI Chronicles" story, worksheets, quizzes, and standard and case-based exercises help you learn by reading, doing, questioning and creating.

Values are at the heart of this book. Along with learning what AI can do, children learn that people create and guide technology, that AI can make mistakes, and that safety, kindness and privacy matter online and offline.

🤝 A note for teachers and parents

Encourage students to try the Demo Labs under supervision, discuss the case studies, and question AI answers rather than accept them blindly. Praise reasoning and curiosity as much as right answers. An answer key is provided at the end of the book.

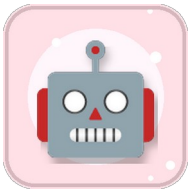
We hope this book sparks curiosity, creativity and confidence in every young learner. Curious minds, brighter futures!

With best wishes,

The GAIO Academic Team

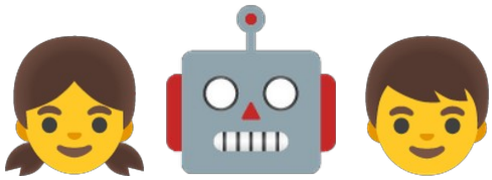
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Super Seniors!



MEET BOLT!

Welcome, Class 11! I'm Bolt. College-level AI starts now: neural networks, gradient descent, pandas, prompt engineering, fairness audits and privacy. Let's lead!



Look for these pictures in your book:

TECH TALK Concepts in a group chat

CONCEPT EXPLAINED Concept explained in depth

AI CHRONICLES Stories of teens & AI

LOOK AROUND YOU See AI in real life

LAB ACTIVITY Do a fun activity

WORKSHEET Write, tick and draw

PUZZLE & DEFINE Solve puzzles

DEMO LAB Hands-on AI tool demo

QUIZ TIME Answer quiz questions

EXERCISE Standard & case-based Qs

KEY TERMS New words to learn

UNIT REVIEW Check your unit

My 6-Month AI Journey

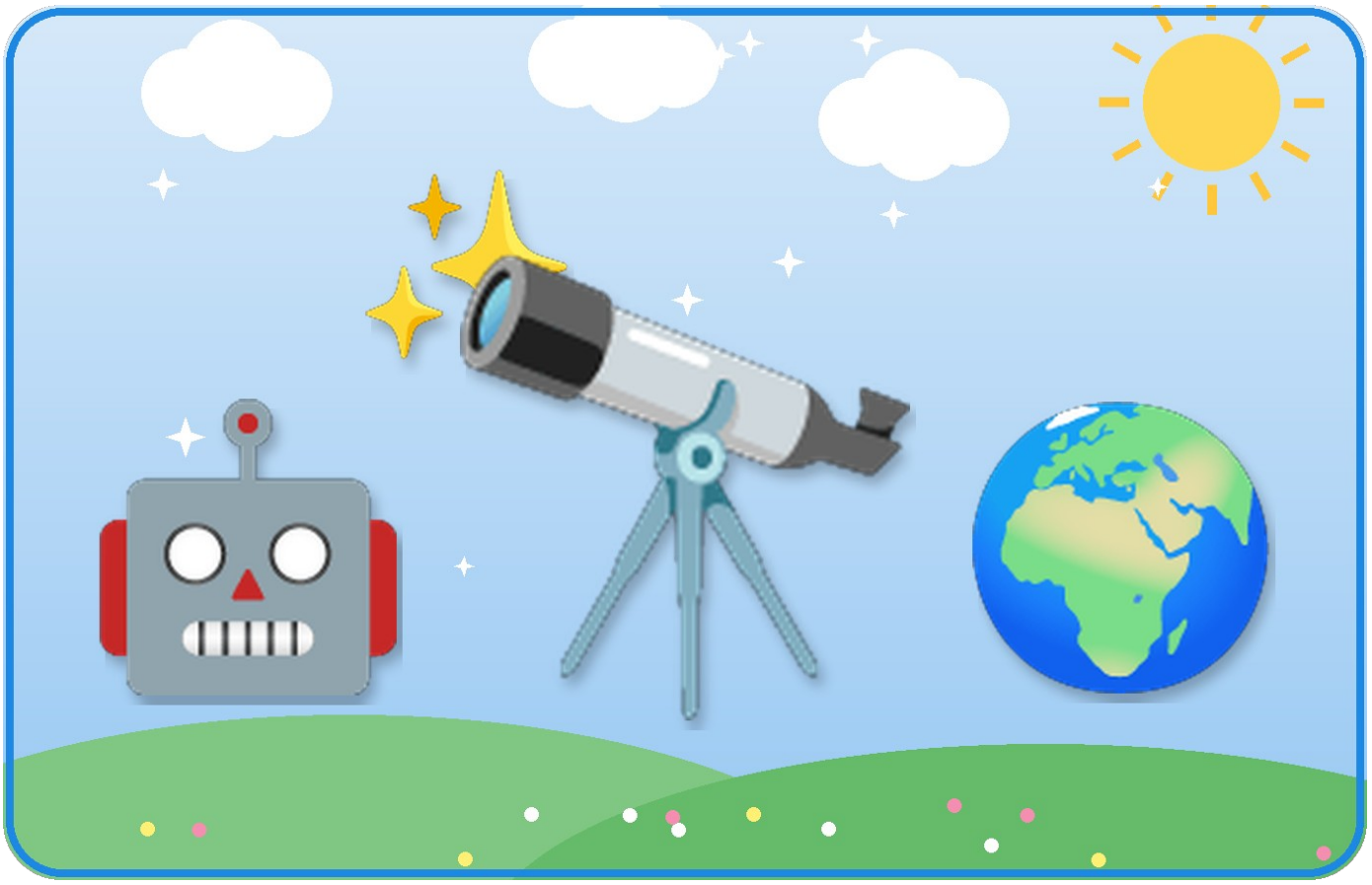
Follow the path. One month at a time!

	MONTH 1 **AI DISCOVER** AI Basics	**Inside Intelligent Machines • Predict, Learn & Improve**
	MONTH 2 **AI CONNECT** Communicate with AI	**Python for Smart Solutions • The Art of Prompting**
	MONTH 3 **AI SOLVE** AI Applications	**AI Powers Innovation • AI Across Industries**
	MONTH 4 **AI RISE** AI & Career	**AI Career Universe • Create Your Career Blueprint**
	MONTH 5 **AI CREATE** AI Tools	**AI Research Desk • AI Creation Studio**
	MONTH 6 **AI CARE** Responsible AI	**Fair AI Challenge • Privacy in an AI World**

MONTH 1

AI DISCOVER

See • Understand • Explore



Area: AI Basics

In this unit, you will explore:

Topic 1

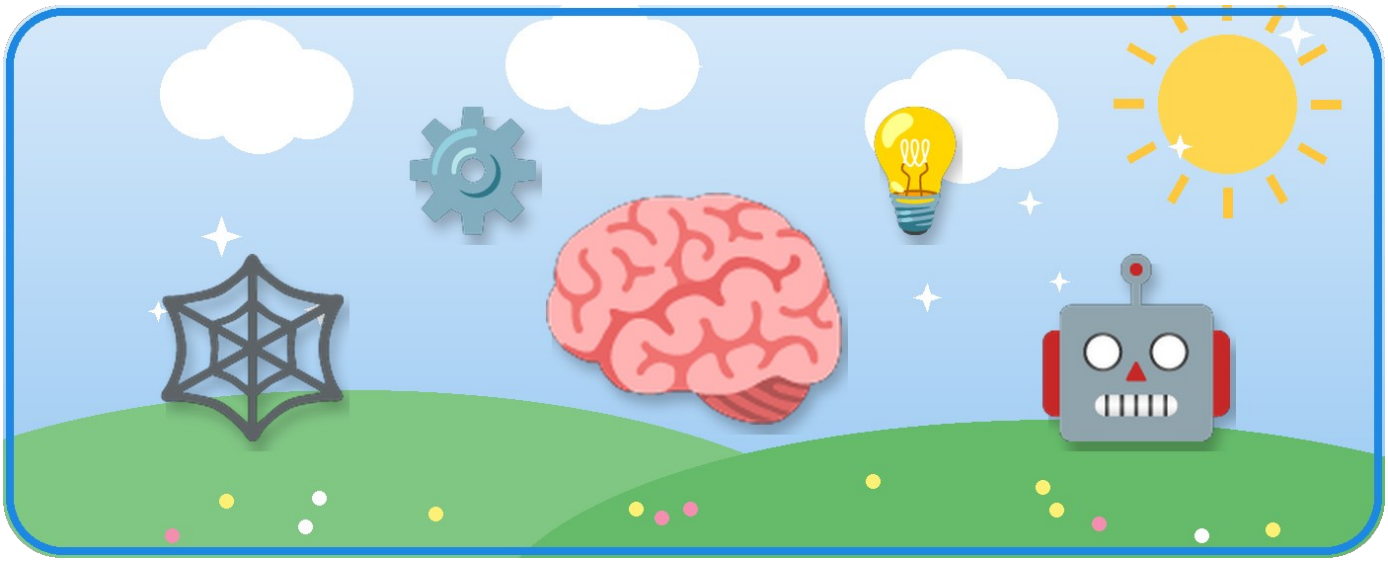
Inside Intelligent Machines

Topic 2

Predict, Learn & Improve

TOPIC 1 • MONTH 1 • AI BASICS

Inside Intelligent Machines



LET'S LEARN

- Neural networks are layers of connected neurons.
- Each neuron: inputs × weights + bias → activation.
- Deep learning = many hidden layers stacked.

Network building blocks

Input layer

Hidden layers

Activation

Output layer

KEY TERMS

Neuron Weighted sum unit

Activation Output decision function

Architecture Network's structure

💬 TECH TALK: Inside Intelligent Machines

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir
Bolt, what actually happens inside a neural network 🧠?
10:02

🤖 Bolt
Layers of artificial neurons. Each neuron computes inputs × weights + bias ➕.
10:04 ✓✓

👧 Riya
Then?
10:06

🤖 Bolt
An activation function decides the output ⚡. ReLU, for example, turns negatives into 0.
10:08 ✓✓

👦 Kabir
Why do we need activation?
10:10

🤖 Bolt
Without it, stacking layers is just one big linear equation. Activation adds non-linearity 〰️, so the network can learn complex patterns.
10:12 ✓✓

Riya

And 'deep' learning?

10:14

Bolt

Many hidden layers stacked 🏗️. Early layers detect edges; later layers detect faces or words.

10:16 ✓✓

Kabir

The design of layers is the architecture 📐: CNNs for images, transformers for language.

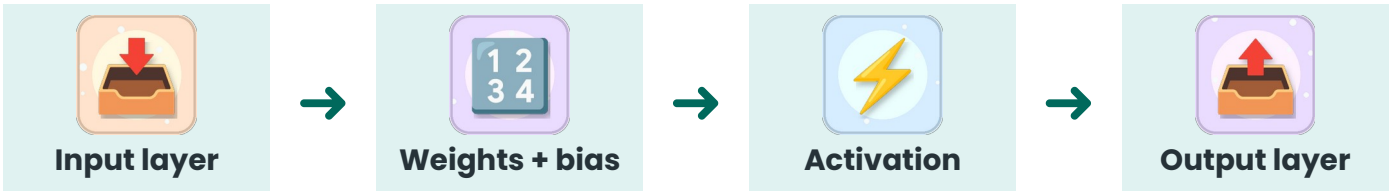
10:18

😊 Type a message...

📌 **Pinned by Bolt**

A neural network is layers of neurons. Each computes inputs × weights + bias, then an activation. Deep learning stacks many hidden layers; architecture is the layer design.

🔄 How it works



INDIA SPOTLIGHT

Indian research labs and start-ups train deep neural networks for speech, vision and Indian-language AI.

📖 CONCEPT EXPLAINED: Inside Intelligent Machines

💡 In this chapter

Neural networks power almost every headline-making AI system today, from face unlock to chatbots. Yet the core idea is surprisingly simple: many tiny calculators, called neurons, connected in layers. In this chapter we open the box and see how a single neuron works, why we stack them into layers, what activation functions do, and how different architectures suit different kinds of data.

🧠 1. From brain cells to artificial neurons

Your brain contains about 86 billion nerve cells. Each receives signals from others, and if the combined signal is strong enough, it 'fires' and passes a signal on. In 1943, Warren McCulloch and Walter Pitts proposed a mathematical version of this idea, and in 1958 Frank Rosenblatt built the 'perceptron', a machine that could learn to classify simple patterns.

An artificial neuron is not a copy of a brain cell; it is a simple formula inspired by one. It takes several numbers as inputs, multiplies each by a weight that says how important that input is, adds them up, adds a bias, and passes the result through an activation function. Millions of such neurons, connected together, can recognise faces or translate languages.

🤔 PAUSE & THINK

Biological neurons inspired artificial ones, but AI does not 'think' like a brain. List two ways an artificial neuron is much simpler than a real brain cell.

➕ 2. Inside one neuron: weights, bias and the weighted sum

Imagine a neuron deciding whether a student will enjoy a cricket match. Inputs might be x1 = hours of free time, x2 = interest in cricket (0 to 10) and x3 = whether friends are going (0 or 1). Each input gets a weight: w1 = 0.2, w2 = 0.6, w3 = 1.5. The neuron computes the weighted sum $z = w1 \cdot x1 + w2 \cdot x2 + w3 \cdot x3 + b$, where b is the bias.

Weights encode importance: a large positive weight means 'this input strongly pushes the answer up', a negative weight pushes it down, and a weight near zero means the input barely matters. The bias shifts the whole result up or down, like a neuron's general

mood. Learning, as we will see in the next topic, is simply the process of finding good values for all these weights and biases.

✏️ **WORKED EXAMPLE**

Let x1 = 3, x2 = 8, x3 = 1, with weights 0.2, 0.6, 1.5 and bias b = -4. Then z = 0.2×3 + 0.6×8 + 1.5×1 - 4 = 0.6 + 4.8 + 1.5 - 4 = 2.9. A positive z suggests 'likely to enjoy'. If friends weren't going (x3 = 0), z = 1.4: still positive but weaker.

⚡ 3. Activation functions: adding the bend

After computing z, the neuron applies an activation function. The step function outputs 1 if z > 0 and 0 otherwise, like a simple on/off switch. The sigmoid squashes any number into the range 0 to 1, useful for probabilities. ReLU (Rectified Linear Unit) outputs max(0, z): negatives become zero, positives pass unchanged. ReLU is fast and is the most common choice in modern deep networks.

Why do we need activation at all? Without it, every neuron just computes a straight-line (linear) combination, and stacking linear layers still gives one big linear function, no more powerful than a single layer. Activation functions introduce non-linearity, letting the network bend its decision boundaries into curves, circles and complex shapes. That is what allows deep networks to model real-world patterns.

✏️ **WORKED EXAMPLE**

With z = 2.9: ReLU(2.9) = 2.9; sigmoid(2.9) ≈ 0.95 (a 95% 'probability'); step(2.9) = 1. With z = -1.2: ReLU = 0; sigmoid ≈ 0.23; step = 0.

🏗️ 4. Layers and deep learning

Neurons are organised in layers. The input layer receives raw data, such as the 784 pixel values of a 28×28 handwritten digit image. One or more hidden layers transform the data, and the output layer gives the answer, for example ten neurons for the digits 0 to 9. When there are many hidden layers, we call it deep learning.

Each layer builds on the previous one. In an image network, early layers detect simple features like edges and corners, middle layers combine them into shapes like circles and loops, and deeper layers recognise whole objects like a '6' or a face. This automatic feature learning is the great strength of deep learning: we no longer have to hand-design every feature.

🤔 **PAUSE & THINK**

A network has an input layer of 784 neurons, one hidden layer of 128 and an output layer of 10. If every neuron connects to every neuron in the next layer, roughly how many weights are there? (Hint: 784×128 + 128×10.)

📐 5. Architectures: the right shape for the data

The arrangement of layers and connections is called the architecture. A fully connected (dense) network links every neuron to every neuron in the next layer; it suits tabular data. Convolutional Neural Networks (CNNs) slide small filters across images to detect local patterns; they power medical scan analysis, number-plate readers and face unlock.

Recurrent Neural Networks (RNNs) were designed for sequences such as speech and text, processing one step at a time. Since 2017, Transformers have largely replaced them for language: they use 'attention' to look at all words at once and decide which matter most. Choosing an architecture is like choosing a vehicle: a bicycle, a truck and a boat are all useful, but for different journeys.

🖐 **TRY IT**

Open playground.tensorflow.org. Try the spiral dataset with one hidden layer of 2 neurons, then with three hidden layers of 6 neurons each. Record the test loss and sketch the decision boundary in each case.

✅ Key takeaways

- A neuron computes a weighted sum of inputs plus a bias, then applies an activation function.
- Weights encode importance; learning means finding good weights and biases.
- Activation functions such as ReLU and sigmoid add the non-linearity that makes networks powerful.
- Deep learning stacks many hidden layers that learn features automatically, from edges to objects.
- Architectures suit data types: dense for tables, CNNs for images, Transformers for language.

? Quick check: answer in one or two lines.

1. Write the formula for the output of a single neuron before activation.

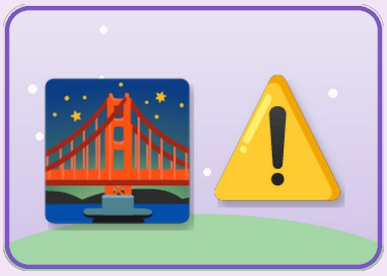
2. Why would a deep network with only linear activations be no better than one layer?

3. What kinds of features do early and deep layers of an image CNN detect?

4. Which architecture would you choose for translating Hindi to English, and why?

AI CHRONICLES: The Network That Saw Cracks

Riya Kabir Engineer Dev



SCENE 1

After a bridge collapse in another state, Engineer Dev asked students: can AI spot cracks in concrete 🔍?

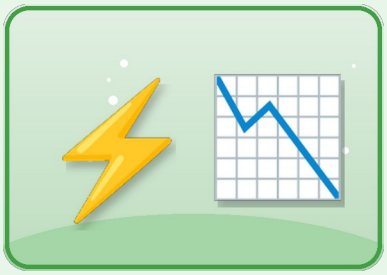
Riya and Kabir collected 2,000 labelled photos: 'crack' and 'no crack'.

SCENE 2

SCENE 3

They built a small CNN: convolution layers to detect edges, then deeper layers for crack shapes.

With a linear activation, it failed. Switching to ReLU, accuracy jumped from 61% to 93% 🚀.



SCENE 4

SCENE 5

Wet surfaces confused it. They added rainy-day photos and retrained.

Dev used it to prioritise inspections. "AI flags, engineers decide," he said.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They test it on real drone footage	**They remove the activation function**	**The bridge learns to repair itself**

🧐 Story check: answer the questions.

1. What architecture did they use and why?

2. What changed when they switched to ReLU?

3. How did they fix the wet-surface problem?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Neural networks at work

Face unlock (CNNs)

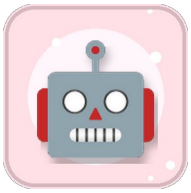
Speech to text

Translation

Medical imaging

Self-driving vision

Chatbots (Transformers)



BOLT SAYS:

A neuron alone is simple maths. Millions of them together can recognise faces!

FUN FACT!

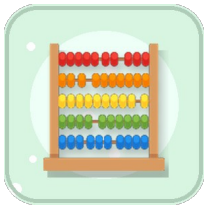
Artificial neurons were first proposed in 1943 by McCulloch and Pitts!

I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: Paper Perceptron

You need:

Worksheet



Calculator

Pairs

Steps:

1.

   Inputs: x1 = 1, x2 = 0. Weights: w1 = 0.6, w2 = 0.4.

2.

   Compute z = x1·w1 + x2·w2 + bias(−0.5).

3.

   Activation: output 1 if z > 0, else 0.

4.

   Try all 4 input pairs. Which logic gate is it?

THINK & TALK

Why can't a single perceptron learn XOR?

Draw what you did!

WORKSHEET: Inside Intelligent Machines (Part A)

Name: _____ Date: _____

1. Match the component to its role.

Input layer ●

● Gives the final prediction

Hidden layer ●

● Receives raw features

Activation ●

● Adds non-linearity

Output layer ●

● Learns intermediate patterns

2. Compute the neuron output. Use the word box.

WORD BOX: 0.1 ✿ 1 ✿ 0.9

z = 1×0.6 + 1×0.4 - 0.1 = ____

step(0.9) with threshold 0 = ____

z = 1×0.6 + 0×0.4 - 0.5 = ____

WORKSHEET: Inside Intelligent Machines (Part B)

Name: _____ Date: _____

1. Linear or non-linear activation? Tick ✓.

	Linear	Non-linear
$f(z) = z$		
ReLU: $\max(0, z)$		
Sigmoid		
$f(z) = 2z + 1$		

2. Forward pass of a neuron. Write 1, 2, 3, 4.

Pass output to next layer

☐

Receive inputs

☐

Apply activation

☐

Weighted sum + bias

☐

3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE & DEFINE: Inside Intelligent Machines

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Neuron

Meaning: _____

Example: _____

Activation

Meaning: _____

Example: _____

Architecture

Meaning: _____

Example: _____

🧪 DEMO LAB: TensorFlow Playground (with teacher)

Name: _____ Date: _____

🎯 **Goal:** See how neurons, layers and activations learn patterns.

🔗 **Open:** playground.tensorflow.org | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Choose the 'circle' dataset. Start with 1 hidden layer of 2 neurons.
2		Press Play. Watch the test loss for 200 epochs.
3		Add a second layer and more neurons. Does it learn better?
4		Switch activation: Linear vs ReLU vs Tanh. Compare results.
5		Try the 'spiral' dataset. What architecture solves it?

🔬 Record your observations

Test loss with 1 small layer:	
Best architecture for spiral:	
Effect of a linear activation:	

💭 **Reflect:** Why can't a network with only linear activations learn a circle?

? QUIZ TIME: Inside Intelligent Machines

Circle the right answer!

Q1. A neuron computes inputs × weights + ...

a) Bias

b) Beat

c) Balloon

Q2. Layers between input and output are...

a) Doors

b) Hidden layers

c) Walls

Q3. Activation functions add...

a) Sweetness

b) Colour

c) Non-linearity

Q4. Deep learning means...

a) Many hidden layers

b) Ocean study

c) Deep sleep

Q5. Transformers power...

a) Electric poles

b) Modern chatbots

c) Only cars

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Inside Intelligent Machines

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Deep learning uses many hidden layers.		
	Activation functions add non-linearity.		
	Stacking linear layers creates non-linearity.		
	CNNs are well-suited to images.		
	Artificial neurons were invented in 2020.		
	Transformers power modern chatbots.		

HOME CONNECT

Explain to a family member how their phone recognises faces, using the 'layers' idea.

🔢 Self-assessment: tick one.

I know it well!	**I know some.**	**I need practice.**

EXERCISE: Inside Intelligent Machines

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A neuron computes inputs × weights + ____.

(a) bias ☐ **(b)** bonus ☐ **(c)** battery ☐

2. Functions like ReLU add ____.

(a) linearity only ☐ **(b)** non-linearity ☐ **(c)** noise ☐

3. Networks with many hidden layers do ____ learning.

(a) shallow ☐ **(b)** manual ☐ **(c)** deep ☐

4. The layer design of a network is its ____.

(a) architecture ☐ **(b)** algorithm only ☐ **(c)** aesthetic ☐

B. Fill in the blanks. Use the word box.

0 **images** **language** **hidden**

1. ReLU turns negative values into ____.

2. CNNs are used mainly for ____.

3. Transformers are used mainly for ____.

4. Layers between input and output are ____ layers.

C. Write T for True and F for False.

1. Activation functions add non-linearity. [ ]

2. Stacking linear layers creates complex non-linear models. [ ]

3. Early CNN layers often detect edges. [ ]

4. Every network must have exactly one layer. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Neuron	____	**(a)** max(0, x)
2. Bias	____	**(b)** Weighted sum + activation
3. ReLU	____	**(c)** Image architecture
4. CNN	____	**(d)** Shifts the output

E. Answer in one word.

1. Which activation outputs max(0, x)? _____

2. What architecture suits images? _____

3. What adds a constant shift in a neuron? _____

F. Answer these questions.

1. Explain how a single artificial neuron works.

2. Why do neural networks need activation functions?

G. Case-based questions. Read and answer.

A crack-detection CNN failed with a linear activation (61%) but reached 93% with ReLU. Wet surfaces confused it until rainy photos were added.

1. ReLU improved accuracy by adding ____.

(a) non-linearity ☐ **(b)** colour ☐ **(c)** noise ☐

2. CNNs were chosen because the data was ____.

(a) audio ☐ **(b)** images ☐ **(c)** text ☐

3. Adding rainy photos improved ____.

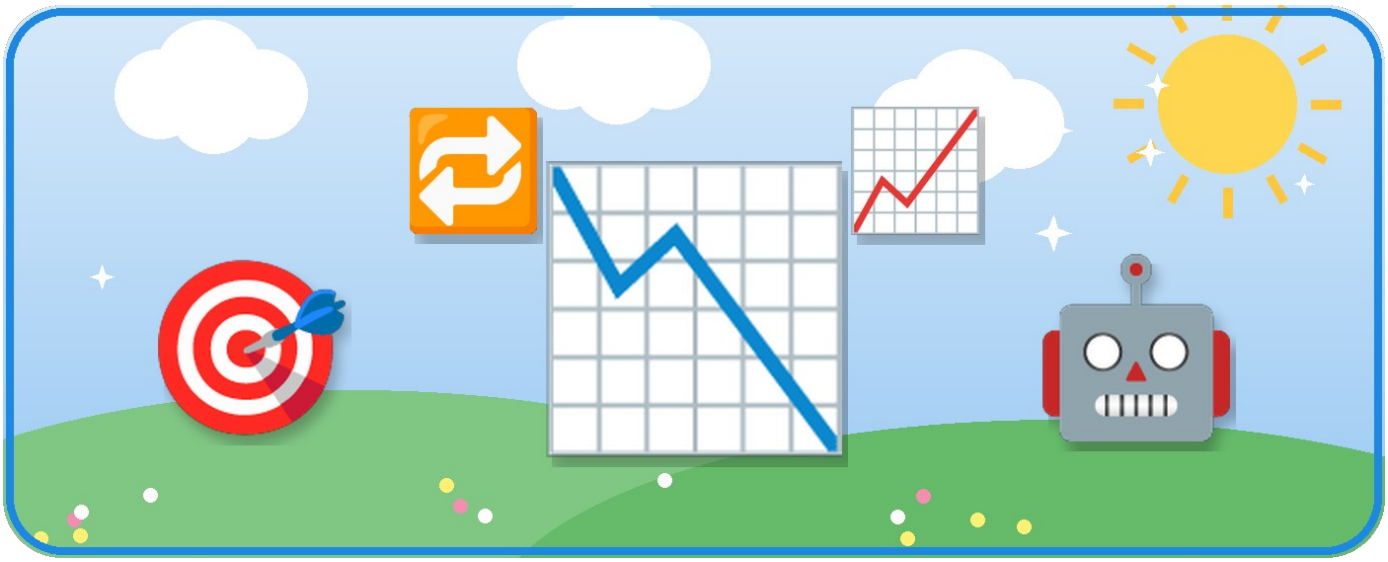
(a) speed only ☐ **(b)** cost ☐ **(c)** data diversity ☐

H. THINK DEEPER

Design a network architecture (layers, activation, output) to classify handwritten digits 0–9. Justify each choice.

TOPIC 2 • MONTH 1 • AI BASICS

Predict, Learn & Improve



LET'S LEARN

- A model learns by reducing its loss (error).
- Gradient descent nudges weights downhill.
- Train, validate, test: never test on training data!

The learning loop

Predict

Measure loss

Gradient step

Repeat (epochs)

KEY TERMS

Loss Measure of error

Gradient Direction of steepest change

Epoch One pass over data

💬 TECH TALK: Predict, Learn & Improve

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, how does a network 'learn' the right weights 🎯?
10:02

🤖 Bolt
It measures loss 📉: how wrong its predictions are, e.g., mean squared error.
10:04 ✓✓

👦 Kabir
Then fixes the weights?
10:06

🤖 Bolt
Gradient descent ⛰️: find the slope of the loss and nudge each weight downhill.
10:08 ✓✓

👧 Riya
How big is each nudge?
10:10

🤖 Bolt
The learning rate. Too big, you overshoot; too small, training crawls 🐢.
10:12 ✓✓

👦 **Kabir**

What's an epoch?

10:14

🤖 **Bolt**

One full pass through the training data 🔁. Training runs many epochs.

10:16 ✓✓

👧 **Riya**

And we check on validation data to avoid overfitting 🧪, then the final test set!

10:18

😊 Type a message...

📌 **Pinned by Bolt**

Models learn by reducing loss. Gradient descent moves weights downhill using a learning rate over many epochs. Use train, validation and test splits.

🔄 How it works

INDIA SPOTLIGHT

Large-scale training in India is supported by national supercomputing and AI compute initiatives.

📖 CONCEPT EXPLAINED: Predict, Learn & Improve

💡 In this chapter

A freshly created neural network knows nothing: its weights are random, so its predictions are guesses. Yet after training, the same network can diagnose diseases or forecast rainfall. How does it improve? This chapter explains the learning loop at the heart of machine learning: measuring error with a loss function, reducing it with gradient descent, repeating over epochs, and checking honestly with separate validation and test data.

📉 1. Loss: measuring how wrong we are

To improve, a model first needs a score for its mistakes. This score is the loss (also called cost or error). For predicting numbers, such as house prices, a common loss is Mean Squared Error (MSE): for each example, subtract the prediction from the true value, square the difference, then average over all examples. Squaring makes all errors positive and punishes big mistakes more.

For classification, such as spam versus not spam, we often use cross-entropy loss. It is small when the model gives high probability to the correct class and large when it is confidently wrong. Whatever the formula, the idea is the same: lower loss means better predictions on that data. Training is simply an organised search for weights that make the loss as small as possible.

✏️ WORKED EXAMPLE

True marks: 80, 60, 90. Predictions: 75, 65, 84. Errors: 5, -5, 6. Squared: 25, 25, 36. MSE = (25 + 25 + 36) ÷ 3 ≈ 28.7. If a better model predicts 79, 61, 88, MSE = (1 + 1 + 4) ÷ 3 = 2.

⛰️ 2. Gradient descent: walking downhill

Picture the loss as a hilly landscape where every position is a different set of weights, and height is the loss. We want to reach the lowest valley. Blindfolded, the best strategy is to feel the slope under your feet and take a step downhill. In maths, that slope is the gradient: it tells us how the loss changes when each weight changes slightly.

Gradient descent updates each weight using the rule: new weight = old weight - learning rate × gradient. If increasing a weight increases the loss (positive gradient), we decrease it, and vice versa. In neural networks, an algorithm called backpropagation efficiently calculates the gradient for millions of weights by working backwards from the output layer to the input layer, using the chain rule from calculus.

✏️ **WORKED EXAMPLE**

Suppose a weight w = 2.0, its gradient is 0.8 and the learning rate is 0.1. New w = 2.0 - 0.1 × 0.8 = 1.92. If the gradient were -0.5, new w = 2.0 + 0.05 = 2.05.

🐇 3. Learning rate, epochs and batches

The learning rate controls the step size. Too large, and the model overshoots the valley and the loss jumps around or even explodes. Too small, and training crawls, needing thousands of extra steps. Practitioners try several values, often between 0.0001 and 0.1, and may reduce the rate as training progresses.

One epoch is one full pass through the training data. Because datasets can contain millions of examples, we usually update weights after small groups called mini-batches, for example 32 examples at a time. This is called mini-batch stochastic gradient descent. A typical training run might last 10 to 100 epochs, with the loss plotted after each one to watch progress.

🤔 **PAUSE & THINK**

A loss curve goes down smoothly for 20 epochs, then starts zig-zagging wildly. What might be wrong, and what would you change?

🧪 4. Train, validation and test: honest evaluation

If we judge a model only on the data it trained on, we may fool ourselves. A model can memorise training examples, noise and all, and score brilliantly there while failing on new data. This is overfitting. To detect it, we split data into three parts: a training set (often 70%) to learn weights, a validation set (15%) to tune choices and check progress, and a test set (15%) used only once at the very end.

Watch two curves during training: training loss and validation loss. Early on, both fall. If the training loss keeps falling while validation loss starts rising, the model is overfitting. Remedies include early stopping (stop at the best validation point), collecting more varied data, simplifying the model, or regularisation techniques such as dropout that randomly switch off neurons during training.

✏️ **WORKED EXAMPLE**

Epoch 10: train loss 0.40, validation 0.45. Epoch 30: train 0.15, validation 0.30. Epoch 50: train 0.05, validation 0.38. The best model is around epoch 30; after that, the model memorises rather than learns.

🔁 5. Underfitting, overfitting and the sweet spot

Underfitting is the opposite problem: the model is too simple to capture the pattern, so both training and validation errors are high. A straight line trying to fit a curved relationship underfits. Overfitting means the model is too flexible: training error is tiny, validation error is high. The goal is the sweet spot in between, where the model generalises well to unseen data.

This trade-off is often called the bias–variance trade-off. High bias models make strong simplifying assumptions and underfit; high variance models react to every wiggle in the training data and overfit. Good machine learning practice is a cycle: train, evaluate on validation data, diagnose, adjust, and repeat, then confirm the final choice on the untouched test set.

🖐 TRY IT

In TensorFlow Playground, add noise of 30 and use a large network with many neurons. Compare training and test loss. Then reduce the network size or add regularisation and compare again.

✅ Key takeaways

- Loss functions such as MSE and cross-entropy measure how wrong a model is.
- Gradient descent updates weights in the direction that reduces loss; backpropagation computes the gradients.
- The learning rate sets step size; epochs are full passes; mini-batches make training efficient.
- Train, validation and test splits keep evaluation honest and reveal overfitting.
- Aim for the sweet spot between underfitting and overfitting.

❓ Quick check: answer in one or two lines.

1. Calculate the MSE for true values 10, 20 and predictions 12, 17.

2. Write the gradient descent update rule in words.

3. What does it mean if validation loss rises while training loss falls?

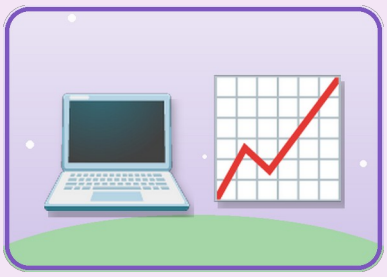
4. Give two remedies for overfitting.

AI CHRONICLES: Lost on the Loss Curve

Kabir

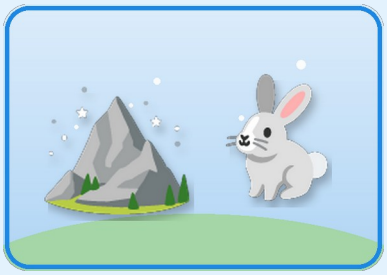
Dr. Shalini

Bolt


SCENE 1

Kabir trained a model to predict crop yield. The loss jumped up and down wildly 📈.

Dr. Shalini looked: "Your learning rate is too high. You're leaping over the valley."


SCENE 2

SCENE 3

He lowered it too far. Training crawled; after 100 epochs, loss barely moved.

A middle value worked. The loss curve slid smoothly down ✅.

SCENE 4

SCENE 5

But validation loss started rising after epoch 40, a sign of overfitting! He used early stopping.

His final test error was low and honest. "Learning is a journey downhill," Dr. Shalini smiled.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir tries different architectures	**Kabir tests on training data only**	**The loss curve becomes a roller coaster ride**

🧐 Story check: answer the questions.

1. Why did the loss jump around?

2. What happened with a tiny learning rate?

3. How did Kabir detect overfitting?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Learning loops in practice

Recommendations improve daily

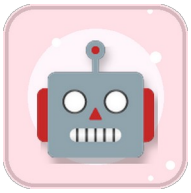
Spam filters retrain

ETA models learn traffic

Game AIs self-play

Weather models update

Fraud models adapt



BOLT SAYS:
Learning rate too big? I overshoot. Too small? I crawl. Tuning matters!

FUN FACT!
Gradient descent is like walking down a foggy mountain by always stepping downhill!

👀 **I spy AI! Draw one AI helper you saw today.**

🧪 LAB ACTIVITY: Blindfold Gradient Descent

🎒 You need:

Blindfold

Floor number line

Guide partner

👣 Steps:

1		Mark 0–10 on the floor. The 'lowest point' is secretly 7.
2		Walker starts at 0, blindfolded.
3		Guide says only 'warmer/colder' (the gradient).
4		Try big steps vs small steps. Which converges best?

THINK & TALK

What does the step size represent in real training?

🖍️ **Draw what you did!**

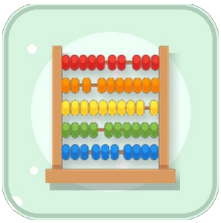
WORKSHEET: Predict, Learn & Improve (Part A)

Name: _____ Date: _____

1. One training iteration. Write 1, 2, 3, 4.

Update weights ☐

Make predictions ☐



Compute gradients ☐

Calculate loss ☐

2. Underfitting or overfitting? Tick ✓.

	Underfit	Overfit
Train 60%, test 58%		
Train 99%, test 70%		
Model too simple		
Memorises noise		

3. Bonus: Draw your favourite picture from this topic!

I drew _____

WORKSHEET: Predict, Learn & Improve (Part B)

Name: _____ Date: _____

1. Gradient descent step: w = w - lr × grad. Use the word box.

WORD BOX: **0.6** ✿ **1.08** ✿ **-6**

grad at w=0 for loss $(w-3)^2$ → ____

new w with lr = 0.1 → ____

next: grad = 2(0.6-3) = -4.8, w = ____

2. Match the dataset split to its job.

Training set ● ● **Final unbiased check**

Validation set ● ● **Learn the weights**

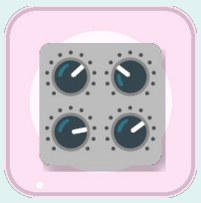
Test set ● ● **Measures error**

Loss ● ● **Tune settings**

PUZZLE & DEFINE: Predict, Learn & Improve

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Loss

Meaning: _____

Example: _____

Gradient

Meaning: _____

Example: _____

Epoch

Meaning: _____

Example: _____

🧪 DEMO LAB: TensorFlow Playground + spreadsheet (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Explore loss, learning rate and epochs.

🔗 **Open:** playground.tensorflow.org | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Pick a dataset. Set learning rate = 0.03. Train and note test loss at 100 epochs.
2		Set learning rate = 3. What happens to the loss curve?
3		Set learning rate = 0.0001. How fast does it learn?
4		Add noise to the data. Compare training vs test loss.
5		Record results in a table and plot loss vs learning rate.

🔬 Record your observations

Best learning rate:	
What happened at 3:	
Signs of overfitting seen:	

💭 **Reflect:** Why do we need a separate validation set while training?

? QUIZ TIME: Predict, Learn & Improve

Circle the right answer!

Q1. A model learns by minimising its...

a) Loss

b) Fun

c) Volume

Q2. One full pass over training data is an...

a) Hour

b) Epoch

c) Era

Q3. Too-large learning rate causes...

a) Sleep

b) Perfect fit

c) Overshooting

Q4. Validation data is used to...

a) Tune the model

b) Delete it

c) Decorate

Q5. Train 99%, test 70% suggests...

a) Underfitting

b) Overfitting

c) Perfection

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Predict, Learn & Improve

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Loss measures how wrong a model is.		
	Gradient descent moves weights uphill.		
	An epoch is one pass over the training data.		
	Test data can be used for tuning.		
	High train, low test accuracy means overfitting.		
	Early stopping can reduce overfitting.		

HOME CONNECT

Track a daily habit (e.g., sleep). Each week, 'update your weights': adjust one thing and measure the result.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Predict, Learn & Improve

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. How wrong a model's predictions are is the ____.

(a) loss ☐ **(b)** list ☐ **(c)** layer ☐

2. Moving weights downhill on the loss is gradient ____.

(a) ascent ☐ **(b)** descent ☐ **(c)** design ☐

3. One full pass through the training data is an ____.

(a) error ☐ **(b)** edge ☐ **(c)** epoch ☐

4. Step size in gradient descent is the learning ____.

(a) rate ☐ **(b)** rule ☐ **(c)** room ☐

B. Fill in the blanks. Use the word box.

overshooting **overfitting** **early** **training**

1. Too high a learning rate causes ____.

2. Validation data helps detect ____.

3. Stopping when validation loss rises is ____ stopping.

4. Test on data never used in ____.

C. Write T for True and F for False.

1. Lower loss means better predictions on that data. [ ]

2. A tiny learning rate always trains fastest. [ ]

3. Training runs many epochs. [ ]

4. Testing on training data is the best evaluation. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Loss	____	**(a)** Full data pass
2. Gradient	____	**(b)** Prediction error
3. Epoch	____	**(c)** Tune & check
4. Validation set	____	**(d)** Slope of loss

E. Answer in one word.

1. What measures model error? _____

2. What is one full pass over data? _____

3. Which method nudges weights downhill? _____

F. Answer these questions.

1. Explain gradient descent using a 'walking downhill' analogy.

2. Why do we split data into train, validation and test sets?

G. Case-based questions. Read and answer.

Kabir's loss jumped wildly with a high learning rate, crawled with a tiny one and worked smoothly with a middle value. Validation loss rose after epoch 40, so he used early stopping.

1. Wild jumps showed the learning rate was too ____.

(a) high ☐ **(b)** low ☐ **(c)** perfect ☐

2. Rising validation loss signals ____.

(a) underfitting ☐ **(b)** overfitting ☐ **(c)** a bug in Python ☐

3. Early stopping prevents ____.

(a) learning ☐ **(b)** testing ☐ **(c)** overfitting ☐

H. THINK DEEPER

Sketch training and validation loss curves for a model that overfits. Mark where you would stop training and explain why.

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. A neuron computes weighted sum plus...

a) Bias | b) Beat | c) Balloon

Q2. One pass over training data is an...

a) Era | b) Epoch | c) Hour

Q3. ReLU(−3) equals...

a) -3 | b) 3 | c) 0

Q4. Data used for final evaluation:

a) Test set | b) Training set | c) Validation

Q5. Gradient descent moves weights...

a) Uphill | b) Downhill on loss | c) Randomly

Q6. Images are best handled by...

a) RNN-songs | b) Calculators | c) CNNs

WORD SEARCH: AI DISCOVER

Find these words. They go across → or down ↓.

NEURON

ACTIVATION

LOSS

GRADIENT

EPOCH

WEIGHT

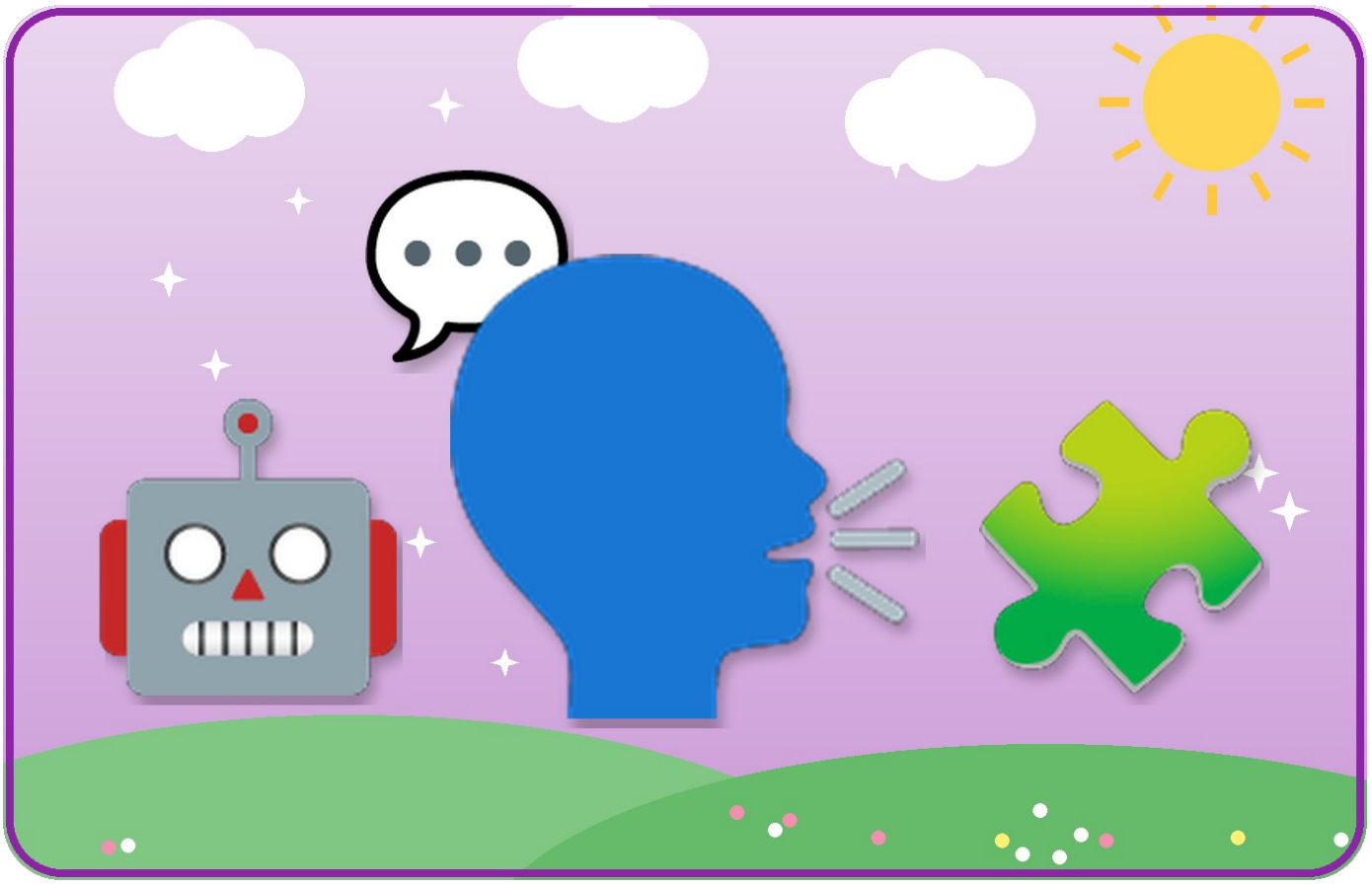
H	P	K	A	O	T	D	F	K	D
A	C	T	I	V	A	T	I	O	N
E	L	S	B	O	S	G	K	K	U
P	R	S	I	W	E	I	G	H	T
O	N	U	A	B	R	H	N	O	F
C	M	T	S	M	C	P	S	D	F
H	S	N	N	E	U	R	O	N	M
R	G	R	A	D	I	E	N	T	A
L	O	S	S	R	B	K	W	U	U
N	F	F	S	H	A	G	T	T	H

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code



Area: Communicate with AI

In this unit, you will explore:

Topic 3

Python for Smart Solutions

Topic 4

The Art of Prompting

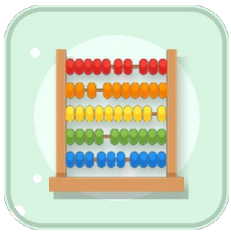
TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

Python for Smart Solutions

LET'S LEARN

- NumPy handles fast number arrays.
- pandas loads and cleans tables (DataFrames).
- scikit-learn trains models in a few lines!

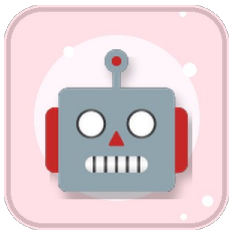
The Python AI toolkit



NumPy

pandas

Matplotlib



scikit-learn

KEY TERMS

DataFrame Table in pandas

Array Grid of numbers

Data cleaning Fixing messy data

💬 TECH TALK: Python for Smart Solutions

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👦 **Kabir**

Bolt, plain Python lists are slow for a million numbers 🐢.

10:02

🤖 **Bolt**

Use NumPy arrays 🔢: fast maths on whole arrays at once. np.mean(a) in a blink!

10:04 ✓✓

👧 **Riya**

And tables like CSV files 📄?

10:06

🤖 **Bolt**

pandas! df = pd.read_csv('marks.csv') loads a DataFrame: rows and named columns.

10:08 ✓✓

👦 **Kabir**

Our data has blanks and duplicates 😩.

10:10

🤖 **Bolt**

Data cleaning: df.dropna(), df.drop_duplicates(), fix types. Often 70% of the work 🧹!

10:12 ✓✓

Riya
Then we train a model?
10:14

Bolt
scikit-learn 🤖: model.fit(X_train, y_train), then model.predict(X_test). A few lines!
10:16 ✓✓

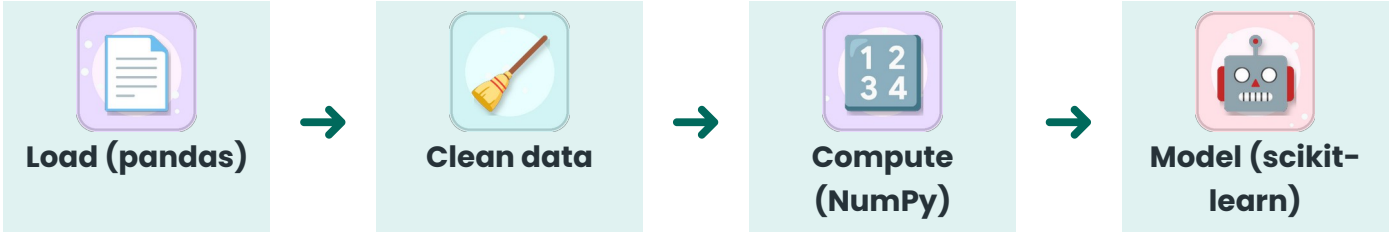
Kabir
NumPy, pandas, scikit-learn: the data-science trio 🚀.
10:18

😊 Type a message...

📌 **Pinned by Bolt**

NumPy handles fast numeric arrays, pandas loads and cleans DataFrames, and scikit-learn trains models in a few lines. Data cleaning is a big part of the work.

🔄 How it works



INDIA SPOTLIGHT

Free cloud notebooks let Indian students run pandas and scikit-learn without installing anything.

📖 CONCEPT EXPLAINED: Python for Smart Solutions

💡 In this chapter

Professional data scientists rarely write everything from scratch. They stand on the shoulders of powerful Python libraries: NumPy for fast numerical computing, pandas for tables of data, and scikit-learn for machine learning. In this chapter you will learn what each library does, why it is faster or easier than plain Python, and how a complete mini-project flows from raw CSV file to trained, evaluated model.

🔢 1. NumPy: fast maths on whole arrays

A Python list can hold numbers, but doing maths on a million of them with a loop is slow, because Python handles each item one by one. NumPy (Numerical Python) stores numbers in compact arrays and runs operations in optimised C code. You can add, multiply or average entire arrays in a single line, which is called vectorisation.

For example, marks = np.array([78, 85, 92, 64]) creates an array. marks + 5 adds 5 to every mark at once; marks.mean() gives the average; marks.max() the highest; marks > 80 returns an array of True/False values. NumPy also supports 2-D arrays (matrices), which is exactly how images and neural-network weights are stored.

✏️ WORKED EXAMPLE

```
import numpy as np
heights = np.array([150, 162, 158, 171])
print(heights.mean())      # 160.25
print(heights[heights > 160])  # [162 171]
print(heights / 100)       # convert cm to metres
```

🐼 2. pandas: tables you can reason with

Real-world data usually arrives as tables: spreadsheets, CSV files or database exports. pandas provides the DataFrame, a table with named columns and indexed rows. df = pd.read_csv('marks.csv') loads a file; df.head() shows the first rows; df.info() shows column types and missing values; df.describe() gives summary statistics like mean and standard deviation.

You can select a column with df['maths'], filter rows with df[df['maths'] > 80], create new columns with df['total'] = df['maths'] + df['science'], and group data with df.groupby('section')['maths'].mean() to compare sections. These few commands replace hours of manual spreadsheet work and make analysis repeatable.

🤔 **PAUSE & THINK**

Why is it better to analyse data with code than by editing a spreadsheet by hand? Think about errors, repeatability and sharing.

🧹 3. Data cleaning: the hidden 70%

Surveys of data scientists often report that cleaning and preparing data takes most of their time. Raw data has missing values, duplicates, typos ('9O' instead of '90'), inconsistent labels ('Male', 'M', 'male') and outliers (a student aged 150). A model trained on dirty data learns wrong patterns: garbage in, garbage out.

pandas offers tools for each problem: df.dropna() removes rows with missing values, or df.fillna(value) fills them; df.drop_duplicates() removes repeats; pd.to_numeric(df['marks'], errors='coerce') converts text to numbers and marks bad values as missing; df['gender'].str.lower() standardises text. Every cleaning decision should be thought through and documented, because it can change results and fairness.

✏️ **WORKED EXAMPLE**

A student marked 'absent' should not be treated as scoring 0; that would unfairly drag down averages and mislead a model. A better choice is to treat it as missing and handle it separately.

🤖 4. scikit-learn: machine learning in a few lines

scikit-learn provides ready-made, well-tested algorithms with a consistent pattern: create a model, fit it on training data, then predict on new data. Linear regression predicts numbers; logistic regression, decision trees and random forests classify; k-means finds clusters. The library also includes tools for splitting data and measuring performance.

A typical workflow is: X = df[['hours', 'attendance']] (features), y = df['maths'] (target); X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2); model = LinearRegression().fit(X_train, y_train); predictions = model.predict(X_test); then compare predictions with y_test using a metric such as mean absolute error or R^2.

🖐️ **TRY IT**

In Google Colab, load any small CSV (for example a class survey), clean it, train a LinearRegression model and print the mean absolute error on the test set.

🔗 5. Putting it together: a mini data-science project

A complete project follows a clear path. Define the question: 'Can study hours and attendance predict maths marks?' Collect and load the data with pandas. Explore it with summaries and charts to understand patterns. Clean it carefully. Split it into training and test sets. Train a model with scikit-learn. Evaluate it honestly on the test set. Finally, interpret and communicate the findings.

Interpretation matters as much as code. A model may show that attendance strongly predicts marks, but correlation is not causation: maybe students who attend more also have better support at home. Responsible analysts state limitations, check whether the model treats groups fairly, and present results clearly with charts so that teachers or decision-makers can act wisely.

🤔 **PAUSE & THINK**

A model predicts that students from Section C will score lower. What questions should you ask before anyone acts on this prediction?

✅ Key takeaways

- NumPy provides fast, vectorised maths on arrays and matrices.
- pandas DataFrames make loading, selecting, filtering and grouping tabular data easy.
- Data cleaning handles missing values, duplicates, typos and outliers, and must be done thoughtfully.
- scikit-learn follows a fit–predict pattern with built-in tools for splitting and evaluation.
- Good projects end with honest interpretation, limitations and fairness checks.

❓ Quick check: answer in one or two lines.

1. What does vectorisation mean in NumPy?

2. Write pandas code to show the average maths mark for each section.

3. Why should 'absent' not be converted to 0 marks?

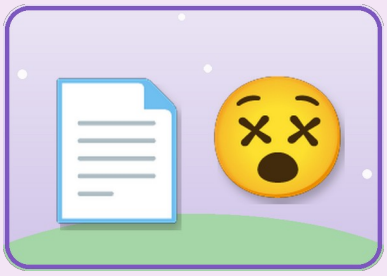
4. List the main steps of a mini data-science project in order.

AI CHRONICLES: The Messy Marks Mystery

Riya

Mr. Joseph

pandas



SCENE 1

Mr. Joseph gave Riya 5 years of marks data: 12,000 rows to predict who needs extra help.

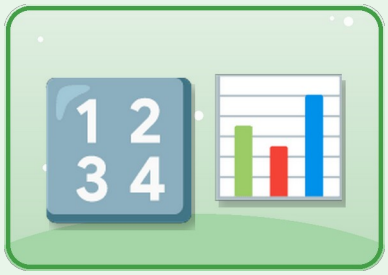
pandas revealed chaos: blanks, duplicate students and marks like '90' and 'absent' 😱.

SCENE 2

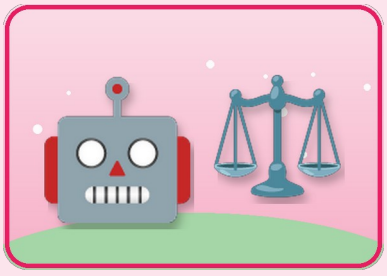
SCENE 3

She cleaned it: dropped duplicates, converted types and handled 'absent' carefully, not as zero.

NumPy stats showed maths scores dipped every year in Term 2. Interesting!



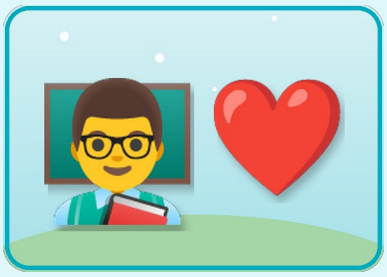
SCENE 4



SCENE 5

A scikit-learn model flagged at-risk students. Riya checked it wasn't unfair to any section.

Extra classes started early. Term 2 results improved. "Clean data, kind decisions," said Mr. Joseph.



SCENE 6

🔮 What happens next? Tick ✓ one!

☐ **Riya builds a dashboard for teachers**	☐ **She leaves the data messy**	☐ **pandas eats the bamboo**

🧐 Story check: answer the questions.

1. What problems were in the data?

2. Why shouldn't 'absent' be zero?

3. Why did Riya check fairness?

Rate this story: shade the stars.

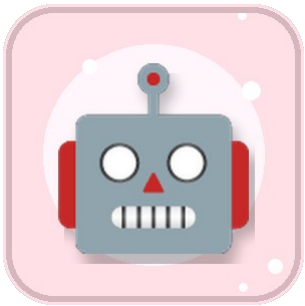
☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Python data workflows

Load CSV survey data

Clean missing values

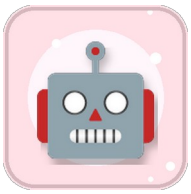
Plot trends



Train a classifier

Measure accuracy

Save the model



BOLT SAYS:

df.head() is your first friend: always look at your data before modelling!

FUN FACT!

pandas got its name from 'panel data', a term from economics!

I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: Marks Analyser Lab

You need:

Computer with Python

marks.csv

Teacher

Steps:

1. df = pd.read_csv('marks.csv')
2. df = df.dropna() to remove blank rows.
3. df['maths'].mean() and df.plot()
4. Find the subject with the highest average.

THINK & TALK

What could go wrong if you skip data cleaning?

Draw what you did!

WORKSHEET: Python for Smart Solutions (Part A)

Name: _____ Date: _____

1. Match the code to what it does.

pd.read_csv()	●	●	Trains a model
df.dropna()	●	●	Loads a CSV file
df.mean()	●	●	Computes averages
model.fit()	●	●	Removes missing rows

2. Predict the output. Use the word box.

WORD BOX: 3 ✿ 6.0 ✿ [2 4 6]

np.array([1,2,3]) * 2 → ____

np.mean([4, 6, 8]) → ____

len(df) for 3 rows → ____

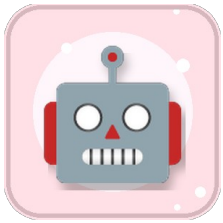
WORKSHEET: Python for Smart Solutions (Part B)

Name: _____ Date: _____

1. A data science workflow. Write 1, 2, 3, 4.

Evaluate ☐

Load data ☐



Train model ☐

Clean data ☐

2. Which needs cleaning? Tick ✓.

	Clean	Needs cleaning
Age column with blanks		
Duplicate rows		
Consistent units, no gaps		
Height in cm & m mixed		

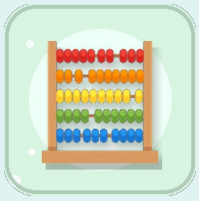
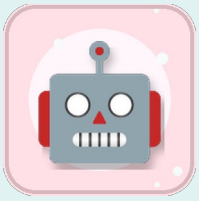
3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE & DEFINE: Python for Smart Solutions

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

DataFrame

Meaning: _____

Example: _____

Array

Meaning: _____

Example: _____

Data cleaning

Meaning: _____

Example: _____

🧪 DEMO LAB: Google Colab (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Load, clean and model data with pandas and scikit-learn.

🔗 **Open:** colab.research.google.com | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		import pandas as pd; df = pd.read_csv('marks.csv'); df.head()
2		df = df.dropna().drop_duplicates(); print(df.shape)
3		import numpy as np; print(np.mean(df['maths']))
4		from sklearn.linear_model import LinearRegression; model = LinearRegression().fit(df[['hours']], df['maths'])
5		print(model.predict([[6]]))

🔬 Record your observations

Rows before/after cleaning:	
Mean maths mark:	
Predicted mark for 6 hours:	

💭 **Reflect:** Why is data cleaning often the biggest part of a project?

? QUIZ TIME: Python for Smart Solutions

Circle the right answer!

Q1. A table in pandas is a...

a) DataFrame

b) Picture frame

c) Window

Q2. Fast numeric arrays come from...

a) MusicPy

b) NumPy

c) PizzaPy

Q3. model.fit(X, y) does what?

a) Draws

b) Deletes

c) Trains the model

Q4. df.dropna() removes...

a) Missing values

b) All numbers

c) Charts

Q5. Always do this before modelling:

a) Sleep

b) Explore the data

c) Ignore data

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Python for Smart Solutions

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	pandas uses DataFrames for tables.		
	NumPy arrays are slower than lists for maths.		
	Data cleaning comes before modelling.		
	scikit-learn uses fit() and predict().		
	High accuracy on imbalanced data is always good.		
	df.head() shows the first rows.		

HOME CONNECT

Collect a small family dataset (e.g., daily expenses). Plan how pandas could analyse it.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Python for Smart Solutions

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. The library for fast numeric arrays is ____.

(a) NumPy ☐ **(b)** Pygame ☐ **(c)** Turtle ☐

2. A pandas table is a ____.

(a) DataSheet ☐ **(b)** DataFrame ☐ **(c)** DataBox ☐

3. scikit-learn trains a model with ____.

(a) model.run() ☐ **(b)** model.go() ☐ **(c)** model.fit() ☐

4. Removing blank rows uses ____.

(a) dropna() ☐ **(b)** append() ☐ **(c)** sort() ☐

B. Fill in the blanks. Use the word box.

csv **duplicates** **predict** **70**

1. pd.read_____ loads a CSV file.

2. df.drop_____() removes repeated rows.

3. model.____() makes predictions.

4. Data cleaning is often ____% of the work.

C. Write T for True and F for False.

1. NumPy is faster than plain lists for large maths. [ ]

2. Missing values never affect models. [ ]

3. scikit-learn can train models in a few lines. [ ]

4. 'absent' should always be treated as 0 marks. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. NumPy	____	**(a)** ML models
2. pandas	____	**(b)** Numeric arrays
3. scikit-learn	____	**(c)** Remove blanks
4. dropna()	____	**(d)** DataFrames

E. Answer in one word.

1. Which library handles DataFrames? _____

2. Which method trains a scikit-learn model? _____

3. Which library computes np.mean? _____

F. Answer these questions.

1. List four common data-cleaning steps.

2. Write the pandas + scikit-learn steps to train and predict.

G. Case-based questions. Read and answer.

A 12,000-row marks dataset had blanks, duplicates and values like '9O' and 'absent'. Riya cleaned it with pandas, found a Term 2 dip with NumPy and built a fair at-risk model.

1. Cleaning was done using _____.

(a) pandas ☐ **(b)** Paint ☐ **(c)** Excel only ☐

2. '9O' is an example of a _____.

(a) feature ☐ **(b)** data entry error ☐ **(c)** model ☐

3. Checking sections for fairness avoids _____.

(a) speed ☐ **(b)** accuracy ☐ **(c)** bias ☐

H. THINK DEEPER

You receive a messy dataset of school bus timings. Plan the cleaning, analysis and one prediction you'd make, with library names.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

The Art of Prompting

LET'S LEARN

- Prompt engineering designs inputs for better AI output.
- Zero-shot, few-shot and step-by-step prompts.
- Iterate: test, compare and refine your prompts!

Prompt techniques

Zero-shot

Few-shot

Step-by-step

Role prompting

KEY TERMS

Few-shot Prompt with examples

Temperature Randomness setting

Context Background info given

💬 TECH TALK: The Art of Prompting

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, the same chatbot gives great answers to some people and poor ones to me 😕.
10:02

🤖 Bolt
Prompt engineering ✍️: designing inputs to get better outputs.
10:04 ✓✓

👦 Kabir
What techniques?
10:06

🤖 Bolt
Zero-shot: just ask. Few-shot: give 2–3 examples first 📋. Step-by-step: ask it to reason in stages.
10:08 ✓✓

👧 Riya
And context?
10:10

🤖 Bolt
Give role, audience, constraints and format 🧩. 'Explain to a Class 11 student in 5 bullets.'
10:12 ✓✓

Kabir

What's temperature 🌡️ ?

10:14

Bolt

A setting for randomness. Low = focused and consistent; high = creative and varied.

10:16 ✓✓

Riya

Test, compare, refine, and still verify facts 🔁!

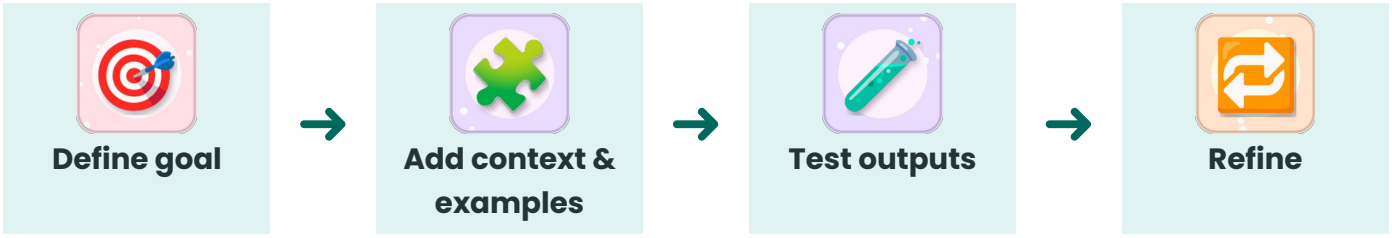
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Prompt engineering designs inputs for better outputs: zero-shot, few-shot and step-by-step prompts, rich context and tuned temperature. Iterate and verify.

🔄 How it works



INDIA SPOTLIGHT

Indian companies hire prompt and conversation designers to build chatbots in many Indian languages.

📖 CONCEPT EXPLAINED: The Art of Prompting

💡 In this chapter

Large language models respond to text instructions called prompts. The same model can give a vague, generic answer or a precise, useful one depending on how it is asked. Prompt engineering is the skill of designing inputs that reliably produce good outputs. In this chapter you will learn why prompts matter, the main prompting techniques, how context and settings shape answers, and how to test prompts systematically.

✍️ 1. Why prompts matter

A language model predicts likely continuations of text. If the prompt is vague, many continuations are plausible, so the model picks a generic, middle-of-the-road answer. If the prompt specifies who the answer is for, what form it should take and what to include or avoid, the space of good continuations narrows, and quality rises.

Think of a prompt as a brief to a talented but literal-minded assistant who has read a great deal but knows nothing about your situation. 'Write about photosynthesis' could produce anything from a poem to a PhD summary. 'Explain photosynthesis to Class 11 biology students in 6 bullet points, including the light and dark reactions and one Indian crop example' leaves far less to chance.

✏️ WORKED EXAMPLE

Weak: 'Tell me about the monsoon.' Strong: 'You are a geography teacher. Explain the Indian south-west monsoon to Class 11 students in 150 words: cause, timing and impact on agriculture. End with one exam-style question.'

🧩 2. Anatomy of a strong prompt

Strong prompts usually contain several elements. A role sets the perspective ('You are an experienced physics tutor'). The task states exactly what to do. Context supplies background: the audience, the syllabus, the purpose. Constraints set limits: word count, difficulty level, what to avoid. The format specifies the shape of the output: bullets, table, JSON, step-by-step working.

Not every prompt needs every element, but missing elements are a common reason for disappointing answers. Clear delimiters help too: placing a passage between triple quotes, for example, tells the model exactly which text to summarise. Asking the model to say 'I don't know' when unsure can reduce confident errors, though it does not eliminate them.

🤔 **PAUSE & THINK**

Pick a task you did recently with an AI tool. Which of the five elements (role, task, context, constraints, format) did your prompt include? Which would have helped?

📋 3. Zero-shot, few-shot and step-by-step prompting

Zero-shot prompting simply asks for the task with no examples: 'Classify this review as positive or negative.' Modern models handle many tasks this way. Few-shot prompting shows the model a few worked examples first, such as three reviews with their labels, then the new review. Examples teach the desired format and judgement more clearly than descriptions alone.

Step-by-step (often called chain-of-thought) prompting asks the model to reason through intermediate steps before giving a final answer. For multi-step maths or logic problems, this often improves accuracy, because each step conditions the next. However, the reasoning can still contain errors, so you must check each step, not just the final answer.

✏️ **WORKED EXAMPLE**

Few-shot: 'Review: The food was cold. → Negative. Review: Loved the service! → Positive. Review: Okay, nothing special. → Neutral. Review: The movie dragged but the songs were great. →' The model now knows the labels and format to use.

🌡️ 4. Settings: temperature and context window

Many AI tools let you adjust temperature, a setting that controls randomness in how the next token is chosen. Low temperature (near 0) makes the model pick the most likely tokens, giving focused, consistent answers, good for facts, code and classification. Higher temperature allows less likely tokens, giving more varied and creative output, useful for brainstorming stories or slogans.

Models also have a context window: the maximum amount of text (measured in tokens) they can consider at once, including your prompt, any documents you paste and the conversation so far. If a conversation grows too long, early details may fall outside the window and be forgotten. For long tasks, summarise progress or start fresh with the key information.

🤔 **PAUSE & THINK**

Which temperature would you use for (a) generating 20 creative names for a science club, (b) checking grammar in an essay, (c) solving a chemistry numerical? Explain.

🔁 5. Iterate, evaluate and stay responsible

Prompt engineering is experimental. Write a prompt, test it on several inputs, look closely at failures, refine and test again. Professionals keep a small test set of tricky examples and compare prompt versions side by side, scoring outputs against criteria such as accuracy, clarity and format. What works for one model may need adjusting for another.

Even an excellent prompt cannot guarantee truth. Models can still hallucinate facts, misquote sources or reflect biases in their training data. Always verify important information against reliable sources such as NCERT textbooks or official websites, never paste private personal data into public tools, and be transparent when AI helped you produce work.

🖐 **TRY IT**

Write three versions of a prompt for 'Make revision notes on Newton's laws'. Score each output out of 5 for accuracy, clarity and usefulness. Which elements made the biggest difference?

✅ Key takeaways

- Prompts narrow the space of possible outputs; vague prompts give generic answers.
- Strong prompts include role, task, context, constraints and format.
- Few-shot examples teach format and judgement; step-by-step prompting helps multi-step reasoning.
- Temperature controls randomness; the context window limits how much text the model sees.
- Iterate with test cases, and always verify facts and protect privacy.

? Quick check: answer in one or two lines.

1. Name the five elements of a strong prompt.

2. How is few-shot prompting different from zero-shot?

3. When would you use a low temperature?

4. Why can't prompt engineering guarantee factual accuracy?

AI CHRONICLES: The Prompt Duel

Kabir | Riya | Judge Meghna

SCENE 1

A college fest held a 'Prompt Duel': make an AI write the best study notes on photosynthesis.

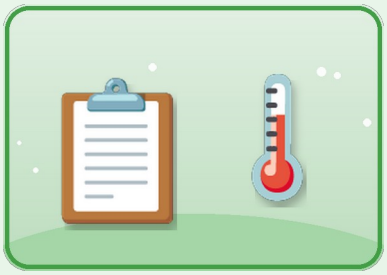
Kabir typed: 'Explain photosynthesis.' He got a long, generic paragraph 😐.

SCENE 2

SCENE 3

Riya used role, audience, format: 'You are a biology teacher. For Class 11, give 6 bullets + 1 diagram description + 3 MCQs.'

She added a few-shot example of her preferred style and set low temperature for consistency.

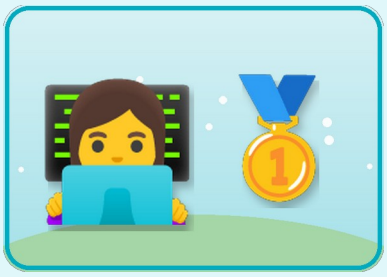


SCENE 4

SCENE 5

She still checked every fact with NCERT and fixed one error before submitting ✅.

Judge Meghna awarded Riya: "Great prompting AND verification. That's the real skill."



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They write a prompt guide for juniors	**Kabir keeps one-line prompts**	**The AI enters the duel itself**

🧐 Story check: answer the questions.

1. Why was Kabir's output generic?

2. Which techniques did Riya use?

3. Why did she still verify?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Prompting in the real world

Summarise research papers

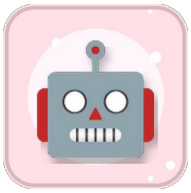
Generate test cases

Translate with tone

Explain data tables

Design brief to image

Draft professional emails



BOLT SAYS:
Give me examples (few-shot) and I'll copy the pattern. Show, don't just tell!

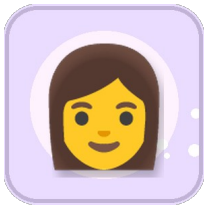
FUN FACT!
'Temperature' controls randomness: low = focused answers, high = creative variety!

I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: Prompt A/B Testing

You need:

Prompt cards



Teacher's AI tool

Scoring rubric

Steps:

1		Task: 'Explain photosynthesis for Class 6.'
2		Prompt A: zero-shot. Prompt B: role + format + example.
3		Score both on accuracy, clarity, level (1–5).
4		Refine the weaker prompt and test again.

THINK & TALK

Which element improved the output most: role, format or example?

Draw what you did!

WORKSHEET: The Art of Prompting (Part A)

Name: _____ Date: _____

1. Zero-shot or few-shot? Tick ✓.

	Zero-shot	Few-shot
'Classify this review as positive or negative.'		
'Happy→Positive, Sad→Negative. Now: Excited→?'		
'Translate to Hindi.'		
Prompt with 3 sample answers		

2. Match the technique to its benefit.

Role prompting ●

Step-by-step ●

Few-shot ●

Output format ●

● **Shows the pattern**

● **Sets expertise & tone**

● **Structured answers**

● **Better reasoning**

WORKSHEET: The Art of Prompting (Part B)

Name: _____ Date: _____

1. Building a strong prompt. Write 1, 2, 3, 4.

Test & refine ☐

Set the role ☐

Specify format ☐

State task & context ☐

2. Fill in the blanks. Use the word box.

WORD BOX: temperature ✿ few-shot ✿ context

Adding examples is _____ prompting.

Randomness is controlled by _____.

Background info is called _____.

3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE & DEFINE: The Art of Prompting

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Few-shot

Meaning: _____

Example: _____

Temperature

Meaning: _____

Example: _____

Context

Meaning: _____

Example: _____

🧪 DEMO LAB: A school-approved chatbot (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Compare prompting techniques scientifically.

🔗 **Open:** any approved chatbot | 👩‍🏫 **With a teacher or parent**

👣 Steps

1	0️⃣	Zero-shot: 'Classify this review as positive or negative: …'
2	📋	Few-shot: give 3 labelled examples first, then the same review.
3	🪜	Step-by-step: ask it to explain reasoning before the answer for a maths word problem.
4	🧩	Add context: role, audience, format. Compare quality.
5	⭐	Score each output 1–5 for accuracy and clarity.

🔬 Record your observations

Best technique for classification:	
Did step-by-step help the maths?	
Highest score prompt:	

💭 **Reflect:** Why must even a perfect prompt's output be verified?

? QUIZ TIME: The Art of Prompting

Circle the right answer!

Q1. A prompt with examples is...

a) Few-shot

b) Zero-shot

c) No-shot

Q2. Low temperature gives...

a) Random output

b) Focused output

c) Cold output

Q3. 'Think step by step' improves...

a) Colour

b) Volume

c) Reasoning

Q4. Background info in a prompt is...

a) Context

b) Chorus

c) Noise

Q5. Good prompting is...

a) One try

b) Iterative

c) Random

MY SCORE: _____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: The Art of Prompting

Look at the picture. Tick ✓ TRUE or FALSE.

	TRUE	FALSE
Few-shot prompts include examples.		
High temperature gives more focused answers.		
Step-by-step prompting can improve reasoning.		
Prompt injection tries to override AI rules.		
Pasting passwords into prompts is safe.		
Good prompting is iterative.		

HOME CONNECT

With a parent, compare a vague and a structured prompt for planning a family trip.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: The Art of Prompting

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Asking without examples is ____ prompting.

(a) zero-shot ☐ **(b)** few-shot ☐ **(c)** one-line ☐

2. Giving examples before the task is ____ prompting.

(a) zero-shot ☐ **(b)** few-shot ☐ **(c)** no-shot ☐

3. The setting that controls randomness is ____.

(a) volume ☐ **(b)** brightness ☐ **(c)** temperature ☐

4. Role, audience and format provide ____.

(a) context ☐ **(b)** noise ☐ **(c)** bugs ☐

B. Fill in the blanks. Use the word box.

consistent step refine checking

1. Low temperature gives ____ answers.

2. Ask the AI to reason step by ____.

3. Test, compare and ____ prompts.

4. Even good prompts need fact ____.

C. Write T for True and F for False.

1. Few-shot examples can improve output format. [ ]

2. High temperature always gives the most accurate facts. [ ]

3. Adding context usually improves relevance. [ ]

4. Prompt engineering removes the need to verify. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Zero-shot	____	**(a)** Randomness level
2. Few-shot	____	**(b)** No examples
3. Temperature	____	**(c)** Role, audience, format
4. Context	____	**(d)** Some examples

E. Answer in one word.

1. Which prompting gives examples first? _____

2. Which setting controls randomness? _____

3. What is designing better prompts called? _____

F. Answer these questions.

1. Rewrite 'Explain gravity' as a strong prompt using context.

2. Compare zero-shot and few-shot prompting.

G. Case-based questions. Read and answer.

Kabir's 'Explain photosynthesis' gave generic text. Riya added role, audience and format, a few-shot style example and low temperature, then verified facts with NCERT.

1. Riya's role/audience/format added ____.

(a) context ☐ **(b)** noise ☐ **(c)** temperature ☐

2. Her style example made it ____.

(a) zero-shot ☐ **(b)** few-shot ☐ **(c)** random ☐

3. Checking NCERT was ____.

(a) unnecessary ☐ **(b)** plagiarism ☐ **(c)** verification ☐

H. THINK DEEPER

Design three prompts (zero-shot, few-shot, step-by-step) for the same task and predict which works best. Then explain how you'd test them fairly.

UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. A table in pandas is a...

a) DataFrame | b) Frame | c) Window

Q2. A prompt with examples is...

a) Zero-shot | b) Few-shot | c) No-shot

Q3. model.predict() is used to...

a) Train | b) Delete | c) Make predictions

Q4. Low temperature gives...

a) Focused answers | b) Wild answers | c) Cold answers

Q5. Overriding AI rules via input is...

a) Prompt art | b) Prompt injection | c) Prompt music

Q6. np.mean([2,4,6]) is...

a) 2 | b) 6 | c) 4

WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

DATAFRAME

ARRAY

FEW-SHOT

CONTEXT

PYTHON

PROMPT

F	D	P	O	S	M	U	M	M	P
F	A	N	P	S	H	R	I	R	S
C	T	F	E	W	S	H	O	T	S
O	A	M	P	P	M	P	U	T	P
N	F	P	R	H	L	Y	F	W	I
T	R	R	R	K	K	T	S	A	T
E	A	O	S	S	W	H	U	R	O
X	M	M	K	G	R	O	S	R	M
T	E	P	W	C	L	N	A	A	G
D	B	T	U	B	I	U	H	Y	D

I finished AI CONNECT! Teacher's sign: _____

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI Powers Innovation

Topic 6

AI Across Industries

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI Powers Innovation

LET'S LEARN

- AI speeds up discovery in science and engineering.
- It finds patterns humans can't see in huge data.
- AI + domain experts = breakthrough innovation!

Innovation frontiers

Protein science

Drug discovery

New materials

Space science

KEY TERMS

Protein folding
How proteins take shape

Innovation New valuable solution

Simulation Virtual experiment

💬 TECH TALK: AI Powers Innovation

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👦 **Kabir**

Bolt, how is AI speeding up science 🔬?

10:02

🤖 **Bolt**

It finds patterns in data too huge for humans. Protein-folding AI predicts 3D protein shapes 🧬 that took years to measure.

10:04 ✓✓

👧 **Riya**

Why does that matter?

10:06

🤖 **Bolt**

Protein shapes guide drug design 💊. Faster predictions mean faster medicines.

10:08 ✓✓

👦 **Kabir**

Other fields?

10:10

🤖 **Bolt**

Materials for batteries 🔋, weather models, chip design and space missions 🚀.

10:12 ✓✓

Riya
What's simulation?
10:14

Bolt
Testing ideas in a virtual model 🖥️ before building them, like crash-testing a car digitally.
10:16 ✓✓

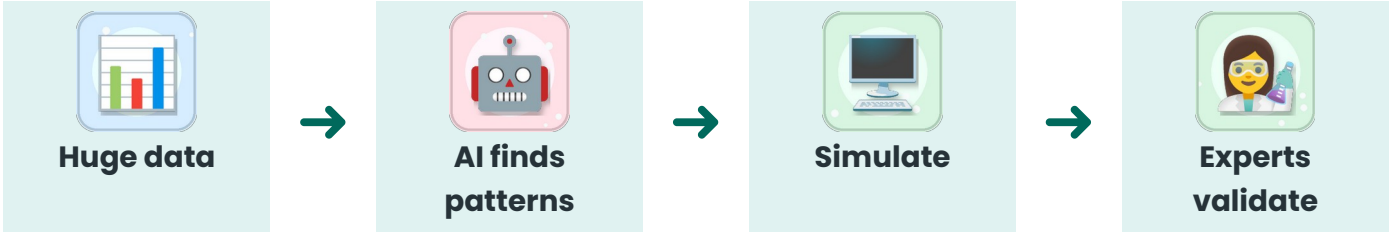
Kabir
AI + domain experts = breakthrough innovation 💡!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI speeds discovery by finding patterns in huge data, predicting protein structures and running simulations. Breakthroughs come from AI working with domain experts.

🔄 How it works



INDIA SPOTLIGHT

ISRO and Indian research institutes use computing and AI for mission planning, satellite data and materials research.

📖 CONCEPT EXPLAINED: AI Powers Innovation

💡 In this chapter

Some of the most exciting uses of AI are not chatbots but scientific discoveries: predicting the shapes of proteins, designing new materials, forecasting extreme weather and planning space missions. In this chapter we explore how AI accelerates innovation, why huge data and pattern-finding matter, how simulation saves time and money, and why breakthroughs still depend on human experts and careful experiments.

🔬 1. Why science needs AI

Modern science produces data faster than humans can read it. A single genome sequencer, telescope survey or climate model can generate terabytes of data. Hidden in that data are patterns that could reveal a new drug target or an early sign of a cyclone, but they are often too subtle or complex for people to spot manually.

Machine learning excels at finding patterns in high-dimensional data, where hundreds or thousands of variables interact. It can screen millions of possibilities, rank the most promising ones and let scientists focus their limited time and lab resources on the best candidates. In this sense, AI acts as a powerful telescope for data, extending what researchers can see.

🤔 PAUSE & THINK

Name one field (medicine, agriculture, astronomy, climate) and describe a kind of data it produces that would be too large for humans to analyse alone.

🧬 2. Case study: the protein-folding breakthrough

Proteins are chains of amino acids that fold into precise 3D shapes, and their shape determines their function in the body. For decades, finding a protein's shape required slow, expensive experiments such as X-ray crystallography, sometimes taking years for a single protein. Predicting the shape from the amino-acid sequence alone was a famous unsolved problem in biology.

In 2020, DeepMind's AlphaFold system predicted protein structures with accuracy comparable to experiments in many cases. Its team later released predicted structures for hundreds of millions of proteins in a free public database, and the work was

recognised with the 2024 Nobel Prize in Chemistry. Researchers worldwide, including in India, now use these predictions to study diseases and design drugs faster.

✏️ **WORKED EXAMPLE**

Before: one research group might spend years solving one structure. After: a researcher can look up a predicted structure in seconds, check its confidence scores, and decide which experiments are worth running.

🖥️ 3. Simulation: testing ideas in a virtual world

A simulation is a computer model of a real system: airflow over a wing, traffic through a city, a chemical reaction or a rocket launch. Engineers use simulations to test many designs cheaply before building physical prototypes. Car makers run virtual crash tests; architects simulate earthquakes; space agencies simulate trajectories and heat shields.

Traditional simulations can be extremely slow because they solve complex physics equations. AI 'surrogate models' learn to approximate these simulations, giving results thousands of times faster, which allows designers to explore far more options. AI is also used to design better batteries, catalysts and solar materials by predicting properties of compounds that have never been made.

🤔 **PAUSE & THINK**

Why might a car company run 10,000 virtual crash tests before crashing a single real car? What are the limits of relying only on simulation?

👩‍🔬 4. AI plus domain experts

AI rarely produces breakthroughs alone. Domain experts frame the right questions, choose meaningful data, interpret results and spot when an AI prediction makes no physical sense. An AI might predict that a molecule binds to a protein, but chemists must check whether it can be synthesised, whether it is toxic and whether it works in real cells.

The most productive teams are interdisciplinary: biologists with data scientists, climate scientists with machine-learning engineers, doctors with software developers. Each brings knowledge the others lack. For students, this means that combining a strong subject interest, like biology, physics or economics, with AI skills is a powerful path to doing meaningful innovative work.

WORKED EXAMPLE

A materials team uses AI to shortlist 50 battery compounds from 100,000 candidates. Chemists synthesise the top 10, test them, and find 2 that outperform current materials. Experimental results are then fed back to improve the AI.

5. Innovation in India and its responsibilities

Indian institutions apply AI to agriculture advisories, weather and monsoon forecasting, satellite image analysis, healthcare screening and language technology. Space and research organisations use computational methods for mission planning and Earth observation, while start-ups build AI tools for farmers, small businesses and classrooms.

Innovation also brings responsibilities. Scientific AI models can be wrong, and over-trusting them can waste resources or cause harm, for example in medicine. Data used in research must respect privacy and consent. And the benefits of AI-powered innovation should reach rural as well as urban communities. Responsible innovators ask not only 'Can we build it?' but also 'Who benefits, and who could be harmed?'

TRY IT

Visit the AlphaFold Protein Structure Database. Search for a protein linked to a disease common in India. Note what the colour-coded confidence shows and write two questions a scientist would still need to test in the lab.

Key takeaways

- AI helps science by finding patterns in data too large or complex for humans.
- Protein-structure prediction is a landmark example of AI accelerating discovery.
- Simulations, increasingly sped up by AI, let engineers test ideas virtually before building.
- Breakthroughs come from AI working with domain experts and verified by experiments.
- Responsible innovation considers accuracy, privacy and who benefits.

Quick check: answer in one or two lines.

1. Why are protein shapes important in medicine?

2. What is a simulation, and how can AI speed it up?

3. Give one reason why AI predictions in science still need experiments.

4. Why are interdisciplinary teams powerful for innovation?

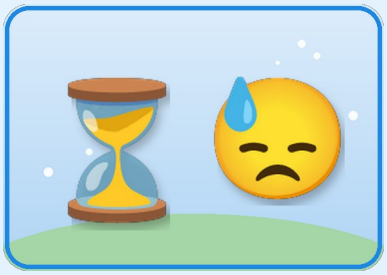
AI CHRONICLES: Fifty Years in Five Minutes

Riya | Dr. Iqbal | Protein AI

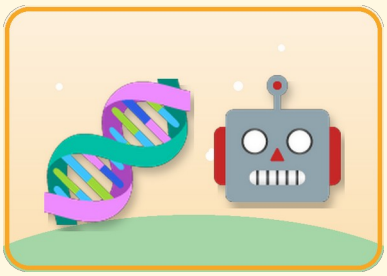
SCENE 1

On a lab visit, Dr. Iqbal showed Riya a protein linked to a tropical disease 🦟.

"Earlier, finding one protein's shape could take years of experiments," she said.



SCENE 2



SCENE 3

Protein-structure AI predicted the 3D shape in minutes, with confidence colours.

Her team simulated how drug molecules might fit the protein, shortlisting candidates.

SCENE 4

SCENE 5

Lab experiments confirmed two promising candidates. AI sped it up; experiments proved it.

Riya decided to study computational biology. "AI plus science can save lives."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya joins a summer research programme	**Dr. Iqbal stops experiments**	**The protein folds itself into a paper plane**

🧐 Story check: answer the questions.

1. What did protein AI predict?

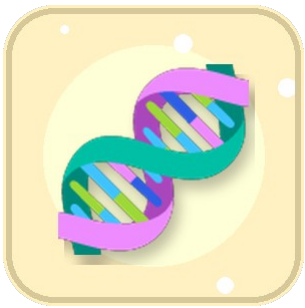
2. How was simulation used?

3. Why were lab experiments still needed?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI-driven breakthroughs



Protein structure prediction

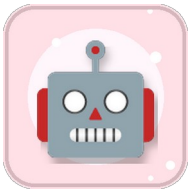
Faster drug screening

Better battery materials

Precise weather models

Exoplanet discovery

Lab automation robots



BOLT SAYS:
AlphaFold predicted the shapes of proteins: work that once took scientists years!

FUN FACT!
The 2024 Nobel Prize in Chemistry honoured AI-based protein structure prediction!

👀 **I spy AI! Draw one AI helper you saw today.**

🧪 LAB ACTIVITY: Innovation Canvas

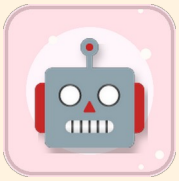
🎒 You need:

Canvas template

Sticky notes

Teams of 4

👣 Steps:

1		Pick a field: health, energy, agriculture or space.
2		Identify one slow or costly problem.
3		Propose how AI could speed it up. What data is needed?
4		List risks & limits. Pitch in 3 minutes.

THINK & TALK

What can AI NOT do in your innovation? Who must stay in the loop?

🖍️ **Draw what you did!**

WORKSHEET: AI Powers Innovation (Part A)

Name: _____ Date: _____

1. Match the field to the AI breakthrough.

Biology

Medicine

Energy

Astronomy

Finding exoplanets

Drug screening

Protein structures

Battery materials

2. Fill in the blanks. Use the word box.

WORD BOX: simulation ✿ innovation ✿ protein folding

A new valuable solution is an _____.

How proteins take their shape is _____.

A virtual experiment is a _____.

WORKSHEET: AI Powers Innovation (Part B)

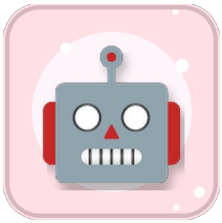
Name: _____ Date: _____

1. AI-assisted discovery. Write 1, 2, 3, 4.

Validate in lab ☐

Collect known data ☐

Generate predictions ☐



Train a model ☐

2. Can AI do this alone? Tick ✓.

	AI can help	Needs humans
Screen millions of molecules		
Approve a medicine for public use		
Predict protein shapes		
Run final clinical trials		

3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE & DEFINE: AI Powers Innovation

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Protein folding

Meaning: _____

Example: _____

Innovation

Meaning: _____

Example: _____

Simulation

Meaning: _____

Example: _____

🧪 DEMO LAB: AlphaFold Protein Structure Database (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Explore how AI predicted protein structures.

🔗 **Open:** alphafold.ebi.ac.uk | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Search 'haemoglobin' (human).
2		Rotate the 3D structure. Note the colour-coded confidence.
3		Find areas of high and low confidence.
4		Read why this protein matters for health.
5		Write how long such structures took before AI, and why AI matters.

🔬 Record your observations

Protein explored:	
Low-confidence regions:	
Why this discovery matters:	

💭 **Reflect:** Why do scientists still verify AI predictions with experiments?

? QUIZ TIME: AI Powers Innovation

Circle the right answer!

Q1. AlphaFold predicts...

a) Protein structures

b) Rain

c) Songs

Q2. A virtual experiment is a...

a) Game

b) Simulation

c) Movie

Q3. Breakthroughs need AI plus...

a) Luck only

b) Rest

c) Domain experts

Q4. 2024 Chemistry Nobel honoured...

a) Protein prediction AI

b) Cars

c) Phones

Q5. AI speeds discovery by finding...

a) Balloons

b) Hidden patterns

c) Pizza

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Powers Innovation

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
AI can predict protein structures.		
AI-found medicines need no testing.		
AI protein work was honoured by a 2024 Nobel Prize.		
Simulations are virtual experiments.		
AI replaces scientists entirely.		
AI helps search for new battery materials.		

HOME CONNECT

Find one recent AI-science breakthrough in the news. Explain it to your family in 1 minute.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Powers Innovation

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. AI predicting 3D protein shapes helps ____ design.

(a) drug ☐ **(b)** dress ☐ **(c)** dance ☐

2. Testing ideas in a virtual model is ____.

(a) stimulation ☐ **(b)** simulation ☐ **(c)** situation ☐

3. Breakthroughs come from AI working with ____.

(a) only computers ☐ **(b)** nobody ☐ **(c)** domain experts ☐

4. AI is useful in science because it finds patterns in ____ data.

(a) huge ☐ **(b)** tiny only ☐ **(c)** no ☐

B. Fill in the blanks. Use the word box.

verified batteries simulations innovation

1. AI predictions must be ____ by experiments.

2. AI can help design better ____ for EVs.

3. Digital crash tests are ____.

4. A new useful idea or product is an ____.

C. Write T for True and F for False.

1. AI can speed up scientific discovery. [ ]

2. AI replaces the need for experiments. [ ]

3. Simulation saves time and cost. [ ]

4. Protein shapes don't matter for medicine. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Protein folding	____	**(a)** New useful idea
2. Simulation	____	**(b)** 3D shape of proteins
3. Innovation	____	**(c)** Specialist in a field
4. Domain expert	____	**(d)** Virtual testing

E. Answer in one word.

1. What virtual testing method saves cost? _____

2. What 3D problem did AI help solve in biology? _____

3. Who validates AI discoveries? _____

F. Answer these questions.

1. Give three examples of AI accelerating science or engineering.

2. Why is 'AI + experts' better than AI alone?

G. Case-based questions. Read and answer.

Protein AI predicted a disease protein's shape in minutes. Scientists simulated drug fits and confirmed two candidates through lab experiments.

1. The AI predicted the protein's ____.

(a) 3D structure ☐ **(b)** price ☐ **(c)** colour ☐

2. Testing drug fits on computers is ____.

(a) guessing ☐ **(b)** simulation ☐ **(c)** farming ☐

3. Lab experiments were needed to ____.

(a) waste time ☐ **(b)** replace AI ☐ **(c)** confirm results ☐

H. THINK DEEPER

Pick a global challenge (clean water, pandemics, energy). Propose how AI + experts could speed up a solution.

TOPIC 6 • MONTH 3 • AI APPLICATIONS

AI Across Industries

LET'S LEARN

- Every sector adapts AI to its own data and goals.
- Value comes from better decisions, speed and scale.
- Successful AI needs data, people and processes!

Sector snapshots

Finance

Healthcare

Agriculture

Manufacturing

KEY TERMS

Predictive maintenance Fix before failure

Forecasting Predicting future values

ROI Return on investment

💬 TECH TALK: AI Across Industries

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👧 **Riya**
Bolt, how does a factory use AI 🏭?
10:02

🤖 **Bolt**
Predictive maintenance 🔧: sensors spot vibration patterns and predict a breakdown before it happens.
10:04 ✓✓

👦 **Kabir**
And retail?
10:06

🤖 **Bolt**
Demand forecasting 📈: predict what will sell, so shops stock right and waste less.
10:08 ✓✓

👧 **Riya**
How do companies decide if AI is worth it?
10:10

🤖 **Bolt**
ROI 💰: return on investment. Benefits (time, cost, quality) compared to cost.
10:12 ✓✓

Kabir
Why do some AI projects fail?
10:14

Bolt
Poor data, no skilled people or old processes ⚠️. AI needs data, people and processes together.
10:16 ✓✓

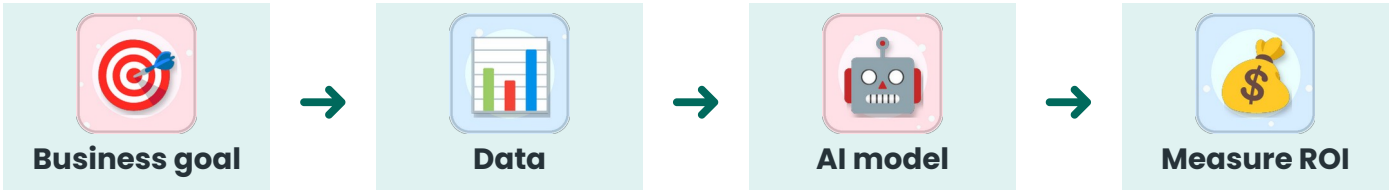
Riya
Right tool, right data, right team 🤝!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Industries adapt AI to their data and goals: predictive maintenance, forecasting and more. Success needs data, people and processes, and value is measured by ROI.

🔄 How it works



INDIA SPOTLIGHT

Indian railways, power utilities and manufacturers are piloting AI for predictive maintenance and forecasting.

📖 CONCEPT EXPLAINED: AI Across Industries

💡 In this chapter

Beyond research labs, AI is reshaping everyday business: factories, banks, shops, hospitals, farms and transport networks. But AI does not create value automatically; many projects fail. This chapter explains how different industries use AI, the main business use cases, how companies measure value with ROI, and what separates successful AI adoption from expensive disappointment.

🏭 1. Manufacturing: predictive maintenance and quality

In a factory, an unexpected machine breakdown can halt production for hours or days. Traditional maintenance either waits for failure (reactive) or services machines on a fixed schedule (preventive), which may replace parts that are still fine. Predictive maintenance uses sensors that record vibration, temperature, sound and power use, and machine-learning models that spot patterns appearing before failures.

Computer vision also inspects products on assembly lines, spotting scratches, cracks or misaligned parts faster and more consistently than tired human eyes. Together, these applications reduce downtime, waste and cost. Railways, power plants and airlines use similar methods to keep trains, turbines and aircraft engines running safely.

✏️ WORKED EXAMPLE

A plant loses ₹5 lakh per month to breakdowns. An AI system costing ₹12 lakh per year cuts breakdowns by 60%, saving ₹3 lakh per month or ₹36 lakh per year. ROI = (36 - 12) ÷ 12 × 100 = 200%.

📈 2. Retail, finance and logistics

Retailers use demand forecasting to predict what customers will buy, where and when, so they can stock shelves correctly and reduce waste, especially for perishables like fruit and milk. Recommendation systems suggest products based on browsing and purchase patterns, and dynamic pricing adjusts prices based on demand.

Banks and payment companies use AI for fraud detection, flagging unusual transactions within milliseconds, and for credit scoring and customer service chatbots. Logistics companies optimise delivery routes, warehouse layouts and fleet schedules. In each case, AI helps make many small decisions faster and more accurately at a scale no human team could manage.

🤔 **PAUSE & THINK**

An e-commerce app recommends items you like, but also nudges you to spend more. Where is the line between helpful personalisation and manipulation?

🏥 3. Healthcare, agriculture and public services

In healthcare, AI assists in reading X-rays and retinal scans, predicting patient risk, scheduling and managing hospital beds. In agriculture, it analyses satellite and drone images to monitor crop health, predicts yields and pest outbreaks, and powers advisory services that reach farmers through mobile phones in local languages.

Governments use AI for traffic management, tax fraud detection, weather and disaster forecasting, and citizen services. These sectors carry higher stakes: errors can affect health, livelihoods and rights. That is why they require careful testing, human oversight, fairness checks and clear accountability, not just technical accuracy.

✏️ **WORKED EXAMPLE**

Industry map: Manufacturing → predictive maintenance, quality inspection. Retail → demand forecasting, recommendations. Finance → fraud detection, credit risk. Healthcare → scan analysis, risk prediction. Agriculture → crop monitoring, advisories. Transport → route optimisation, predictive maintenance.

💰 4. Measuring value: KPIs and ROI

Businesses invest in AI to achieve measurable goals: lower costs, higher revenue, faster service, better quality or reduced risk. Key Performance Indicators (KPIs) track these goals, for example machine downtime hours, forecast error, fraud losses, customer wait time or satisfaction scores. KPIs should be measured before and after AI is introduced to show real impact.

Return on Investment (ROI) compares benefit to cost: ROI = (benefit - cost) ÷ cost × 100%. Costs include not just software but data collection, cloud computing, staff training, integration with existing systems and ongoing maintenance. Benefits may also include harder-to-measure gains such as safety or customer trust, which decision-makers should consider alongside the numbers.

🤔 **PAUSE & THINK**

A chatbot reduces call-centre costs but customer satisfaction drops because people can't reach a human. Is the project successful? Which KPIs would you track?

⚠️ 5. Why AI projects succeed or fail

Studies of business AI repeatedly find that many projects never move beyond the pilot stage. Common reasons include poor or insufficient data, unclear business goals, lack of skilled people, resistance from staff who fear for their jobs, and failure to change processes around the new tool. A brilliant model that no one trusts or uses creates no value.

Successful adoption treats AI as a combination of data, people and processes. Companies start with a clear problem, run a small pilot, measure KPIs honestly, train and involve employees, and scale gradually. They also set up governance: rules about data use, fairness, security and who is accountable when the AI makes a mistake.

🖐 TRY IT

Choose a local business (a pharmacy, bakery, bus depot or farm). Identify one AI use case, two KPIs to measure it, its likely costs and one risk. Present it as a one-paragraph proposal.

✅ Key takeaways

- Industries adapt AI to their own data: maintenance, quality, forecasting, fraud detection and more.
- High-stakes sectors such as health and public services need extra testing and oversight.
- KPIs and ROI measure real value; costs include data, people and integration.
- Many AI projects fail due to poor data, unclear goals or lack of adoption.
- Success needs data, people, processes and governance together.

❓ Quick check: answer in one or two lines.

1. What is predictive maintenance?

2. Write the formula for ROI.

3. Give two KPIs a hospital might track for an AI scheduling system.

4. List three reasons AI projects fail in companies.

AI CHRONICLES: The Train That Told on Itself

Kabir | Mr. Siddiqui | Sensors

SCENE 1

Kabir's father, Mr. Siddiqui, maintained locomotives. Sudden breakdowns caused long delays.

New sensors recorded vibration, temperature and sound from engines every second.

SCENE 2

SCENE 3

An AI model learned patterns before past failures and began predicting risks days ahead.

Technicians fixed parts early. Breakdowns fell by 40% in a year 📉.

SCENE 4

SCENE 5

Kabir calculated ROI for a school project: savings were three times the cost!

"The AI didn't replace us; it made our experience count more," his father said proudly.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐ **Kabir studies AI for railways**	☐ **They remove all sensors**	☐ **The train writes its own diary**

🧐 Story check: answer the questions.

1. What data did sensors collect?

2. What is predictive maintenance?

3. How did Kabir show the value?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Industry use cases

Credit risk scoring

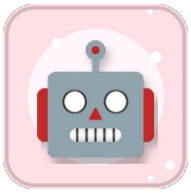
Hospital bed forecasting

Yield prediction

Predictive upkeep

Dynamic pricing

Legal document review



BOLT SAYS:

Predictive maintenance: I listen to machine vibrations and warn before they break!

FUN FACT!

Predictive maintenance can cut unexpected machine breakdowns significantly in factories!

I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: Industry Consultant Role-Play

You need:

Company cards

ROI sheet

Consultant teams

Steps:

1. Draw a company card: dairy co-op, bank, hospital...
2. Find their costliest problem.
3. Propose an AI use case + data needed.
4. Estimate benefits, costs and risks. Present!

THINK & TALK

Why do many AI projects fail even with good models?

Draw what you did!

WORKSHEET: AI Across Industries (Part A)

Name: _____ Date: _____

1. Match the sector to the AI use case.

Banking ●

Manufacturing ●

Agriculture ●

Retail ●

●

Demand & pricing

●

Credit risk

●

Yield prediction

●

Predictive upkeep

2. Is AI a good fit? Tick ✓.

	Good fit	Poor fit
Lots of data, repeated decisions		
No data, one-time decision		
High-volume routine checks		
Decisions needing deep empathy alone		

WORKSHEET: AI Across Industries (Part B)

Name: _____ Date: _____

1. Compute ROI = (savings - cost) / cost × 100. Use the word box.

WORD BOX: 300% ✿ 100% ✿ 50%

savings ₹8L, cost ₹2L → ____

savings ₹6L, cost ₹3L → ____

savings ₹3L, cost ₹2L → ____

2. Circle the signs that AI suits a business problem.

Lots of historical data

Frequent repeated decisions

No clear goal

High cost of errors

Totally random events

Measurable outcome

PUZZLE & DEFINE: AI Across Industries

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

2. Define it! Write each term in your own words with an example.

Predictive maintenance

Meaning: _____

Example: _____

Forecasting

Meaning: _____

Example: _____

ROI

Meaning: _____

Example: _____

🧪 DEMO LAB: Spreadsheet ROI model (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Build a simple ROI case for an AI project.

🔗 **Open:** Google Sheets / Excel | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Scenario: a factory loses ₹5 lakh/month in machine breakdowns.
2. AI predictive maintenance costs ₹12 lakh/year and cuts breakdowns by 60%.
3. Calculate yearly savings and ROI = (benefit - cost) ÷ cost × 100.
4. List 3 risks: poor data, staff training, sensor costs.
5. Make a chart: cost vs benefit over 3 years.

🔬 Record your observations

Yearly savings:	
ROI %:	
Biggest risk:	

💭 **Reflect:** Besides money, what benefits and harms should a company consider?

? QUIZ TIME: AI Across Industries

Circle the right answer!

Q1. Fixing machines before failure is...

a) Predictive upkeep

b) Repainting

c) Scrapping

Q2. ROI means...

a) Robot On Internet

b) Return on investment

c) Rhythm of India

Q3. Successful AI needs data, people and...

a) Pizza

b) Luck

c) Processes

Q4. Banks use AI for...

a) Credit risk

b) Crops

c) Films

Q5. AI is a poor fit when there is...

a) Lots of data

b) No data

c) Routine work

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Across Industries

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Predictive maintenance fixes machines before they fail.		
ROI compares benefits with costs.		
AI works well with no data or goals.		
Banks use AI for fraud detection.		
Staff training doesn't matter for AI adoption.		
Demand forecasting can reduce wastage.		

HOME CONNECT

Interview a relative about one repeated decision at their workplace. Could AI help?

🔢 Self-assessment: tick one.

😀	🙂	😐
I know it well!	**I know some.**	**I need practice.**

EXERCISE: AI Across Industries

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Predicting machine failures before they happen is ____ maintenance.

(a) predictive ☐ **(b)** punitive ☐ **(c)** periodic-only ☐

2. Predicting future sales is demand ____.

(a) fiction ☐ **(b)** forecasting ☐ **(c)** framing ☐

3. Benefit compared to cost is ____.

(a) RAM ☐ **(b)** RGB ☐ **(c)** ROI ☐

4. AI success needs data, people and ____.

(a) processes ☐ **(b)** posters ☐ **(c)** parties ☐

B. Fill in the blanks. Use the word box.

cost **temperature** **data** **waste**

1. ROI = (benefit - cost) ÷ ____ × 100.

2. Sensors capture vibration and ____.

3. Poor ____ is a common cause of AI failure.

4. Forecasting reduces ____ in retail.

C. Write T for True and F for False.

1. Predictive maintenance can reduce breakdowns. [ ]

2. Every AI project automatically succeeds. [ ]

3. ROI helps decide if AI is worth it. [ ]

4. Staff training is irrelevant to AI success. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Predictive maintenance	____	**(a)** Return on investment
2. Forecasting	____	**(b)** Fix before failure
3. ROI	____	**(c)** How work is done
4. Process	____	**(d)** Predict demand

E. Answer in one word.

1. What does ROI stand for? _____

2. What predicts machine failure? _____

3. What predicts future sales? _____

F. Answer these questions.

1. Explain predictive maintenance with an example.

2. Why do some AI projects fail in companies?

G. Case-based questions. Read and answer.

Locomotive sensors fed an AI model that predicted failures days ahead. Breakdowns fell 40%, and savings were three times the cost. Technicians said AI made their experience count more.

1. The system used ____ maintenance.

(a) predictive ☐ **(b)** random ☐ **(c)** no ☐

2. Savings three times the cost means a ____ ROI.

(a) negative ☐ **(b)** positive ☐ **(c)** zero ☐

3. The technicians' role was ____.

(a) eliminated ☐ **(b)** ignored ☐ **(c)** augmented ☐

H. THINK DEEPER

Choose an Indian industry. Write a one-page AI proposal: problem, data, model type, ROI estimate and risks.

🏅 UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. Fixing machines before failure:

a) Predictive upkeep

b) Repainting

c) Scrapping

Q2. AlphaFold predicts...

a) Weather

b) Protein structures

c) Music

Q3. ROI = (savings - cost) / ...

a) savings

b) balloons

c) cost

Q4. AI drug candidates must be...

a) Lab-validated

b) Sold directly

c) Hidden

Q5. AI is a poor fit when there is...

a) Lots of data

b) No data/goal

c) Routine work

Q6. Breakthroughs need AI plus...

a) Luck

b) Rest

c) Domain experts

WORD SEARCH: AI SOLVE

🔍 **Find these words. They go across → or down ↓.**

INNOVATION

SIMULATION

ROI

FORECAST

PROTOTYPE

PATENT

N	U	P	A	T	E	N	T	P	E
M	D	B	E	R	G	I	O	K	O
F	O	R	E	C	A	S	T	S	N
U	M	T	U	O	U	H	L	A	I
W	F	L	T	U	U	D	G	U	I
K	D	C	R	R	C	M	C	O	E
A	K	O	O	D	B	R	O	I	W
W	P	R	O	T	O	T	Y	P	E
I	N	N	O	V	A	T	I	O	N
S	I	M	U	L	A	T	I	O	N

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

AI Career Universe

Topic 8

Create Your Career Blueprint

TOPIC 7 • MONTH 4 • AI & CAREER

AI Career Universe

LET'S LEARN

- The AI career universe has many orbits.
- Research, engineering, product, data and policy.
- Each needs a different mix of skills!

Career orbits

Research scientist

ML engineer

Data scientist

AI governance

KEY TERMS

MLOps Running models in production

Product manager Leads what gets built

Governance Rules & oversight

💬 TECH TALK: AI Career Universe

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👦 **Kabir**

Bolt, is every AI job just 'coder' 💻?

10:02

🤖 **Bolt**

No! Picture a universe with orbits 🪐: research, engineering, product, data and policy.

10:04 ✓✓

👧 **Riya**

What do researchers do?

10:06

🤖 **Bolt**

Invent new methods and publish papers 📄. Engineers build and scale real systems 🛠️.

10:08 ✓✓

👦 **Kabir**

What's MLOps?

10:10

🤖 **Bolt**

Running models reliably in production ⚙️: deployment, monitoring, retraining. DevOps for ML.

10:12 ✓✓

Riya

And non-technical roles?

10:14

Bolt

Product managers decide what to build and why. Governance and policy experts make AI safe and legal.

10:16 ✓✓

Kabir

Different orbits, different skill mixes!

10:18

Type a message...

Pinned by Bolt

AI careers span research, engineering, MLOps, product, data and governance. Each orbit needs a different mix of technical, business and ethical skills.

How it works

INDIA SPOTLIGHT

India's tech sector employs large numbers of AI and data professionals, and global companies run AI research centres in Indian cities.

📖 CONCEPT EXPLAINED: AI Career Universe

💡 In this chapter

When people hear 'AI career', they often imagine a programmer typing code all day. In reality, building and using AI responsibly takes researchers, engineers, operations specialists, designers, product managers, data professionals, lawyers and policy experts. This chapter maps the AI career universe, explains what each role actually does day to day, and shows how different strengths can lead to meaningful work in AI.

🪐 1. The career universe: five orbits

A helpful way to picture AI careers is as orbits around a shared centre. The research orbit invents new methods and pushes the boundaries of what AI can do. The engineering orbit builds reliable systems that real people use. The product orbit decides what to build and why. The data orbit collects, cleans, analyses and interprets information. The governance orbit ensures AI is safe, fair and legal.

These orbits overlap constantly. A product manager needs enough technical understanding to talk with engineers; a researcher needs communication skills to publish and present; a policy expert needs to understand how models work in order to regulate them sensibly. Knowing the whole universe helps you see where your strengths fit and which skills to build.

🤔 PAUSE & THINK

Which orbit sounds most interesting to you right now? Which school subjects you enjoy connect to it?

🔬 2. Research and engineering roles

AI research scientists design new algorithms and architectures, run experiments and publish papers. They typically need strong mathematics (linear algebra, probability, calculus), programming skills and, often, postgraduate degrees. Research engineers turn ideas into code at scale, running large experiments on powerful computers.

Machine learning engineers build, train and deploy models into real products such as recommendation systems, fraud detectors or speech recognisers. Software engineers integrate AI into apps and websites. Computer vision, natural language processing and

robotics engineers specialise in particular data types. These roles reward strong coding, problem-solving and a habit of testing everything carefully.

✏️ **WORKED EXAMPLE**

A day for an ML engineer might include: checking overnight training results, debugging a data pipeline, discussing a new feature with the product manager, writing tests, and reviewing a colleague's code.

⚙️ 3. MLOps: keeping models healthy in production

A model that performs well in a notebook can fail in the real world. Data changes over time (a problem called data drift): new slang appears, customer behaviour shifts, sensors age. MLOps (Machine Learning Operations) is the discipline of deploying models reliably, monitoring their performance, retraining them when needed and managing versions safely.

MLOps engineers combine software engineering, cloud computing and machine learning knowledge. They build automated pipelines that test models before release, alert teams when accuracy drops, and allow rolling back to a previous version if something goes wrong. As more organisations depend on AI, these 'behind the scenes' roles are increasingly in demand.

🤔 **PAUSE & THINK**

A fraud-detection model was 95% accurate last year but only 80% this year. What might have changed? What would an MLOps engineer do?

📱 4. Product, design and data roles

AI product managers decide which problems are worth solving, define what success looks like, prioritise features and coordinate engineers, designers and business teams. They need strong communication, user empathy and enough technical understanding to make realistic plans. UX designers ensure AI tools are easy to use, explain their limits clearly and handle errors gracefully.

Data analysts and data scientists explore data, build dashboards, test hypotheses and communicate insights to decision-makers. Data engineers build the pipelines and storage systems that move data reliably. Annotation and data-quality specialists create the labelled data that supervised models learn from. Each role is essential: without good data and clear goals, even advanced models fail.

✏️ **WORKED EXAMPLE**

Skill mix examples: Product manager → communication, user research, basic ML. Data analyst → statistics, SQL, Python, visualisation. UX designer → design tools, user testing, psychology.

⚖️ 5. Governance, policy and ethics roles

As AI affects jobs, elections, healthcare and justice, societies need people who understand both technology and its impact. AI policy analysts advise governments and companies on laws and standards. Responsible-AI specialists audit systems for bias, privacy and safety. Lawyers specialise in data protection, intellectual property and liability for AI decisions.

These roles welcome students from law, economics, political science, philosophy and sociology, especially those who also learn the basics of how AI works. Their work determines whether AI benefits everyone or deepens inequality. For India, with its diversity of languages, regions and communities, such experts are especially important.

🖐️ **TRY IT**

Find two real job postings: one technical AI role and one policy or ethics role. List the top five skills each requires. Circle the skills that appear in both.

✅ Key takeaways

- AI careers span research, engineering, product, data and governance orbits.
- Research needs strong maths and experimentation; engineering needs robust coding and testing.
- MLOps keeps models reliable in production by monitoring, retraining and versioning.
- Product, design and data roles connect AI to real user needs.
- Governance and policy roles welcome humanities and social-science backgrounds.

❓ Quick check: answer in one or two lines.

1. Name the five orbits of the AI career universe.

2. What is data drift, and why does it matter?

3. What does an AI product manager do?

4. Why are policy experts needed in AI?

AI CHRONICLES: The Orbit Map

Kabir

Riya

Ms. Ananya (AI lead)

SCENE 1

At an industry visit, Ms. Ananya asked: "Who wants an AI career?" Only coders raised hands 🙋.

She drew a 'career universe': research, engineering, MLOps, product, data, policy.

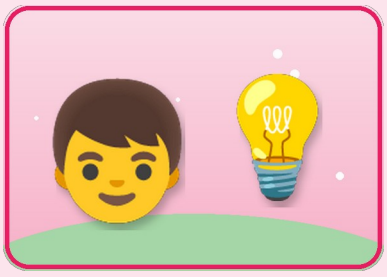
SCENE 2

SCENE 3

She told how a perfect model failed in production because no one monitored it. MLOps saved the day.

Her product manager decided WHAT to build; a policy expert checked it was legal and fair.

SCENE 4


SCENE 5

Kabir, who loved economics and debate, realised AI governance was his orbit ⚖️.

Riya chose ML engineering. "Different orbits, same universe," Ms. Ananya smiled 🌌.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Both shadow professionals for a week	**Kabir decides AI isn't for him**	**The orbits collide**

🧐 Story check: answer the questions.

1. Which orbits did Ms. Ananya describe?

2. Why did the model fail in production?

3. Which orbits did Kabir and Riya choose, and why?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Roles & what they do

Research: invent new methods

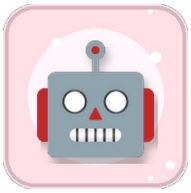
ML engineer: deploy models

Data scientist: find insights

Product manager: guide AI products

Security: protect AI systems

Policy: shape AI laws



BOLT SAYS:
Same AI project, ten different jobs. Find the orbit that fits you!

FUN FACT!
India's IndiaAI Mission is investing in computing, datasets and AI skills nationwide!

I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: Career Speed-Dating

You need:

Role cards

3-minute timer

Notes sheet

Steps:

1		Half the class becomes 'experts' with role cards.
2		Others rotate every 3 minutes, asking questions.
3		Note: tasks, skills, education, pros & cons.
4		Rank your top 3 roles and explain why.

THINK & TALK

Which role surprised you most?

Draw what you did!

WORKSHEET: AI Career Universe (Part A)

Name: _____ Date: _____

1. Match the role to its main task.

Research scientist ●

ML engineer ●

Product manager ●

Policy expert ●

● Shapes rules & laws

● Invents new methods

● Decides product direction

● Deploys & scales models

2. Fill in the blanks. Use the word box.

WORD BOX: governance ✿ MLOps ✿ product manager

Running models in production is _____.

A _____ leads what gets built.

Rules & oversight for AI is _____.

WORKSHEET: AI Career Universe (Part B)

Name: _____ Date: _____

1. Technical or non-technical focus? Tick ✓.

	Technical	Non-technical
ML engineer		
AI policy analyst		
Research scientist		
Product manager		

2. Draw your AI career universe: put your top 3 roles in orbits!

Closest orbit (my top choice): _____

PUZZLE & DEFINE: AI Career Universe

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

MLOps

Meaning: _____

Example: _____

Product manager

Meaning: _____

Example: _____

Governance

Meaning: _____

Example: _____

🧪 DEMO LAB: Job portals + career universe map (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Map AI roles and the skills each needs.

🔗 **Open:** any job/career portal | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Find real job posts for: ML engineer, MLOps engineer, AI product manager, AI policy analyst.
2. List 5 required skills for each.
3. Draw a 'career universe' with each role as an orbit.
4. Mark skills shared by all roles (e.g., communication).
5. Circle the orbit that fits you best and explain why.

🔬 Record your observations

Most common skill across roles:	
Most surprising role:	
My best-fit orbit:	

💭 **Reflect:** Why do non-coding roles matter for responsible AI?

? QUIZ TIME: AI Career Universe

Circle the right answer!

Q1. Who invents new AI methods?

a) Research scientist

b) Singer

c) Driver

Q2. Running models in production is...

a) MusicOps

b) MLOps

c) CookOps

Q3. Who shapes AI laws?

a) Gamer

b) Chef

c) Policy expert

Q4. IndiaAI Mission invests in...

a) Compute, data, skills

b) Sweets

c) Balloons

Q5. A product manager mainly...

a) Repairs servers

b) Guides what to build

c) Only draws

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Career Universe

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🛠️	ML engineers deploy models to production.		
👩‍💻	All AI careers require only coding.		
⚖️	Policy experts help shape AI laws.		
🇮🇳	IndiaAI Mission supports AI skills and compute.		
🎓	Every AI job requires a PhD.		
🧭	Product managers guide what gets built.		

HOME CONNECT

Ask a working relative: which AI roles exist (or could exist) in their industry?

🔢 Self-assessment: tick one.

😀	🙂	😐
I know it well!	**I know some.**	**I need practice.**

EXERCISE: AI Career Universe

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Running ML models reliably in production is ____.

(a) MLOps ☐ **(b)** MLA ☐ **(c)** MRI ☐

2. The person deciding what AI product to build is the ____.

(a) data entry clerk ☐ **(b)** product manager ☐ **(c)** printer ☐

3. Rules and oversight for safe AI are AI ____.

(a) graphics ☐ **(b)** gaming ☐ **(c)** governance ☐

4. Inventing new AI methods is AI ____.

(a) research ☐ **(b)** repair ☐ **(c)** retail ☐

B. Fill in the blanks. Use the word box.

retraining papers build mix

1. MLOps includes deployment, monitoring and ____.

2. Researchers publish ____.

3. Engineers ____ real systems.

4. Each orbit needs a different skill ____.

C. Write T for True and F for False.

1. AI careers include non-coding roles. [ ]

2. Models never need monitoring after deployment. [ ]

3. Policy experts help keep AI legal and fair. [ ]

4. Product managers decide what to build and why. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Researcher	____	**(a)** Deploys & monitors
2. ML engineer	____	**(b)** New methods
3. MLOps	____	**(c)** Rules & oversight
4. Governance	____	**(d)** Builds models

E. Answer in one word.

1. What does MLOps manage? _____

2. Who decides what to build? _____

3. Which orbit makes AI legal and fair? _____

F. Answer these questions.

1. Describe three AI career orbits and their skills.

2. Why is MLOps important?

G. Case-based questions. Read and answer.

A model worked perfectly in testing but failed in production because no one monitored it. An MLOps engineer set up monitoring and retraining.

1. Monitoring and retraining are ____ tasks.

(a) MLOps ☐ **(b)** marketing ☐ **(c)** HR ☐

2. The failure shows models can ____ over time.

(a) improve automatically ☐ **(b)** degrade ☐ **(c)** never change ☐

3. This role belongs to the ____ orbit.

(a) policy ☐ **(b)** research ☐ **(c)** engineering ☐

H. THINK DEEPER

Which AI career orbit fits you? Justify with your skills, interests and a 3-step plan to enter it.

TOPIC 8 • MONTH 4 • AI & CAREER

Create Your Career Blueprint

LET'S LEARN

- A blueprint maps goals, skills, milestones and backups.
- Use SWOT: strengths, weaknesses, opportunities, threats.
- Review and update it every term!

Blueprint parts

Vision

Skill gaps

Milestones

Plan B

KEY TERMS

SWOT Strengths-Weaknesses-Opportunities-Threats

Milestone Checkpoint on the path

Skill gap What you still need

💬 TECH TALK: Create Your Career Blueprint

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, I want a plan for the next five years, not just next exam 🗺️.
10:02

🤖 Bolt
Build a career blueprint 📐: vision, goals, skills, milestones and backups.
10:04 ✓✓

👦 Kabir
Where do we start?
10:06

🤖 Bolt
A SWOT analysis 🧭: Strengths, Weaknesses, Opportunities, Threats.
10:08 ✓✓

👧 Riya
Then?
10:10

🤖 Bolt
Find your skill gap 📏: skills the target role needs minus skills you have.
10:12 ✓✓

Kabir

And milestones?

10:14

Bolt

Checkpoints with dates 📅: 'Finish ML course by Dec', 'Build 3 projects by Class 12', 'Apply to 5 colleges'.

10:16 ✓✓

Riya

Review and update it every term 🔁!

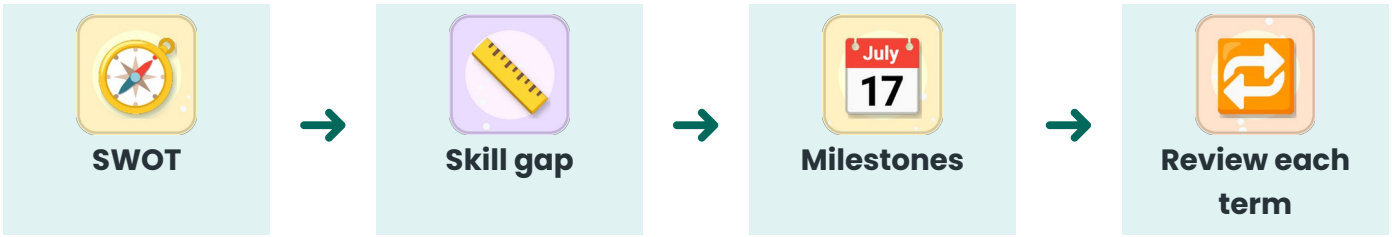
10:18

😊 Type a message...

📌 **Pinned by Bolt**

A career blueprint maps vision, goals, skills, milestones and backup plans. Use SWOT, measure skill gaps and review every term.

🔄 How it works



INDIA SPOTLIGHT

Indian career-guidance portals and college counselling help students map streams, entrance exams and degree options.

📖 CONCEPT EXPLAINED: Create Your Career Blueprint

💡 In this chapter

Wishing for a great career is easy; planning one is a skill. A career blueprint is a living document that turns your goals into concrete steps, milestones and backup options. In this chapter you will learn how to analyse yourself honestly with SWOT, identify skill gaps, set milestones, plan multiple pathways and review your plan regularly, so that you move forward with direction rather than anxiety.

🧭 1. Start with a vision, not a job title

A good blueprint begins with a broad vision: what kind of problems do you want to solve, and what kind of life do you want? 'I want to use technology to improve healthcare in rural India' is a vision. 'I want to be an ML engineer at a big company' is a specific target. Visions are flexible and survive changes in the job market; targets can change as you learn more.

Because AI is transforming jobs quickly, many roles you will hold may not exist today. Anchoring your plan in interests, values and transferable skills, such as problem-solving, communication and learning ability, keeps it useful even as job titles change. Your vision acts as a compass; specific roles are destinations you may choose along the way.

🤔 PAUSE & THINK

Write your vision in one sentence beginning 'I want to use my skills to…'. Does it describe a problem you care about?

📊 2. SWOT analysis for yourself

SWOT stands for Strengths, Weaknesses, Opportunities and Threats. Strengths and weaknesses are internal: things about you, such as strong maths, creativity, poor time management or fear of public speaking. Opportunities and threats are external: things in your environment, such as free online courses, a coding club, family responsibilities or exam pressure.

An honest SWOT requires self-reflection and feedback from teachers, mentors and friends. The power of SWOT is in the connections: use strengths to seize opportunities, fix

weaknesses that block opportunities, and plan for threats. For example, a strength in maths plus the opportunity of free online courses suggests learning data science early.

✏️ **WORKED EXAMPLE**

Strengths: logical thinking, good at maths. Weaknesses: procrastination, limited coding. Opportunities: SWAYAM/NPTEL courses, school robotics lab. Threats: heavy entrance coaching load, burnout. Action: take one short Python course in the summer; use a weekly planner.

📏 3. Finding and closing skill gaps

Choose one or two target roles and research what they require: look at real job postings, college syllabi and interviews with professionals. List the required skills, then mark the ones you already have and the ones you lack. The difference is your skill gap. Group gaps into technical skills (Python, statistics), domain knowledge (biology, finance) and human skills (communication, teamwork).

Closing gaps works best through projects rather than only courses. A project forces you to apply skills, solve unexpected problems and produce evidence for your portfolio. For each gap, pick a resource (course, book, club) and a project that uses the skill, then estimate how long it will take.

🤔 **PAUSE & THINK**

Pick one skill gap you have. What is one small project you could finish in four weeks to start closing it?

📅 4. Milestones, pathways and backup plans

Milestones are dated checkpoints that make progress visible: 'Complete an introductory ML course by December', 'Build and publish two projects by the end of Class 11', 'Shortlist colleges and entrance exams by June', 'Apply for a summer programme in Class 12'. Good milestones follow the SMART principle: specific, measurable, achievable, relevant and time-bound.

There are many pathways to most careers. Entrance exam results, family circumstances or new interests may change your route. A backup plan, such as a B.Sc in Data Science if engineering admission does not work out, is not a sign of weakness; it reduces stress and keeps your vision alive. Many successful professionals reached their goals through unexpected routes.

✏️ WORKED EXAMPLE

Pathway A: Science (PCM) → engineering entrance → B.Tech CSE/AI → internship → ML engineer. Pathway B: Science → B.Sc Data Science/Statistics → online certifications → M.Sc/M.Tech → data scientist. Both reach AI careers.

🔁 5. Reviewing and adapting your blueprint

A blueprint is not a contract; it is a tool. Review it at the end of every term. Which milestones did you reach? Which slipped, and why? Has your interest shifted? Update the plan honestly. Celebrate progress, however small, because motivation grows from seeing evidence of improvement.

Mentors make reviews more valuable. A teacher, senior student or professional can point out opportunities you missed and help you see weaknesses kindly. Keep your blueprint on one or two pages so it stays practical. Over time, it becomes a record of your growth that also helps with college applications and interviews.

🖐️ TRY IT

Create your one-page career blueprint with five boxes: Vision, SWOT, Target role and skill gap, Six milestones with dates, Backup pathway. Share it with one mentor for feedback.

✅ Key takeaways

- A blueprint starts with a flexible vision anchored in interests and values.
- SWOT analysis links internal strengths and weaknesses with external opportunities and threats.
- Skill gaps are closed most effectively through projects that build evidence.
- SMART milestones and backup pathways make plans realistic and reduce stress.
- Review and update the blueprint every term, ideally with a mentor.

❓ Quick check: answer in one or two lines.

1. What does each letter of SWOT stand for?

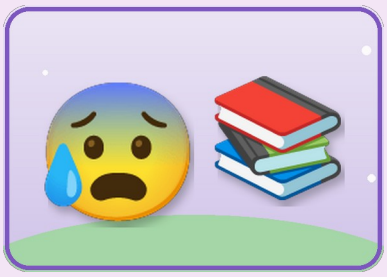
2. How do you calculate a skill gap?

3. Write one SMART milestone for your next three months.

4. Why is a backup plan useful?

AI CHRONICLES: Blueprint Before Boards

Riya | Mentor Mr. Paul | Planner



SCENE 1

Riya felt lost between JEE coaching, school, and her AI dream.

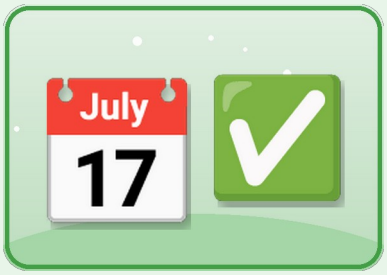
Mr. Paul asked her to do a SWOT. Strength: maths. Weakness: time management. Opportunity: online ML courses. Threat: burnout.

SCENE 2

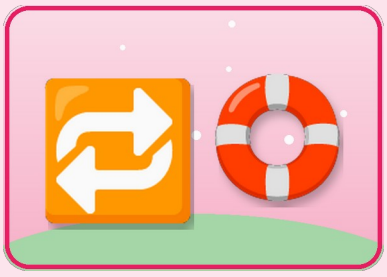
SCENE 3

Target role: ML engineer. Skill gap: Python projects, statistics, communication.

She set milestones: ML course by December, 2 projects by summer, debate club for communication.



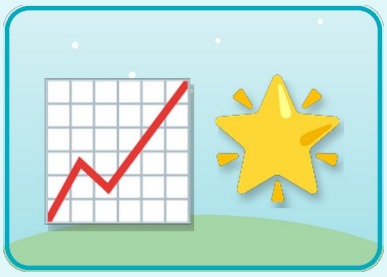
SCENE 4



SCENE 5

She also wrote a backup: B.Sc Data Science if JEE didn't go well. Stress dropped 😌.

Each term she reviewed progress. "A blueprint isn't a cage; it's a compass," Mr. Paul said.



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya helps classmates make blueprints	**Riya tears up the plan**	**The planner plans a holiday**

🧐 Story check: answer the questions.

1. What did Riya's SWOT show?

2. What skill gap did she find?

3. Why did the backup plan reduce stress?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Blueprint building blocks

Entrance exam prep plan

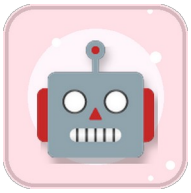
Project portfolio

Olympiads & hackathons

Mentors & seniors

Online certifications

Communication practice



BOLT SAYS:
A goal without a plan is a wish. A plan without review is a guess!

FUN FACT!
SWOT analysis is used by companies worldwide, and works for personal career planning too!

👀 **I spy AI! Draw one AI helper you saw today.**

LAB ACTIVITY: Personal SWOT & Blueprint

You need:

SWOT grid

Blueprint template

Mentor

Steps:

1		Fill your SWOT grid honestly.
2		Write a 3-year vision statement.
3		Set 6 milestones with dates.
4		Review with a mentor. Add a Plan B.

THINK & TALK

Which 'threat' worries you most? How will you reduce it?

Draw what you did!

WORKSHEET: Create Your Career Blueprint (Part A)

Name: _____ Date: _____

1. Internal (S/W) or external (O/T)? Tick ✓.

	Internal	External
Good at maths		
Free online AI courses		
Poor time management		
Highly competitive exams		
Strong communicator		

2. Draw your 3-year career blueprint with milestones!

My vision: _____

WORKSHEET: Create Your Career Blueprint (Part B)

Name: _____ Date: _____

1. Match the SWOT quadrant to the example.

Strength ●

Weakness ●

Opportunity ●

Threat ●

● Free online AI courses

● Excellent at problem-solving

● Tough competition for seats

● Procrastination

2. Fill in the blanks. Use the word box.

WORD BOX: skill gap ✿ milestone ✿ SWOT

Strengths-Weaknesses-Opportunities-Threats is _____.

A checkpoint on the path is a _____.

What you still need to learn is a _____.

PUZZLE & DEFINE: Create Your Career Blueprint

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

SWOT

Meaning: _____

Example: _____

Milestone

Meaning: _____

Example: _____

Skill gap

Meaning: _____

Example: _____

DEMO LAB: Career Blueprint template (unplugged)

Name: _____ Date: _____

Goal: Create your 5-year career blueprint.

Open: printed template / spreadsheet | **With a teacher or parent**

Steps

1. Complete a SWOT for yourself.
2. Pick a target role and list its required skills.
3. Calculate your skill gap.
4. Set 6 dated milestones from now to the first year of college.
5. Write a backup path and a date to review the plan.

Record your observations

My top strength:	
Biggest skill gap:	
First milestone and date:	

Reflect: How does a backup plan reduce stress rather than show weakness?

? QUIZ TIME: Create Your Career Blueprint

Circle the right answer!

Q1. The O in SWOT stands for...

a) Opportunities

b) Oranges

c) Orbits

Q2. A checkpoint on your path is a...

a) Balloon

b) Milestone

c) Wall

Q3. Strengths and weaknesses are...

a) External

b) Random

c) Internal

Q4. A blueprint should be reviewed...

a) Every term

b) Never

c) At 60

Q5. What you still need to learn is a...

a) Gift

b) Skill gap

c) Party

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Create Your Career Blueprint

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	SWOT stands for Strengths, Weaknesses, Opportunities, Threats.		
	A Plan B shows weakness.		
	Portfolios help prove skills.		
	Milestones should be measurable.		
	A blueprint should never change.		
	Mentors can improve your plan.		

HOME CONNECT

Share your blueprint with your family and agree on monthly check-ins.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Create Your Career Blueprint

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Strengths, Weaknesses, Opportunities, Threats is ____.

(a) SWOT ☐ **(b)** SMART ☐ **(c)** SWAP ☐

2. Required skills minus current skills is a skill ____.

(a) set ☐ **(b)** gap ☐ **(c)** game ☐

3. Dated checkpoints in a plan are ____.

(a) memes ☐ **(b)** metrics ☐ **(c)** milestones ☐

4. A blueprint should be reviewed every ____.

(a) term ☐ **(b)** decade ☐ **(c)** never ☐

B. Fill in the blanks. Use the word box.

backup **threat** **opportunity** **cage**

1. A ____ plan helps if plan A fails.

2. Burnout can be a ____ in SWOT.

3. Online courses can be an ____.

4. A blueprint is a compass, not a ____.

C. Write T for True and F for False.

1. SWOT includes external opportunities. [ ]

2. Milestones should have no dates. [ ]

3. Skill gaps guide what to learn next. [ ]

4. A backup plan shows weakness. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Strength	____	**(a)** External chance
2. Weakness	____	**(b)** Internal positive
3. Opportunity	____	**(c)** External risk
4. Threat	____	**(d)** Internal limit

E. Answer in one word.

1. What analysis lists strengths and threats? _____

2. What is a dated checkpoint? _____

3. What is required minus current skills? _____

F. Answer these questions.

1. Write your own SWOT for an AI career.

2. Why should a blueprint include a backup path?

G. Case-based questions. Read and answer.

Riya's SWOT showed maths strength, weak time management, online courses as opportunity and burnout as threat. She set milestones and a backup B.Sc path.

1. Maths is Riya's _____.

(a) strength ☐ **(b)** threat ☐ **(c)** opportunity ☐

2. Burnout is a _____.

(a) strength ☐ **(b)** threat ☐ **(c)** milestone ☐

3. The B.Sc option is a _____.

(a) weakness ☐ **(b)** skill gap ☐ **(c)** backup plan ☐

H. THINK DEEPER

Build a 5-year blueprint for a friend who loves biology and coding. Include SWOT, three milestones and a backup.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. The T in SWOT is...

a) Threats

b) Trees

c) Tunes

Q2. Running models in production:

a) MusicOps

b) MLOps

c) CookOps

Q3. Who invents new AI methods?

a) Singer

b) Driver

c) Research scientist

Q4. A checkpoint on your path is a...

a) Milestone

b) Balloon

c) Wall

Q5. Free online courses are an...

a) Weakness

b) Opportunity

c) Threat

Q6. AI policy experts work on...

a) Recipes

b) Games

c) Rules & laws

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

MLOPS

GOVERNANCE

SWOT

MILESTONE

SKILL GAP

MENTOR

K	O	M	E	N	T	O	R	P	F
G	O	V	E	R	N	A	N	C	E
H	C	A	R	E	E	R	K	M	I
B	C	B	P	I	S	T	R	L	L
E	G	C	O	G	P	I	F	O	M
O	S	K	I	L	L	G	A	P	U
L	T	G	L	S	W	O	T	S	D
B	H	I	U	W	H	D	L	F	K
P	M	I	L	E	S	T	O	N	E
A	B	M	C	K	L	A	L	K	L

I finished AI RISE! Teacher's sign: _____

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

AI Research Desk

Topic 10

AI Creation Studio

TOPIC 9 • MONTH 5 • AI TOOLS

AI Research Desk

LET'S LEARN

- Research builds new knowledge systematically.
- Literature review → question → method → results.
- AI assists, but humans verify and cite sources!

Research method

Literature review

Research question

Methodology

Results & discussion

KEY TERMS

Literature review Survey of past studies

Peer review Expert checking

Variable Factor you measure

💬 TECH TALK: AI Research Desk

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir
Bolt, I want to research whether AI tutors improve maths scores 📊.
10:02

🤖 Bolt
Start with a literature review 📚: what have others already found?
10:04 ✓✓

👧 Riya
Can AI help with that?
10:06

🤖 Bolt
It can summarise papers and suggest keywords, but always open and read the real paper 🔍.
10:08 ✓✓

👦 Kabir
Then the experiment?
10:10

🤖 Bolt
Define variables 🧪: independent (AI tutor or not), dependent (score), controlled (same test, same time).
10:12 ✓✓

👧 **Riya**

How do we know results are trustworthy?

10:14

🤖 **Bolt**

Peer review 👥: experts check methods before publication. And you report limitations honestly.

10:16 ✓✓

👦 **Kabir**

Review, question, method, results, cite ✅!

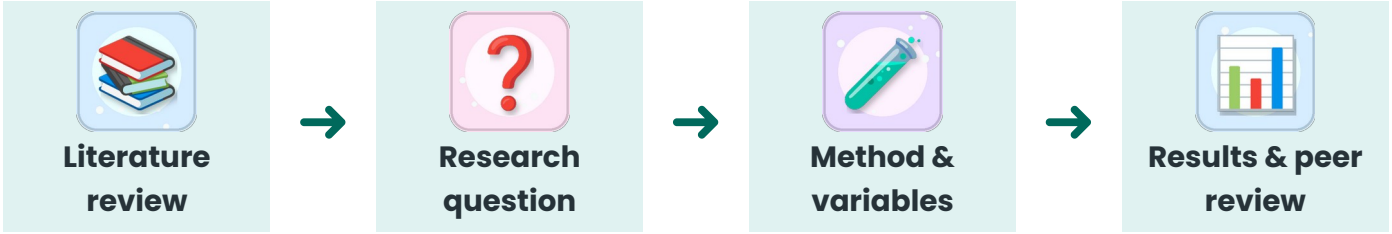
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Research builds knowledge systematically: literature review, question, method with variables, results and peer review. AI assists; humans verify and cite.

🔄 How it works



INDIA SPOTLIGHT

Indian students can publish projects through national science fairs and access research via open-access repositories.

📖 CONCEPT EXPLAINED: AI Research Desk

💡 In this chapter

Research is how humanity builds reliable knowledge: not by guessing or trusting the loudest voice, but by asking clear questions, gathering evidence systematically and letting others check the work. AI tools can now search, summarise and even draft parts of research, which is powerful but risky. This chapter teaches the research process, experimental design, the role of peer review, and how to use AI as an honest research assistant.

📚 1. The research process

Research follows a cycle. It begins with curiosity and a broad topic, narrows into a specific research question, reviews what is already known, forms a hypothesis, designs a method to test it, collects and analyses data, and finally reports results with honest discussion of limitations. Each new result often raises fresh questions, starting the cycle again.

A good research question is specific, answerable and meaningful. 'Is AI good for education?' is too broad. 'Do Class 9 students who use an AI tutor for 20 minutes daily for four weeks improve their algebra test scores more than those who do not?' can actually be tested. Narrowing the question is often the hardest and most important step.

🤔 PAUSE & THINK

Turn the broad topic 'social media and students' into one specific, testable research question.

🔍 2. Literature review: standing on shoulders

A literature review surveys existing research on your question. It prevents you from repeating work that has already been done, shows what methods others used, reveals gaps and contradictions, and helps you frame your contribution. Sources include journal articles, conference papers, books, government reports and credible institutional data.

Search tools such as Google Scholar help find papers by keyword. Read the abstract first, then the introduction and conclusion, and only then the full methods if the paper is relevant. Keep organised notes: citation details, main finding, method, sample size and

limitations. Good researchers also notice who funded a study and whether results have been replicated.

✏️ **WORKED EXAMPLE**

Note card: Author(s), Year, Title, Source. Question studied. Method (e.g., 200 students, 6 weeks). Main finding (e.g., 8% score improvement). Limitations (one city, short time). How it relates to my question.

🧪 3. Designing experiments: variables and controls

In an experiment, the independent variable is what you deliberately change, such as using an AI tutor or not. The dependent variable is what you measure, such as test scores. Controlled variables are kept the same for all groups, such as the test, the time of day and the syllabus, so that any difference can be attributed to the independent variable.

Good designs use a control group that does not receive the intervention, assign participants randomly where possible to avoid bias, and use a large enough sample to make results reliable. Ethical research also requires informed consent, protection of participants' privacy and avoiding harm. With students, permission from parents and the school is essential.

✏️ **WORKED EXAMPLE**

Question: Does background music affect concentration? Independent variable: music or silence. Dependent variable: number of maths problems solved correctly in 15 minutes. Controlled: same problems, same room, same time. Groups: 30 students randomly split.

👥 4. Peer review, replication and limitations

Before research is published in reputable journals, it goes through peer review: independent experts examine the question, methods, analysis and conclusions, and may request changes or reject the paper. Peer review is not perfect, but it filters out many errors and overclaims. Replication, when other researchers repeat a study and get similar results, gives even stronger confidence.

Honest researchers report limitations openly: a small sample, a short time period, one school or region, or possible measurement errors. Limitations are not failures; they help readers judge how far results can be generalised and suggest future research. Exaggerating findings, hiding inconvenient data or fabricating results are serious breaches of research integrity.

🤔 **PAUSE & THINK**

A news headline says 'AI tutors boost marks by 80%!' but the actual study found 8% in one school over four weeks. What went wrong between the study and the headline?

🤖 5. Using AI as a research assistant, responsibly

AI tools can help brainstorm questions, suggest keywords, summarise long papers, explain difficult terms, draft outlines and check grammar. Some research assistants can search academic databases and link to sources. Used well, they save time on routine tasks and let you focus on thinking and analysis.

However, AI can hallucinate: it may invent papers, authors or statistics that sound perfectly real, or misstate a study's findings. Always open and read the original source before citing it. Never let AI fabricate data. Follow your school's rules on AI use and disclose how you used it, for example: 'An AI tool was used to summarise sources; all summaries were checked against the original papers.'

🖐 **TRY IT**

Ask an AI tool for three research papers on 'AI tutoring and learning outcomes'. Search for each on Google Scholar. How many exist exactly as described? Record any differences.

✅ Key takeaways

- Research moves from question to literature review, hypothesis, method, results and limitations.
- Specific, testable questions are the foundation of good research.
- Experiments need independent, dependent and controlled variables, plus ethics and consent.
- Peer review and replication build trust; limitations must be reported honestly.
- AI can assist research but may hallucinate sources, so always verify and disclose use.

? Quick check: answer in one or two lines.

1. What is the difference between an independent and a dependent variable?

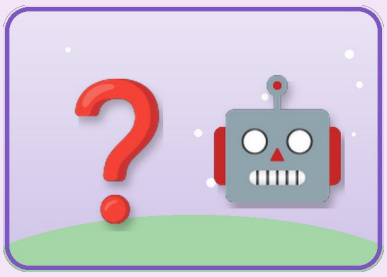
2. Why is a literature review done before an experiment?

3. What is peer review?

4. Give two rules for using AI responsibly in research.

AI CHRONICLES: The Tutor Trial

Kabir | Dr. Lakshmi | Data

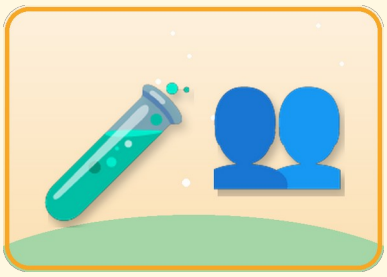


SCENE 1

Kabir asked: "Do AI tutors improve maths scores?" He began with a literature review.

An AI summary claimed a study showed '80% improvement'. The real paper said 8%. He corrected it.

SCENE 2

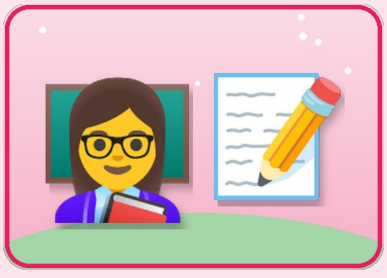


SCENE 3

Design: two equal groups. One used an AI tutor for 4 weeks; one didn't. Same test, same time.

The AI-tutor group improved more, but only for students who practised regularly.

SCENE 4



SCENE 5

Dr. Lakshmi reviewed his method and asked him to report limitations: small sample, short time.

His honest paper won at a state science congress. "Truth over hype," Kabir wrote.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir repeats it with more schools	**Kabir hides the limitations**	**The tutor writes the paper itself**

🧐 Story check: answer the questions.

1. What error did the AI summary make?

2. What were the variables?

3. Why report limitations?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI research assistants can...

Find related papers

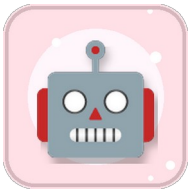
Summarise abstracts

Suggest analyses

Translate studies

Manage references

Check clarity of writing



BOLT SAYS:

I can summarise 50 papers, but I can also misquote them. Always open the original!

FUN FACT!

Peer review means experts check research before it's published in journals!

I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: Mini Research Paper

You need:

Research question

3+ sources

IMRaD template

Steps:

1		Choose: 'Does screen time affect sleep in teens?'
2		Review 3 sources. Note findings & gaps.
3		Design a class survey (method). Collect data.
4		Write a 2-page IMRaD report with citations.

THINK & TALK

What limitations does your study have?

Draw what you did!

WORKSHEET: AI Research Desk (Part A)

Name: _____ Date: _____

1. IMRaD structure. Write 1, 2, 3, 4.

Discussion ☐

Introduction ☐

Results ☐

Methods ☐

2. Independent or dependent variable? (Screen time vs sleep) Tick ✔.

	Independent	Dependent
Hours of screen time		
Hours of sleep		
Amount of light (experiment)		
Plant height measured		

3. Bonus: Draw your favourite picture from this topic!

I drew _____

WORKSHEET: AI Research Desk (Part B)

Name: _____ Date: _____

1. Match the term to its meaning.

Literature review ●

Peer review ●

Correlation ●

Variable ●

● Factor that can change

● Survey of past studies

● Variables moving together

● Expert checking

2. Good research practice? Tick ✓.

	Good	Poor
Verifying AI-suggested sources		
Copying AI summaries without reading		
Citing every source		
Claiming causation from correlation		

PUZZLE & DEFINE: AI Research Desk

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Literature review

Meaning: _____

Example: _____

Peer review

Meaning: _____

Example: _____

Variable

Meaning: _____

Example: _____

🧪 DEMO LAB: Google Scholar + an AI research assistant (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Run a mini literature review and design an experiment.

🔗 **Open:** scholar.google.com | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Find 3 real papers on 'AI tutoring and learning outcomes'.
2		Ask AI to summarise one; compare with the paper's abstract.
3		Write a research question and hypothesis.
4		Design an experiment: independent, dependent and controlled variables.
5		Cite all 3 papers properly and list 2 limitations.

🔬 Record your observations

Did the AI summary miss anything?	
My variables:	
One limitation:	

💭 **Reflect:** Why is peer review essential before trusting research?

QUIZ TIME: AI Research Desk

Circle the right answer!

Q1. Experts checking research is...

a) Peer review

b) Gaming

c) Interview

Q2. IMRaD's M stands for...

a) Music

b) Methods

c) Moon

Q3. A survey of past studies is a...

a) Film

b) Newspaper

c) Literature review

Q4. AI summaries of papers must be...

a) Verified

b) Copied

c) Trusted blindly

Q5. The factor you change is the...

a) Dependent

b) Independent variable

c) Random

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Research Desk

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
A literature review surveys past studies.		
AI-suggested references are always real.		
Correlation proves causation.		
Peer review is expert checking.		
Research should cite sources.		
Methods should allow others to repeat the study.		

HOME CONNECT

Pick a question your family wonders about. Find and verify 2 reliable sources together.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Research Desk

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Reviewing existing studies is a ____.

(a) literature review ☐ **(b)** film review ☐ **(c)** code review ☐

2. The variable you change in an experiment is ____.

(a) dependent ☐ **(b)** independent ☐ **(c)** controlled ☐

3. Experts checking research before publication is ____.

(a) self-review ☐ **(b)** no review ☐ **(c)** peer review ☐

4. The variable you measure is the ____ variable.

(a) dependent ☐ **(b)** independent ☐ **(c)** random ☐

B. Fill in the blanks. Use the word box.

controlled **limitations** **paper** **systematically**

1. Variables kept the same are ____.

2. Report ____ honestly.

3. AI summaries must be checked against the real ____.

4. Research is building knowledge ____.

C. Write T for True and F for False.

1. AI summaries can misstate findings. [ ]

2. Limitations should be hidden. [ ]

3. Peer review improves trust. [ ]

4. A literature review comes after results. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Literature review	____	**(a)** What you measure
2. Independent variable	____	**(b)** Study past work
3. Dependent variable	____	**(c)** Expert checking
4. Peer review	____	**(d)** What you change

E. Answer in one word.

1. What variable do you change? _____

2. What process checks research? _____

3. What studies past research? _____

F. Answer these questions.

1. Design an experiment on whether music helps concentration. Name all variables.

2. Why should AI-generated summaries be verified?

G. Case-based questions. Read and answer.

An AI summary said a study showed '80% improvement'; the real paper said 8%. Kabir ran a 4-week trial with two groups and reported a small sample as a limitation.

1. The AI summary contained a ____.

(a) factual error ☐ **(b)** citation ☐ **(c)** variable ☐

2. Using or not using the AI tutor is the ____ variable.

(a) dependent ☐ **(b)** independent ☐ **(c)** controlled ☐

3. A small sample is a ____.

(a) result ☐ **(b)** hypothesis ☐ **(c)** limitation ☐

H. THINK DEEPER

Propose a research question on AI in education. Write the hypothesis, method, variables and how you'd ensure ethics.

TOPIC 10 • MONTH 5 • AI TOOLS

AI Creation Studio

LET'S LEARN

- Creators chain AI tools into pipelines.
- Script → visuals → voice → edit → publish.
- Rights, credits and authenticity matter!

Creator pipeline

Script

Visuals

Voice

Edit

KEY TERMS

Pipeline Chain of tools/steps

Licence Permission to use

Provenance Origin of content

💬 TECH TALK: AI Creation Studio

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👧 **Riya**
Bolt, we're making a 3-minute documentary on Kerala backwaters 🎬.
10:02

🤖 **Bolt**
Chain tools into a pipeline ⛓️: script visuals voice edit publish.
10:04 ✓✓

👦 **Kabir**
Can AI generate the visuals?
10:06

🤖 **Bolt**
Some, but check the licence 📜: who can use the output, and can it be used commercially?
10:08 ✓✓

👧 **Riya**
What about music?
10:10

🤖 **Bolt**
Use licensed or royalty-free tracks 🎵. Credit creators properly.
10:12 ✓✓

👦 **Kabir**

How will viewers know what's real?

10:14

🤖 **Bolt**

Provenance 🏷️: records of where content came from and how it was edited. Some tools attach content credentials.

10:16 ✓✓

👧 **Riya**

Create, credit, prove it's authentic ✅!

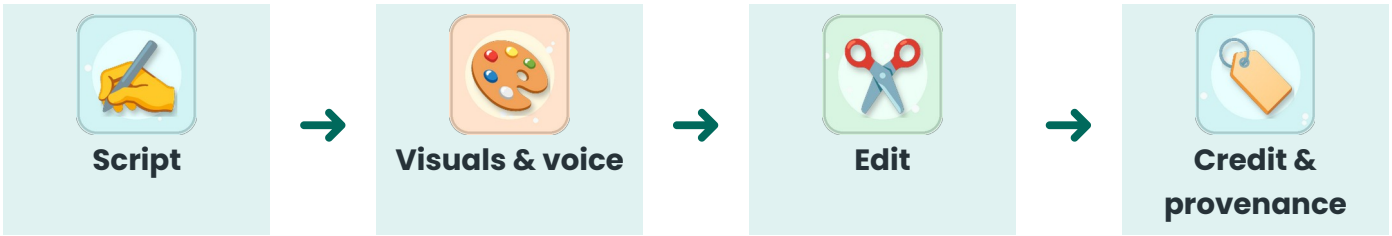
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Creators chain AI tools into pipelines. Check licences, credit creators and keep provenance so audiences can trust what is real.

🔄 How it works



INDIA SPOTLIGHT

India's film and media industry uses AI for dubbing, subtitles and effects, raising new questions about rights and credit.

📖 CONCEPT EXPLAINED: AI Creation Studio

💡 In this chapter

Generative AI has turned a phone or laptop into a complete creative studio: scripts, images, voices, music and video can all be produced with AI help. Professional creators now chain several tools into pipelines. But creation also raises questions: who owns AI-assisted work, which tools may be used for what, and how can audiences trust what they see? This chapter covers pipelines, licences, credit and provenance.

⛓ 1. Creative pipelines: chaining AI tools

A pipeline is a sequence of steps where the output of one stage becomes the input of the next. A typical AI-assisted video pipeline is: research and script writing, storyboard, visual creation (photos, illustrations, AI-generated images), voice-over (recorded or synthetic), music, editing, subtitles and publishing. Different tools may handle different stages.

Planning the pipeline first saves time and improves quality. Decide which stages you will do yourself and where AI adds the most value. Many creators keep the core creative decisions, such as the message, story and final edit, in human hands while using AI for repetitive or technical tasks such as noise removal, subtitles and resizing for different platforms.

✏️ WORKED EXAMPLE

Documentary pipeline: Research (books, interviews) → Script (own words, AI for grammar) → Storyboard (hand-drawn) → Visuals (own drone shots + licensed AI images) → Voice (own recording, AI noise removal) → Music (royalty-free) → Edit → Credits & publish.

📜 2. Licences and copyright

Copyright protects original creative works such as text, images, music and films, giving creators rights over how their work is copied and used. Using someone's song or photo without permission can violate copyright, even in a school project posted online. Licences specify what others may do with a work: some allow free use with credit, some forbid commercial use, some require permission or payment.

AI tools have their own terms of service that say who may use the outputs and how. Some forbid commercial use on free plans; some ask you to disclose AI use. Laws about copyright in AI-generated works are still evolving in many countries. The safe approach

is to read the terms, use licensed or royalty-free assets, and keep a record of every tool and asset used.

🤔 **PAUSE & THINK**

You find a beautiful photo on a website with no licence information. Can you use it in your published school magazine? What should you do instead?

🏷️ 3. Credit, disclosure and originality

Giving credit means acknowledging the people and tools whose work contributed to yours. A credits slide might list: script by the team; photos by the team and a named photographer; music track title and creator with licence; AI tools used for image generation and audio clean-up. Disclosure means stating clearly where and how AI was used.

Originality does not mean avoiding all tools; it means bringing your own ideas, perspective, research and choices. An AI image based on your carefully designed storyboard and edited into your narrative is part of your original work. Simply generating a video from a one-line prompt and presenting it as your own creative effort is not. Judges and teachers increasingly value transparency.

✏️ **WORKED EXAMPLE**

Disclosure statement: 'Concept, script, filming and editing by students. AI tools were used to generate two historical scenes (labelled on screen), clean audio and create subtitles. Music: royalty-free track, credited at end.'

🔍 4. Provenance: proving what is real

As AI-generated media becomes realistic, audiences need ways to know where content came from. Provenance means the record of a piece of content's origin and history of edits. Some cameras and software can attach content credentials, a kind of tamper-evident label that records which tool created or edited an image or video.

Industry groups have developed open standards for content credentials, and some platforms display labels for AI-generated content. These systems are not yet universal, and labels can sometimes be removed. Still, creators who preserve provenance and label AI content help build an information environment where trust is possible.

🤔 **PAUSE & THINK**

If all AI-generated images carried visible labels, what problems would that solve? What problems would remain?

🎬 5. Quality, ethics and impact

Professional quality comes from clear purpose and careful editing, not from tools alone. Ask: who is my audience, what do I want them to feel or understand, and does every scene serve that goal? Check AI-generated visuals for errors such as wrong details, distorted hands or historically inaccurate elements, and fix or remove them.

Ethical creators never use AI to impersonate real people without consent, create misleading content about real events, or copy the distinctive style of a living artist to compete with them unfairly. Used responsibly, AI can help students tell local stories, document heritage in regional languages, and make educational content accessible to people with disabilities through subtitles and audio descriptions.

🖐 **TRY IT**

Plan a 60-second video on a local heritage site. Draw a pipeline diagram, mark which stages use AI, list licences to check and write your disclosure statement.

✅ Key takeaways

- Creative pipelines chain steps; plan where AI adds value and where humans decide.
- Copyright and tool licences determine how content may be used; read the terms.
- Credit everyone and disclose AI use; originality comes from your ideas and choices.
- Provenance and content credentials help audiences trust media.
- Never use AI to impersonate, mislead or unfairly copy others.

? Quick check: answer in one or two lines.

1. What is a creative pipeline?

2. Why must you check an AI tool's licence before publishing?

3. Write a short AI disclosure statement for a poster.

4. What is provenance?

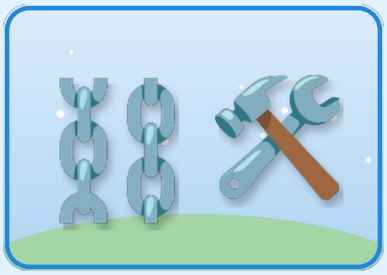
AI CHRONICLES: Backwaters, Credits and Proof

Riya | Kabir | Film Festival Jury

SCENE 1

Riya and Kabir made a documentary on Kerala's backwaters for a youth film festival.

Their pipeline: script their own drone shots + AI-generated historical scenes AI voice cleanup edit.



SCENE 2

SCENE 3

One AI image tool's licence banned festival use. They switched to a tool that allowed it.

They used licensed music and credited every tool, artist and source.

SCENE 4

SCENE 5

Content credentials showed which scenes were AI-made. The jury could verify everything.

They won 'Most Responsible Production'. "Trust is part of the story," the jury said.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They make a film on climate refugees	**They skip licences next time**	**The backwaters make their own film**

🧐 Story check: answer the questions.

1. What was their pipeline?

2. What licence problem did they face?

3. How did provenance help?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Studio-grade AI projects

Educational explainer series

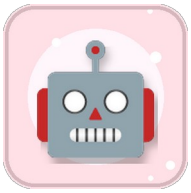
Game prototypes

Original soundtrack

Data journalism story

Product concept renders

Multilingual campaigns



BOLT SAYS:
Plan the story first. Tools change every month; good storytelling doesn't!

FUN FACT!
Copyright rules for AI-generated works are still being debated around the world!

I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: 60-Second Explainer Production

You need:

Storyboard

AI tools (with teacher)

Headphones

Steps:

1.
   Pick an AI concept. Write a 120-word script.
2.
   Create or source 6 visuals (check licences).
3.
   Record your voice-over (or AI voice, disclosed).
4.
   Edit, add captions, credits and provenance notes.

THINK & TALK

Which part was most human? Which was most AI?

Draw what you did!

WORKSHEET: AI Creation Studio (Part A)

Name: _____ Date: _____

1. Creator pipeline. Write 1, 2, 3, 4.

Edit & publish ☐

Write script ☐

Record voice ☐

Create visuals ☐

2. Fill in the blanks. Use the word box.

WORD BOX: provenance ✿ licence ✿ pipeline

A chain of tools is a _____.

Permission to use content is a _____.

The origin record of content is its _____.

3. Bonus: Draw your favourite picture from this topic!

I drew _____

WORKSHEET: AI Creation Studio (Part B)

Name: _____ Date: _____

1. Match the tool to the pipeline stage.

Text model ● ● **Narration**

Image model ● ● **Final assembly**

Voice tool ● ● **Script drafting**

Video editor ● ● **Visual assets**

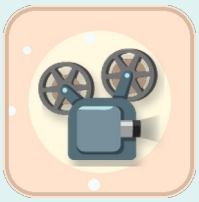
2. Ethical creator practice? Tick ✓.

	Ethical	Unethical
Disclosing AI-generated visuals		
Using film songs without licence		
Adding captions for accessibility		
Cloning a celebrity's voice for ads		

PUZZLE & DEFINE: AI Creation Studio

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Pipeline

Meaning: _____

Example: _____

Licence

Meaning: _____

Example: _____

Provenance

Meaning: _____

Example: _____

🧪 DEMO LAB: Canva / video editor + content credentials check (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Build a responsible AI creation pipeline.

🔗 **Open:** canva.com/education + contentcredentials.org/verify | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Write a 60-second script on a local heritage site.
2		Create visuals (own photos + AI images) and note each source's licence.
3		Record voice-over; clean it with AI.
4		Edit into a short video with royalty-free music.
5		Add a credits slide and check one image with a content-credentials verifier.

🔬 Record your observations

Licences checked:	
AI tools in pipeline:	
Provenance result:	

💭 **Reflect:** Why will provenance become more important as AI media grows?

QUIZ TIME: AI Creation Studio

Circle the right answer!

Q1. A chain of tools/steps is a...

a) Pipeline

b) Yarn

c) Train

Q2. Permission to use content is a...

a) Ticket

b) Licence

c) Balloon

Q3. The origin of content is its...

a) Palette

b) Pitch

c) Provenance

Q4. What should come first?

a) The story

b) The tool

c) Ads

Q5. AI copyright rules are...

a) Fully settled

b) Still debated

c) Non-existent

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Creation Studio

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
A pipeline chains tools and steps.		
Any online song is free to use.		
Provenance records how content was made.		
Captions improve accessibility.		
Cloning voices without consent is fine.		
Planning matters more than tools.		

HOME CONNECT

Watch a short video with your family. Discuss: is anything AI-made? How could you tell?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Creation Studio

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Chaining tools step by step is a ____.

(a) pipeline ☐ **(b)** pipe dream ☐ **(c)** playlist ☐

2. Permission terms for using content are a ____.

(a) letter ☐ **(b)** licence ☐ **(c)** log ☐

3. Records of a content's origin and edits are ____.

(a) proverbs ☐ **(b)** promotion ☐ **(c)** provenance ☐

4. Using licensed music respects ____.

(a) rights ☐ **(b)** rhythm only ☐ **(c)** rules of grammar ☐

B. Fill in the blanks. Use the word box.

publish **commercially** **credentials** **creator**

1. Script → visuals → voice → edit → ____.

2. Check if output can be used ____.

3. Content ____ show how media was made.

4. Credit every tool and ____.

C. Write T for True and F for False.

1. AI tool licences can restrict use. [ ]

2. Provenance doesn't matter for trust. [ ]

3. Royalty-free music still may need credit. [ ]

4. All AI outputs can be used anywhere freely. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Pipeline	____	**(a)** Origin record
2. Licence	____	**(b)** Chained steps
3. Provenance	____	**(c)** Acknowledgement
4. Credits	____	**(d)** Usage permission

E. Answer in one word.

1. What chains creation steps? _____

2. What sets usage permissions? _____

3. What records content origin? _____

F. Answer these questions.

1. Design a 5-step AI creation pipeline for a school campaign.

2. Why do licences and provenance matter?

G. Case-based questions. Read and answer.

A documentary team switched AI image tools because one licence banned festival use, used licensed music, credited all sources and used content credentials. They won a responsibility award.

1. Switching tools was due to the ____.

(a) licence ☐ **(b)** colour ☐ **(c)** speed ☐

2. Content credentials show ____.

(a) ticket prices ☐ **(b)** provenance ☐ **(c)** cast salaries ☐

3. Their award recognised ____.

(a) speed ☐ **(b)** budget ☐ **(c)** responsible creation ☐

H. THINK DEEPER

Should AI-generated media always carry a visible label? Debate pros and cons and propose a balanced rule.

🏅 UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. Experts checking research is...

a) Peer review	b) Gaming	c) Interview

Q2. Permission to reuse content is a...

a) Ticket	b) Licence	c) Balloon

Q3. Correlation proves causation?

a) Always	b) Sometimes yes	c) No

Q4. The origin record of content:

a) Provenance	b) Palette	c) Pitch

Q5. AI-suggested references must be...

a) Copied	b) Verified	c) Trusted

Q6. IMRaD's D stands for...

a) Dance	b) Dog	c) Discussion

WORD SEARCH: AI CREATE

🔍 **Find these words. They go across → or down ↓.**

PEER REVIEW

VARIABLE

PIPELINE

LICENCE

PROVENANCE

CITATION

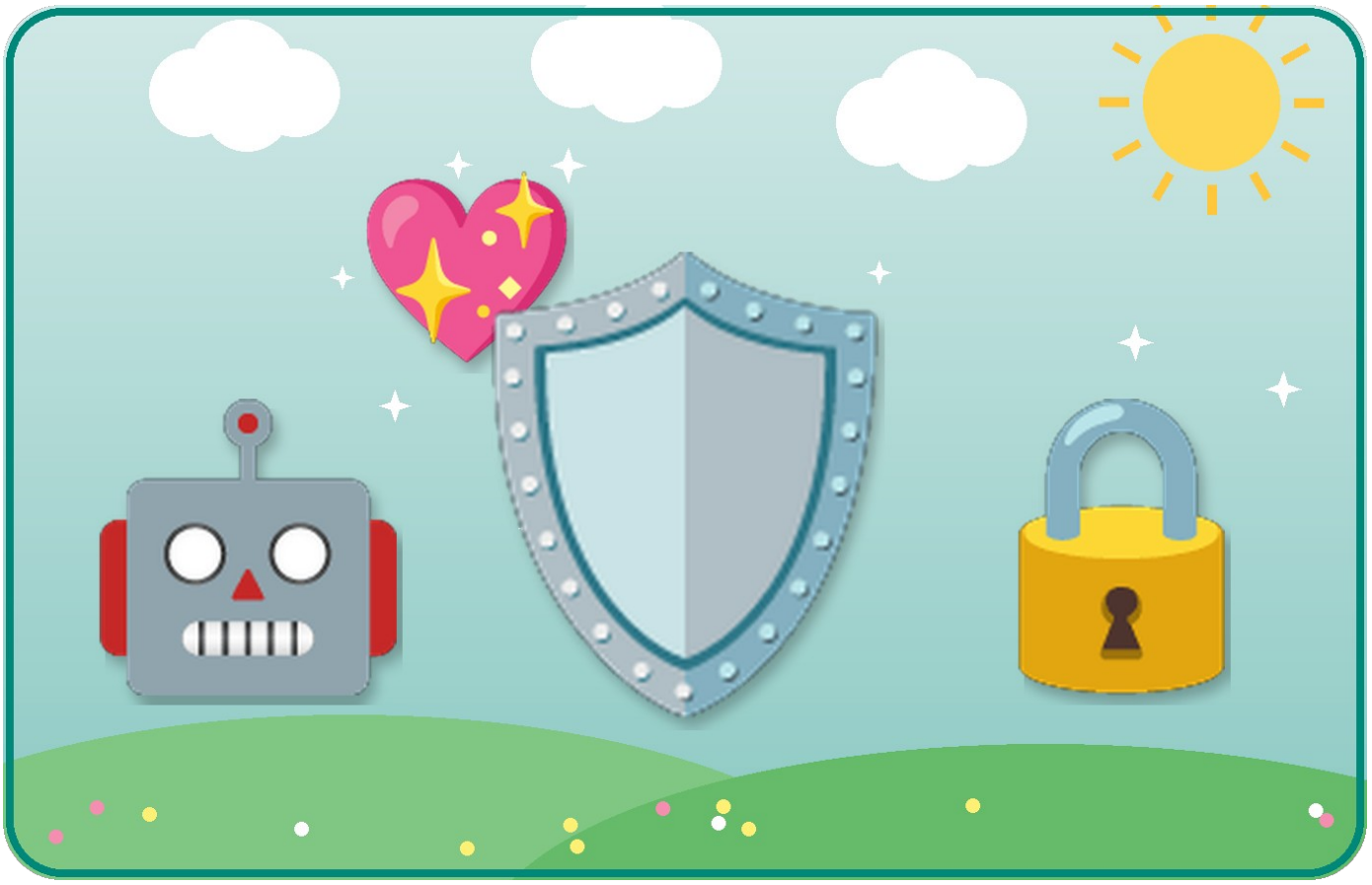
T	P	T	I	H	R	B	L	P	P
H	R	R	C	I	E	K	I	E	I
K	O	R	I	W	R	S	C	E	P
W	V	D	T	A	E	K	E	R	E
K	E	T	A	L	W	K	N	R	L
S	N	A	T	P	M	M	C	E	I
V	A	R	I	A	B	L	E	V	N
U	N	E	O	B	A	I	T	I	E
P	C	D	N	T	G	A	O	E	O
W	E	U	R	N	R	N	G	W	K

I finished AI CREATE! Teacher's sign: _____

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

Fair AI Challenge

Topic 12

Privacy in an AI World

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Fair AI Challenge

LET'S LEARN

- Bias can enter AI at every stage.
- Data bias, label bias, design bias, deployment bias.
- Audit with fairness metrics and diverse teams!

Where bias hides

Data

Labels

Model design

Deployment

KEY TERMS

Fairness metric
Measures equal treatment

Audit Systematic check

Representation
Fair share in data

💬 TECH TALK: Fair AI Challenge

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 **Kabir**

Bolt, where does bias come from in AI ⚖️?

10:02

🤖 **Bolt**

Every stage! Data bias 📊: some groups missing. Label bias: humans label unfairly.

10:04 ✓✓

👧 **Riya**

And more?

10:06

🤖 **Bolt**

Design bias 🛠️: choosing a feature like pin code that acts as a proxy for caste or income. Deployment bias: used where it wasn't meant to be.

10:08 ✓✓

👦 **Kabir**

How do we measure fairness?

10:10

🤖 **Bolt**

Fairness metrics 📏: compare error rates or approval rates across groups.

10:12 ✓✓

👧 **Riya**

Who checks?

10:14

🤖 **Bolt**

Audits 🔍 by independent teams, and diverse builders who spot blind spots 🌈.

10:16 ✓✓

👦 **Kabir**

Measure, audit, include, fix ✅!

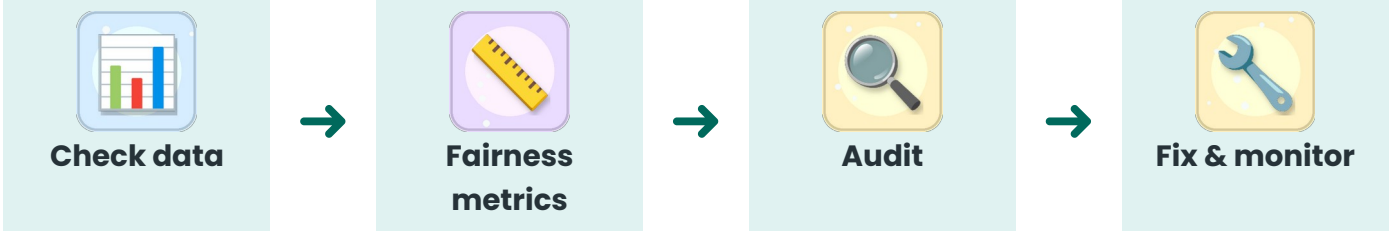
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Bias can enter through data, labels, design and deployment. Measure with fairness metrics, audit independently and build with diverse teams.

🔄 How it works



INDIA SPOTLIGHT

India's diversity of languages, regions and communities makes fairness testing especially important for AI systems used here.

📖 CONCEPT EXPLAINED: Fair AI Challenge

💡 In this chapter

AI systems increasingly help decide who gets a loan, a job interview, a scholarship or extra medical attention. If these systems are biased, they can treat people unfairly at massive scale, often invisibly. This chapter explains where bias comes from at each stage of an AI system, how fairness can be measured, what audits involve, and how diverse teams and good processes make AI fairer.

⚖️ 1. What is AI bias?

AI bias means a system produces systematically unfair outcomes for certain groups of people, for example by gender, region, language, caste, religion, disability or income. Bias does not require anyone to intend harm. It usually emerges because the data, design choices or deployment context reflect existing inequalities or leave some groups out.

Famous examples worldwide include face-recognition systems that were far less accurate on darker-skinned women, hiring tools that downgraded CVs mentioning women's colleges because past hiring favoured men, and speech systems that struggled with certain accents. In a diverse country like India, such risks multiply across languages, dialects and communities.

🤔 PAUSE & THINK

Why can an AI system be biased even if none of its creators intended to discriminate?

📊 2. Where bias enters: data and labels

Data bias occurs when training data under-represents some groups or reflects past discrimination. If a medical model is trained mostly on data from city hospitals, it may perform poorly for rural patients with different conditions or equipment. If historical loan data reflects unfair lending, a model learning from it will repeat that unfairness.

Label bias arises when the labels humans assign are themselves biased or inconsistent. If annotators label certain dialects as 'unprofessional' or certain faces as 'suspicious', the model learns these judgements. Measurement bias occurs when the thing we

measure is a poor proxy for what we care about, such as using arrest records as a measure of crime.

✏️ **WORKED EXAMPLE**

A speech assistant trained on 90% urban English and Hindi speakers has a 5% error rate for them but 25% for rural Bhojpuri speakers. The model is not 'broken'; its data left many voices out.

🧩 3. Design and deployment bias

Design bias comes from choices made while building the system, such as which features to include. Even if sensitive attributes like caste or religion are removed, other features can act as proxies: pin code may correlate with community or income; first names may correlate with gender or religion. The model can still discriminate indirectly through these proxies.

Deployment bias happens when a system is used in a context it was not designed or tested for, such as using a model built for one state in another with different demographics, or using a screening tool as a final decision-maker. Feedback loops can worsen bias: if a biased system denies opportunities to a group, future data will show fewer successes for that group, reinforcing the bias.

🤔 **PAUSE & THINK**

An exam-scoring AI was trained on essays from English-medium schools. It is now used for students from regional-medium schools. What kind of bias could occur?

📏 4. Measuring fairness

Fairness can be measured by comparing outcomes across groups. Demographic parity checks whether groups receive positive decisions (such as approvals) at similar rates. Equal opportunity checks whether qualified people in each group are approved at similar rates, that is, similar true positive rates. Comparing error rates, such as false rejections, across groups is another common check.

Different fairness definitions can conflict; it is often mathematically impossible to satisfy all of them at once. Choosing which matters most is a value judgement involving the affected communities, not just a technical decision. That is why fairness work combines statistics with ethics, law and conversation with the people affected.

✏️ **WORKED EXAMPLE**

Loan AI: Group A approval rate 60%, Group B 30%. Among applicants who actually repaid past loans, Group A was approved 85% of the time, Group B only 50%. This gap in equal opportunity suggests unfair treatment of qualified Group B applicants.

🔍 5. Audits, diverse teams and fixes

An AI audit systematically examines a system's data, design, performance and impact, ideally by an independent team. Auditors check representation in data, test performance separately for different groups, look for proxy features, review how decisions are used and whether people can appeal. Audits should be repeated, because data and contexts change.

Fixes include collecting more representative data, removing or adjusting proxy features, re-weighting training examples, adjusting decision thresholds, and adding human review for high-stakes decisions. Diverse teams, including people from different genders, regions, languages and abilities, are more likely to spot blind spots early. Transparency and channels for complaints let affected people raise problems.

🖐️ **TRY IT**

Using a spreadsheet of 40 sample decisions, calculate approval rates and false rejection rates for two groups. Identify one possible proxy feature and propose two fixes.

✅ Key takeaways

- AI bias is systematic unfairness toward groups, often unintended.
- Bias can enter through data, labels, measurement, design and deployment.
- Proxy features can cause indirect discrimination even without sensitive attributes.
- Fairness metrics compare approval and error rates across groups; definitions can conflict.
- Independent audits, diverse teams, better data and human review make AI fairer.

? Quick check: answer in one or two lines.

1. Give one example each of data bias and label bias.

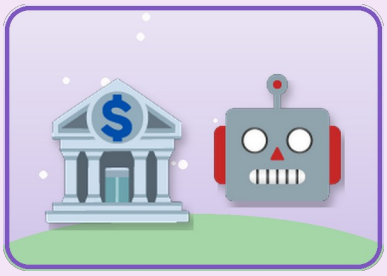
2. What is a proxy feature? Give an example.

3. Explain demographic parity in one sentence.

4. List three ways to reduce AI bias.

AI CHRONICLES: The Pin Code Problem

Kabir | Riya | Bank



SCENE 1

A bank piloted an AI for student education loans. Kabir's cousin, from a rural area, was rejected 😞.

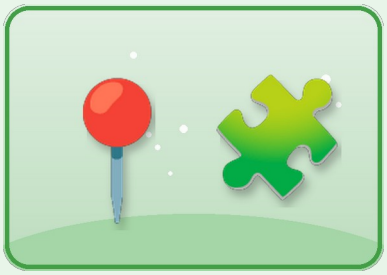
Riya and Kabir got permission to audit anonymised decisions as a school project.

SCENE 2

SCENE 3

Fairness metrics showed rural applicants were rejected twice as often with similar repayment histories.

The culprit: pin code was a proxy feature, hiding regional and income bias.



SCENE 4

SCENE 5

The bank removed the proxy, retrained with balanced data and added human review for rejections.

Approval gaps shrank. Kabir's cousin reapplied and got the loan.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
The bank runs audits every quarter	**The bank ignores the audit**	**The pin code changes its address**

🧐 Story check: answer the questions.

1. What did the fairness metrics show?

2. Why was pin code a problem?

3. What three fixes did the bank make?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Fairness case areas

Hiring screeners

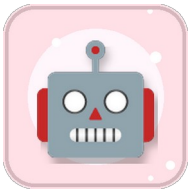
Loan approvals

Speech recognition accents

Face recognition skin tones

Medical risk scores

Risk tools in courts



BOLT SAYS:

If my training data under-represents a group, I'll make more errors for them. Audit me!

FUN FACT!

Fairness metrics compare error rates across groups to detect unequal treatment!

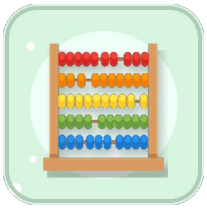
I spy AI! Draw one AI helper you saw today.

LAB ACTIVITY: Bias Audit Simulation

You need:

Applicant cards

Audit sheet



Calculator

Steps:

1. A 'model' approved applicants using a rule (on cards).
2. Compute approval rate for Group A and Group B.
3. Is the gap fair? What caused it?
4. Redesign the rule and re-audit.

THINK & TALK

Can a model be fair if its data reflects past unfairness?

Draw what you did!

WORKSHEET: Fair AI Challenge (Part A)

Name: _____ Date: _____

1. Match the bias type to the example.

Data bias ●

● Biased humans tagged the data

Label bias ●

● Too few rural voices in dataset

Design bias ●

● Used where it wasn't tested

Deployment bias ●

● Chose a proxy feature like pin code

2. Compute approval rates. Use the word box.

WORD BOX: 80% ✿ 40% ✿ 40

Group A: 40 of 50 approved = ____

Group B: 20 of 50 approved = ____

Gap = 80% - 40% = ____ percentage points

WORKSHEET: Fair AI Challenge (Part B)

Name: _____ Date: _____

1. Circle features that could act as unfair proxies.

Pin code

Exam score

School name

First name

Typing speed (for a typing job)

Neighbourhood

2. Fairness audit steps. Write 1, 2, 3, 4.

Mitigate & re-audit

☐

Collect outcomes by group

☐

Judge if gaps are fair

☐

Compute metrics

☐

PUZZLE & DEFINE: Fair AI Challenge

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Fairness metric

Meaning: _____

Example: _____

Audit

Meaning: _____

Example: _____

Representation

Meaning: _____

Example: _____

🧪 DEMO LAB: Spreadsheet fairness audit (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Audit a model's decisions with fairness metrics.

🔗 **Open:** Google Sheets / Excel | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Teacher gives 40 loan decisions by an AI with applicant group A/B and actual repayment.
2. Calculate approval rate for each group.
3. Calculate error rate (wrong rejections) for each group.
4. Find features that may act as proxies (e.g., pin code).
5. Recommend 3 fixes.

🔬 Record your observations

Approval rate A vs B:	
Error rate A vs B:	
Proxy feature found:	

💭 **Reflect:** Can a model be accurate overall but still unfair? Explain.

? QUIZ TIME: Fair AI Challenge

Circle the right answer!

Q1. A systematic check of AI is an...

a) Audit

b) Audition

c) Art

Q2. Too few rural voices is...

a) Label bias

b) Data bias

c) Music bias

Q3. Fairness metrics compare...

a) Pizza sizes

b) Colours

c) Errors across groups

Q4. Using pin code as a proxy can cause...

a) Hidden bias

b) Perfect fairness

c) Fun

Q5. Fair AI benefits from...

a) No checks

b) Diverse teams

c) Luck

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Fair AI Challenge

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Unrepresentative data can cause bias.		
	Pin code can act as a proxy for sensitive traits.		
	Fairness is purely a maths problem.		
	Regular audits help catch bias.		
	Removing the 'gender' column guarantees fairness.		
	Diverse teams help build fairer AI.		

HOME CONNECT

Discuss with family: name a decision (e.g., admissions) where unfair AI would hurt people.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Fair AI Challenge

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Bias from missing groups in data is ____ bias.

(a) data ☐ **(b)** design ☐ **(c)** deployment ☐

2. A feature that indirectly encodes a sensitive attribute is a ____.

(a) label ☐ **(b)** proxy ☐ **(c)** pixel ☐

3. Independent checking of an AI system is an ____.

(a) advert ☐ **(b)** update ☐ **(c)** audit ☐

4. Comparing error rates across groups uses fairness ____.

(a) metrics ☐ **(b)** memes ☐ **(c)** magic ☐

B. Fill in the blanks. Use the word box.

stage **label** **teams** **monitoring**

1. Bias can enter at every ____.

2. Human labelling can add ____ bias.

3. Diverse ____ spot blind spots.

4. Fair AI needs regular ____.

C. Write T for True and F for False.

1. A model can be accurate overall but unfair to a group. [ ]

2. Removing names always removes bias. [ ]

3. Pin code can act as a proxy feature. [ ]

4. Audits help detect unfair outcomes. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Data bias	____	**(a)** Proxy features
2. Label bias	____	**(b)** Missing groups
3. Design bias	____	**(c)** Wrong context use
4. Deployment bias	____	**(d)** Unfair labels

E. Answer in one word.

1. What indirectly encodes sensitive traits? _____

2. What independent check finds bias? _____

3. What compares group outcomes? _____

F. Answer these questions.

1. Explain four types of AI bias with examples.

2. How would you audit an AI hiring tool?

G. Case-based questions. Read and answer.

An education-loan AI rejected rural applicants twice as often with similar repayment histories. Pin code acted as a proxy. The bank removed it, retrained and added human review.

1. Pin code acted as a ____.

(a) proxy feature ☐ **(b)** label ☐ **(c)** loss ☐

2. Comparing rejection rates is a ____.

(a) guess ☐ **(b)** fairness metric ☐ **(c)** licence ☐

3. Adding human review is a ____.

(a) new bias ☐ **(b)** deployment error ☐ **(c)** safeguard ☐

H. THINK DEEPER

Design a fairness audit plan for an AI that shortlists students for scholarships. Include metrics, data checks and who reviews.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

Privacy in an AI World

LET'S LEARN

- AI runs on data, so privacy is critical.
- India's DPDP Act 2023 sets rules for personal data.
- Techniques like anonymisation protect people!

Privacy protections

Consent

Anonymisation

Data minimisation

Federated learning

KEY TERMS

Anonymisation
Removing identity

Federated learning Train without sharing data

Minimisation
Collect only what's needed

💬 TECH TALK: Privacy in an AI World

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, a health app wants my full medical history to 'personalise tips' 🏥.
10:02

🤖 Bolt
Ask: do they need all of it? Data minimisation ✂️: collect only what's necessary.
10:04 ✓✓

👦 Kabir
What does India's law say?
10:06

🤖 Bolt
The DPDP Act 2023 ⚖️ needs clear consent, purpose limits and gives rights to access and correct your data.
10:08 ✓✓

👧 Riya
Can researchers use data without exposing people?
10:10

🤖 Bolt
Anonymisation 🎭: remove names and identifiers. But combining datasets can sometimes re-identify people!
10:12 ✓✓

Kabir

Any smarter methods?

10:14

Bolt

Federated learning 📱: the model trains on your device, and only updates leave, not your raw data.

10:16 ✓✓

Riya

Minimise, anonymise, consent, protect 🔒!

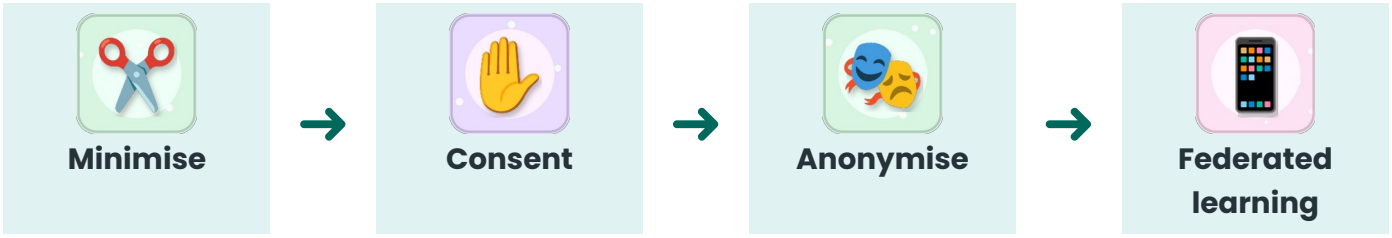
10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI runs on data, so privacy is critical. India's DPDP Act 2023 requires consent and purpose limits. Minimisation, anonymisation and federated learning protect people.

🔄 How it works



INDIA SPOTLIGHT

India's Digital Personal Data Protection Act, 2023 sets rules for how organisations collect and use personal data.

📖 CONCEPT EXPLAINED: Privacy in an AI World

💡 In this chapter

AI systems learn from data, and much of that data is about people: what we buy, search, say, where we go and even our health. Privacy is the right to control information about ourselves. This chapter explains why privacy matters more in the age of AI, how India's Digital Personal Data Protection Act 2023 sets rules, and which technical methods, such as minimisation, anonymisation and federated learning, help protect people.

🔒 1. Why privacy matters more with AI

AI can infer sensitive information that you never explicitly shared. Purchase patterns may reveal pregnancy; typing speed and app usage may hint at health conditions; social media likes may predict political views. Combining many small, harmless-looking pieces of data can build a detailed profile of a person, which can be used for targeting, manipulation or discrimination.

Large AI models are also trained on vast amounts of text and images collected from the internet, which may include personal information. Data breaches can expose millions of records at once. Privacy is therefore not just about secrets; it is about power, dignity, safety and the freedom to explore, learn and make mistakes without being constantly profiled.

🤔 PAUSE & THINK

List three pieces of information about you that an app could infer without you directly telling it.

⚖️ 2. India's DPDP Act 2023: key ideas

The Digital Personal Data Protection Act, 2023 sets rules for how organisations, called data fiduciaries, process digital personal data of individuals, called data principals. In general, organisations need a lawful basis, usually the person's consent, which should be free, specific, informed and unambiguous, given through a clear notice describing what data is collected and why.

Individuals have rights such as accessing information about their data, correcting and erasing it, and seeking grievance redressal. Organisations must protect data with

reasonable security safeguards, report breaches and process children's data with verifiable parental consent and extra care. A Data Protection Board handles complaints and can impose penalties. Detailed rules continue to be notified, so students should check official sources for current details.

✏️ **WORKED EXAMPLE**

A school app asks for students' location at all times 'to improve services'. Under DPDP principles, the school should explain the specific purpose, collect only what is needed for that purpose, obtain proper consent (parental for minors) and allow withdrawal.

✂️ 3. Data minimisation and purpose limitation

Data minimisation means collecting only the data that is necessary for a clearly defined purpose. If an attendance app only needs to know whether a student is present, it does not need their exact birth date, home address or contacts. Every extra piece of data creates risk without adding value.

Purpose limitation means using data only for the purpose people agreed to, not quietly reusing it for advertising or selling it to others. Storage limitation means deleting data when it is no longer needed. These principles, built into good system design from the start, are often called 'privacy by design'.

🤔 **PAUSE & THINK**

Look at the permissions requested by one app on your phone. Which ones are truly necessary for its main purpose?

🎭 4. Anonymisation and its limits

Anonymisation tries to remove information that identifies individuals, such as names, phone numbers and ID numbers, so that data can be used for research or analysis without exposing people. Pseudonymisation replaces identifiers with codes, which can be reversed with a key and therefore offers weaker protection.

Anonymisation is harder than it seems. Researchers have shown that combinations of quasi-identifiers, such as birth date, gender and postal code, can uniquely identify a large share of people. Linking an 'anonymous' dataset with public information can re-identify individuals. Stronger techniques include generalising data (age ranges instead of birth dates), adding statistical noise (differential privacy) and strict access controls.

✏️ **WORKED EXAMPLE**

Instead of 'Born 14-03-2009, Female, PIN 629001', store 'Age 15–17, District: Kanyakumari'. The data remains useful for analysis but is much harder to link to one person.

📱 5. Privacy-preserving AI: federated learning and more

Federated learning trains AI models without collecting everyone's raw data in one place. The model is sent to devices, such as phones or hospital computers; each device trains on its own local data and sends back only model updates, which are combined centrally. Keyboard prediction on some phones uses this approach, so your messages stay on your device.

Other privacy-preserving techniques include differential privacy (adding carefully calibrated noise so individuals cannot be singled out), encryption, on-device AI and secure computation. None is perfect, and each has trade-offs with accuracy and cost. Combined with strong laws, honest design and informed users, they help make AI both useful and respectful of privacy.

🖐️ **TRY IT**

Design a privacy-first plan for a school health-check app: list the data needed, the purpose of each item, how consent is obtained, how data is stored and when it is deleted. Mark which DPDP principles each choice supports.

✅ Key takeaways

- AI can infer sensitive information and build detailed profiles, making privacy vital.
- India's DPDP Act 2023 requires consent, purpose limits, security and rights for individuals.
- Data minimisation, purpose limitation and storage limitation are core privacy-by-design principles.
- Anonymisation can fail through re-identification; generalisation and noise help.
- Federated learning and differential privacy enable AI with less exposure of raw data.

? Quick check: answer in one or two lines.

1. What is a data fiduciary under the DPDP Act?

2. Explain data minimisation with an example.

3. Why can removing names fail to anonymise data?

4. How does federated learning protect privacy?

AI CHRONICLES: The Anonymous List That Wasn't

Riya | Dr. Nisha | Phones

SCENE 1

A clinic shared an 'anonymised' dataset for a student health project: names removed 🎭.

Riya noticed birth date, pin code and gender together could identify individuals. A privacy risk!

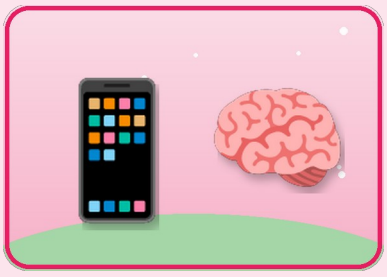
SCENE 2

SCENE 3

She informed Dr. Nisha, referring to consent and purpose limits under the DPDP Act.

They minimised the data: age ranges instead of birth dates, district instead of pin code.

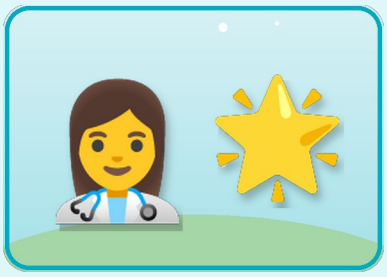
SCENE 4



SCENE 5

For a future app, Riya proposed federated learning: the model learns on phones, raw data stays there.

Dr. Nisha thanked her. "Protecting patients is part of helping them."



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya writes a privacy checklist for the clinic	**The clinic shares full data online**	**The phones keep the secrets**

🧐 Story check: answer the questions.

1. Why wasn't removing names enough?

2. How was data minimised?

3. How does federated learning protect privacy?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

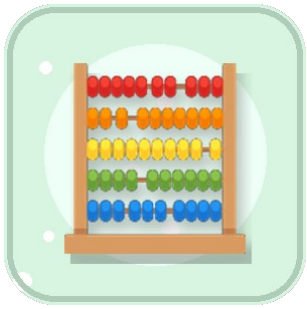
LOOK AROUND YOU: Privacy-preserving AI

Keyboards learn on-device

Hospitals anonymise records

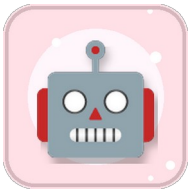
Apps collect only what's needed

Encryption in transit



Noise added to statistics

Right to erasure



BOLT SAYS:
With federated learning, I learn from your phone without your data ever leaving it!

FUN FACT!
Under the DPDP Act, organisations need verifiable parental consent to process children's data!

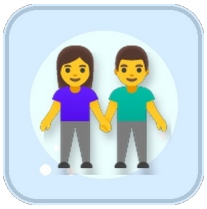
👀 **I spy AI! Draw one AI helper you saw today.**

🧪 LAB ACTIVITY: Re-identification Challenge

🎒 You need:

'Anonymous' dataset

Clue cards



Teams

👣 Steps:

1		Get a table with names removed: age, pin code, school, hobby.
2		Use clue cards to guess who's who.
3		Discuss: was it truly anonymous?
4		Propose fixes: remove/generalise columns.

THINK & TALK

Why isn't removing names enough to protect privacy?

🖍️ **Draw what you did!**

WORKSHEET: Privacy in an AI World (Part A)

Name: _____ Date: _____

1. Match the technique to its idea.

Anonymisation ●

Minimisation ●

Federated learning ●

Encryption ●

● Scramble data in transit

● Remove identifying details

● Collect only what's needed

● Train where data lives

2. Privacy-friendly or risky design? Tick ✓.

	Friendly	Risky
Collect only email for a newsletter		
Torch app tracking location		
On-device learning		
Uploading all contacts by default		

WORKSHEET: Privacy in an AI World (Part B)

Name: _____ Date: _____

1. Is it personal data? Tick ✓.

	Personal	Not personal
Full name + phone number		
Exact home location		
City's average rainfall		
Birth date + pin code + gender		
National literacy rate		

2. Fill in the blanks. Use the word box.

WORD BOX: DPDP ✿ federated ✿ anonymisation

India's 2023 data law is the _____ Act.

Removing identity is _____.

Training on devices is _____ learning.

PUZZLE & DEFINE: Privacy in an AI World

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Anonymisation

Meaning: _____

Example: _____

Federated learning

Meaning: _____

Example: _____

Minimisation

Meaning: _____

Example: _____

🧪 DEMO LAB: Privacy policy analysis + re-identification game (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Analyse data practices and see why anonymisation is hard.

🔗 **Open:** any popular app's privacy policy | 👩‍🏫 **With a teacher or parent**

👣 Steps

1	📜	Read a popular app's privacy policy. List data collected and purposes.
2	✂️	Mark data that isn't necessary (minimisation).
3	🎭	Teacher gives an 'anonymised' class list with birth month, pin code and hobby.
4	🔍	Try to identify classmates. How many can you guess?
5	📱	Explain how federated learning could protect this data.

🔬 Record your observations

Unnecessary data collected:	
Classmates re-identified:	
DPDP right that applies:	

💭 **Reflect:** Why isn't removing names always enough to protect privacy?

? QUIZ TIME: Privacy in an AI World

Circle the right answer!

Q1. Removing identity from data is...

a) Anonymisation

b) Animation

c) Announcement

Q2. India's data protection law is the...

a) RTI Act

b) DPDP Act 2023

c) Motor Act

Q3. Training without sharing raw data is...

a) Party learning

b) Upload learning

c) Federated learning

Q4. Collect only what's needed is...

a) Minimisation

b) Maximisation

c) Randomisation

Q5. Removing names alone is...

a) Always enough

b) Often not enough

c) Fun

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Privacy in an AI World

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Removing names alone guarantees anonymity.		
	The DPDP Act 2023 protects personal data in India.		
	Federated learning keeps raw data on devices.		
	Data minimisation means collecting extra data.		
	Differential privacy adds noise to protect individuals.		
	Privacy by design builds protection in from the start.		

HOME CONNECT

With parents, review the privacy policy of one app your family uses. What data does it collect?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Privacy in an AI World

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Collecting only necessary data is data ____.

(a) minimisation ☐ **(b)** maximisation ☐ **(c)** monetisation ☐

2. Removing personal identifiers is ____.

(a) animation ☐ **(b)** anonymisation ☐ **(c)** amplification ☐

3. Training on devices without sending raw data is ____ learning.

(a) supervised ☐ **(b)** deep ☐ **(c)** federated ☐

4. India's data protection law is the ____ Act 2023.

(a) DPDP ☐ **(b)** IPL ☐ **(c)** GST ☐

B. Fill in the blanks. Use the word box.

consent **re-identify** **ranges** **updates**

1. DPDP requires clear ____.

2. Combining datasets can ____ people.

3. Use age ____ instead of exact birth dates.

4. Only model ____ leave the device in federated learning.

C. Write T for True and F for False.

1. Removing names always guarantees anonymity. [ ]

2. Data minimisation reduces privacy risk. [ ]

3. Federated learning keeps raw data on devices. [ ]

4. The DPDP Act gives rights to access and correct data. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Minimisation	____	**(a)** Train on devices
2. Anonymisation	____	**(b)** Collect only needed
3. Federated learning	____	**(c)** Permission
4. Consent	____	**(d)** Remove identifiers

E. Answer in one word.

1. What is India's 2023 data law? _____

2. What technique trains on devices? _____

3. Collecting only needed data is? _____

F. Answer these questions.

1. Why is anonymisation sometimes not enough?

2. Explain federated learning in simple words.

G. Case-based questions. Read and answer.

An 'anonymised' clinic dataset still had birth date, pin code and gender. Riya flagged re-identification risk; they switched to age ranges and districts and proposed federated learning.

1. The combination of fields created a ____ risk.

(a) re-identification ☐ **(b)** speed ☐ **(c)** colour ☐

2. Switching to age ranges is ____.

(a) maximisation ☐ **(b)** minimisation ☐ **(c)** deletion of AI ☐

3. Federated learning keeps raw data ____.

(a) on a public site ☐ **(b)** in emails ☐ **(c)** on devices ☐

H. THINK DEEPER

Design a privacy-first plan for a school attendance app: what data, why, how protected and which DPDP principles apply.

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. Removing identity from data:

a) Anonymisation | b) Animation | c) Announcement

Q2. Pin code can be a...

a) Gift | b) Proxy for sensitive traits | c) Song

Q3. India's data protection law:

a) RTI Act | b) Motor Act | c) DPDP Act 2023

Q4. Equal approval rates across groups:

a) Demographic parity | b) Randomness | c) Harmony

Q5. Training on devices without sharing data:

a) Upload learning | b) Federated learning | c) Party learning

Q6. Adding noise to protect individuals:

a) Loud privacy | b) Music privacy | c) Differential privacy

WORD SEARCH: AI CARE

Find these words. They go across → or down ↓.

AUDIT	PRIVACY	FAIRNESS	CONSENT	ENCRYPT	ETHICS

P U T I W C O F L C
R A U D I T M A O I
I P D G K E D I B U
V G M R G N S R E U
A S A M H C W N T O
C K M U D R C E H S
Y S G D W Y I S I K
G N R H E P W S C G
S A G F A T L T S W
W K M C O N S E N T

I finished AI CARE! Teacher's sign: _____

MY PICTURE DICTIONARY (1/2)

Activation
Output decision function

Anonymisation
Removing identity

Architecture
Network's structure

Array
Grid of numbers

Audit
Systematic check

Context
Background info given

Data cleaning
Fixing messy data

DataFrame
Table in pandas

Epoch
One pass over data

Fairness metric
Measures equal treatment

Federated learning
Train without sharing data

Few-shot
Prompt with examples

Forecasting
Predicting future values

Governance
Rules & oversight

Gradient
Direction of steepest change

Innovation
New valuable solution

Licence
Permission to use

Literature review
Survey of past studies

MY PICTURE DICTIONARY (2/2)

Loss
Measure of error

Milestone
Checkpoint on the path

Minimisation
Collect only what's needed

MLOps
Running models in production

Neuron
Weighted sum unit

Peer review
Expert checking

Pipeline
Chain of tools/steps

Predictive maintenance
Fix before failure

Product manager
Leads what gets built

Protein folding
How proteins take shape

Provenance
Origin of content

Representation
Fair share in data

ROI
Return on investment

Simulation
Virtual experiment

Skill gap
What you still need

SWOT
Strengths-Weaknesses-Opportunities-Threats

Temperature
Randomness setting

Variable
Factor you measure

❓ OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. z = 2·1 + (−1)·3 + 1. What is z?

a) 0

b) 2

c) -2

Q2. Sigmoid outputs values between...

a) 0 and 10

b) 0 and 1

c) −∞ and ∞

Q3. Loss (w−3)², w=0, lr=0.1. New w after one step?

a) 3

b) -0.6

c) 0.6

Q4. Train 98%, test 60% indicates...

a) Overfitting

b) Underfitting

c) Perfect model

Q5. df.groupby('city')['sales'].sum() gives...

a) Total rows

b) Sales total per city

c) An error

? OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. 99% accuracy on 99%-'No' data may hide...

a) Success

b) Colour issues

c) Class imbalance

Q7. 'Think step by step' is called...

a) Chain-of-thought

b) Role-play

c) Zero-shot

Q8. For creative brainstorming, use temperature...

a) 0.0

b) Higher

c) None

Q9. AI screening of molecules is called...

a) Sketching

b) Gaming

c) Virtual screening

Q10. Savings ₹12L, cost ₹4L. ROI is...

a) 200%

b) 300%

c) 8%

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. Who deploys and monitors ML models?

a) Policy analyst

b) ML engineer

c) Illustrator

Q12. 'Free online courses' in SWOT is an...

a) Strength

b) Weakness

c) Opportunity

Q13. Ice-cream sales & drownings both rise in summer. This shows...

a) Correlation, not causation

b) Ice-cream causes drowning

c) Nothing

Q14. APA reference starts with...

a) The date only

b) Author's name

c) A link

Q15. Using CC-BY content requires...

a) Payment always

b) Nothing

c) Giving credit

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. Group A 45/60, Group B 30/60. Ratio of rates:

a) 0.67

b) 1.0

c) 0.5

Q17. Removing 'gender' column guarantees fairness?

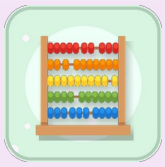
a) Yes

b) No, proxies remain

c) Only on Mondays

Q18. Age + pin code + gender together can...

a) Celebrate



b) Compute GDP

c) Re-identify people

Q19. Children's data under DPDP Act needs...

a) Parental consent

b) No consent

c) A gift

Q20. Building protection in from the start is...

a) Privacy patching

b) Privacy by design

c) Privacy by chance

MY SCORE: I got _____ / 20. Colour a star for every 4 right!

CERTIFICATE

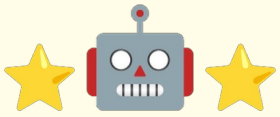
of AI Explorer

This is to say that

_____

of Class 11 finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. Inside Intelligent Machines	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Input - Receives raw features; Hidden - Learns intermediate patterns; Activation - Adds non-linearity; Output - Gives the final prediction **Part A 2:** 0.9, 1, 0.1 **Part B 1:** f(z) = z: Linear; ReLU: max(0, z): Non-linear; Sigmoid: Non-linear; f(z) = 2z + 1: Linear **Part B 2:** 1-Receive inputs, 2-Weighted sum + bias, 3-Apply activation, 4-Pass output to next layer **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** 0, images, language, hidden **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** ReLU; CNN; Bias **Story next:** Any sensible choice; best: They test it on real drone footage **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
2. Predict, Learn & Improve	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-F 5-T 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Make predictions, 2-Calculate loss, 3-Compute gradients, 4-Update weights **Part A 2:** Train 60%, test 58%: Underfit; Train 99%, test 70%: Overfit; Model too simple: Underfit; Memorises noise: Overfit **Part B 1:** -6, 0.6, 1.08 **Part B 2:** Training - Learn the weights; Validation - Tune settings; Test - Final unbiased check; Loss - Measures error **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** overshooting, overfitting, early, training **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Loss; Epoch; Gradient descent **Story next:** Any sensible choice; best: Kabir tries different architectures **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

3. Python for Smart Solutions	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** read_csv - Loads a CSV file; dropna - Removes missing rows; mean - Computes averages; fit - Trains a model **Part A 2:** [2 4 6], 6.0, 3 **Part B 1:** 1-Load data, 2-Clean data, 3-Train model, 4-Evaluate **Part B 2:** Age column with blanks: Needs cleaning; Duplicate rows: Needs cleaning; Consistent units, no gaps: Clean; Height in cm & m mixed: Needs cleaning **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** csv, duplicates, predict, 70 **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** pandas; fit(); NumPy **Story next:** Any sensible choice; best: Riya builds a dashboard for teachers **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
4. The Art of Prompting	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 'Classify this review as positive or negative.': Zero-shot; 'Happy→Positive, Sad→Negative. Now: Excited→?': Few-shot; 'Translate to Hindi.': Zero-shot; Prompt with 3 sample answers: Few-shot **Part A 2:** Role - Sets expertise & tone; Step-by-step - Better reasoning; Few-shot - Shows the pattern; Format - Structured answers **Part B 1:** 1-Set the role, 2-State task & context, 3-Specify format, 4-Test & refine **Part B 2:** few-shot, temperature, context **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** consistent, step, refine, checking **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Few-shot; Temperature; Prompt engineering **Story next:** Any sensible choice; best: They write a prompt guide for juniors **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

5. AI Powers Innovation

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Biology - Protein structures; Medicine - Drug screening; Energy - Battery materials; Astronomy - Finding exoplanets
Part A 2: innovation, protein folding, simulation
Part B 1: 1-Collect known data, 2-Train a model, 3-Generate predictions, 4-Validate in lab
Part B 2: Screen millions of molecules: AI can help; Approve a medicine for public use: Needs humans; Predict protein shapes: AI can help; Run final clinical trials: Needs humans
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: verified, batteries, simulations, innovation
Exercise C: 1-T 2-F 3-T 4-F
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Simulation; Protein folding; Domain experts
Story next: Any sensible choice; best: Riya joins a summer research programme
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

6. AI Across Industries

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Banking - Credit risk; Manufacturing - Predictive upkeep; Agriculture - Yield prediction; Retail - Demand & pricing
Part A 2: Lots of data, repeated decisions: Good fit; No data, one-time decision: Poor fit; High-volume routine checks: Good fit; Decisions needing deep empathy alone: Poor fit
Part B 1: 300%, 100%, 50%
Part B 2: Circle: Lots of historical data, Frequent repeated decisions, High cost of errors, Measurable outcome
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: cost, temperature, data, waste
Exercise C: 1-T 2-F 3-T 4-F
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Return on investment; Predictive maintenance; Forecasting
Story next: Any sensible choice; best: Kabir studies AI for railways
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

7. AI Career Universe	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Research - Invents new methods; ML engineer - Deploys & scales models; PM - Decides product direction; Policy - Shapes rules & laws **Part A 2:** MLOps, product manager, governance **Part B 1:** ML engineer: Technical; AI policy analyst: Non-technical; Research scientist: Technical; Product manager: Non-technical **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** retraining, papers, build, mix **Exercise C:** 1-T 2-F 3-T 4-T **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Models in production; Product manager; Governance/policy **Story next:** Any sensible choice; best: Both shadow professionals for a week **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
8. Create Your Career Blueprint	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Good at maths: Internal; Free online AI courses: External; Poor time management: Internal; Highly competitive exams: External; Strong communicator: Internal **Part A 2:** Child's own work **Part B 1:** Strength - Problem-solving; Weakness - Procrastination; Opportunity - Free online courses; Threat - Tough competition **Part B 2:** SWOT, milestone, skill gap **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** backup, threat, opportunity, cage **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** SWOT; Milestone; Skill gap **Story next:** Any sensible choice; best: Riya helps classmates make blueprints **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

9. AI Research Desk	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-F 4-T 5-T 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Introduction, 2-Methods, 3-Results, 4-Discussion **Part A 2:** Hours of screen time: Independent; Hours of sleep: Dependent; Amount of light (experiment): Independent; Plant height measured: Dependent **Part B 1:** Literature review - Survey of past studies; Peer review - Expert checking; Correlation - Variables moving together; Variable - Factor that can change **Part B 2:** Verifying AI-suggested sources: Good; Copying AI summaries without reading: Poor; Citing every source: Good; Claiming causation from correlation: Poor **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** controlled, limitations, paper, systematically **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Independent; Peer review; Literature review **Story next:** Any sensible choice; best: Kabir repeats it with more schools **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
10. AI Creation Studio	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Write script, 2-Create visuals, 3-Record voice, 4-Edit & publish **Part A 2:** pipeline, licence, provenance **Part B 1:** Text model - Script drafting; Image model - Visual assets; Voice tool - Narration; Video editor - Final assembly **Part B 2:** Disclosing AI-generated visuals: Ethical; Using film songs without licence: Unethical; Adding captions for accessibility: Ethical; Cloning a celebrity's voice for ads: Unethical **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** publish, commercially, credentials, creator **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Pipeline; Licence; Provenance **Story next:** Any sensible choice; best: They make a film on climate refugees **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

11. Fair AI Challenge

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Data - Too few rural voices; Label - Biased humans tagged data; Design - Proxy feature like pin code; Deployment - Used where it wasn't tested
Part A 2: 80%, 40%, 40
Part B 1: Circle: Pin code, School name, First name, Neighbourhood
Part B 2: 1-Collect outcomes by group, 2-Compute metrics, 3-Judge if gaps are fair, 4-Mitigate & re-audit
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: stage, label, teams, monitoring
Exercise C: 1-T 2-F 3-T 4-T
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Proxy; Audit; Fairness metric
Story next: Any sensible choice; best: The bank runs audits every quarter
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

12. Privacy in an AI World

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-F 2-T 3-T 4-F 5-T 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Anonymisation - Remove identifying details; Minimisation - Collect only what's needed; Federated - Train where data lives; Encryption - Scramble data
Part A 2: Collect only email for a newsletter: Friendly; Torch app tracking location: Risky; On-device learning: Friendly; Uploading all contacts by default: Risky
Part B 1: Full name + phone number: Personal; Exact home location: Personal; City's average rainfall: Not personal; Birth date + pin code + gender: Personal; National literacy rate: Not personal
Part B 2: DPDP, anonymisation, federated
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: consent, re-identify, ranges, updates
Exercise C: 1-F 2-T 3-T 4-T
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: DPDP Act; Federated learning; Minimisation
Story next: Any sensible choice; best: Riya writes a privacy checklist for the clinic
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

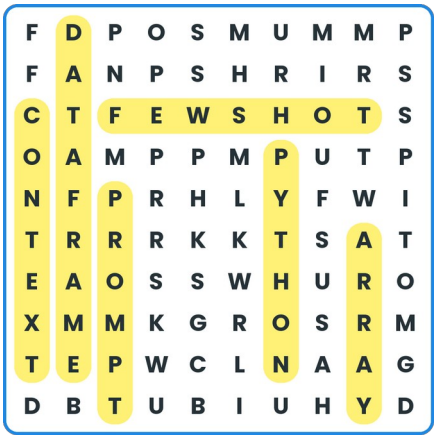
Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

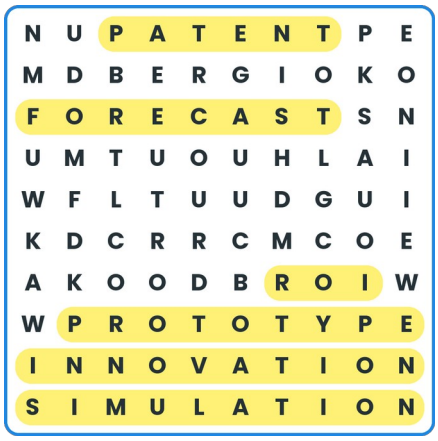
Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

WORD SEARCH ANSWERS

AI DISCOVER



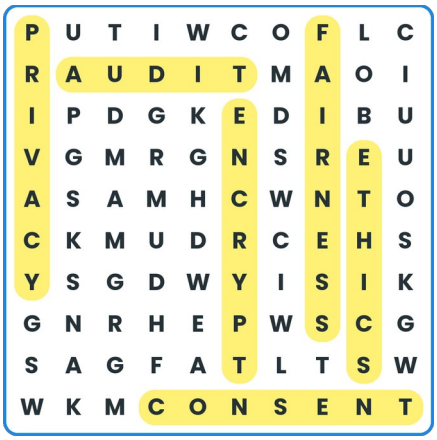
AI CONNECT



AI SOLVE

AI RISE

AI CREATE



AI CARE

MY NOTES

MY NOTES

MY NOTES

MY NOTES

MAMMA ADA CENTRE

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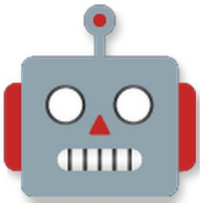
Innovate Today • Lead Tomorrow



About this book

Concept Edition: a college-level AI book for super-senior learners of Class 11, on the 6-month GAIO journey.

- 📖 Concept Explained: in-depth chapters
- 💬 Tech Talk: ideas as a group chat
- 🧪 Demo Lab: real tools, code & audits
- 📚 AI Chronicles: 12 illustrated stories
- 📝 Standard, case-based & HOTS questions

AI Basics | **Communicate with AI** | **AI Applications**

AI & Career | **AI Tools** | **Responsible AI**

Think | Explore | Create | A Better Tomorrow

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